Getting started with the STEVAL-WLC38RX wireless power receiver evaluation board for 5 W Qi BPP and 15 W Qi EPP applications

Introduction

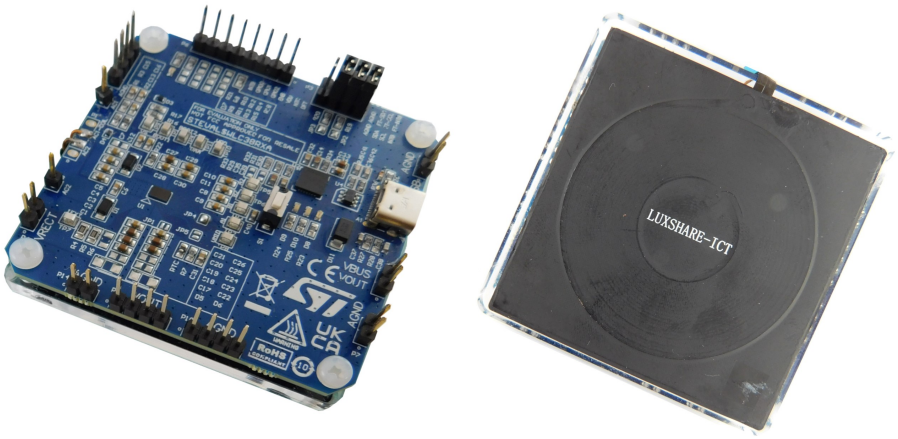
The STEVAL-WLC38RX evaluation board is based on STWLC38 and is designed for wireless power receiver applications. It allows users to quickly start their 5W Qi-BPP and 15W Qi-EPP compatible wireless charging receiver projects.

The STWLC38 wireless power receiver can manage up to 15 W of power according to the Wireless Power Consortium's Extended Power Profile (EPP).

The integrated circuit requires only a few external components and offers high design flexibility. Using an on-board USB-to-I^2C bridge, the user can monitor and control the STWLC38 using the STSW-WPSTUDIO graphical user interface (GUI).

STEVAL-WLC38RX includes several safety mechanisms providing overtemperature (OVTP), overcurrent (OCP) and overvoltage (OVP) protections as well as foreign object detection (FOD) for reliable designs.

Figure 1. STEVAL-WLC38RX evaluation board



1 Get started

To get started with STEVAL-WLC38RX, you will need the following items to use the reference design kit:

- Evaluation kit components:
  - STEVAL-WLC38RX board
- Additional hardware
  - STEVAL-WBC86TX for the best results for a 5 W turnkey wireless charging system or any other Qi BPP or Qi EPP compliant transmitter available on the market
  - Electronic load
  - Windows PC for measurements and configurations
- Software:
  - STSW-WPSTUDIO Wireless Power Studio PC GUI installation package
  - I²C drivers
- Application notes:
  - Coil selection guide: AN5961
  - GUI guide: UM3164

2 Reference design specifications

Target specifications of STEVAL-WLC38RX evaluation board are as follows:

Table 1. reference design specifications

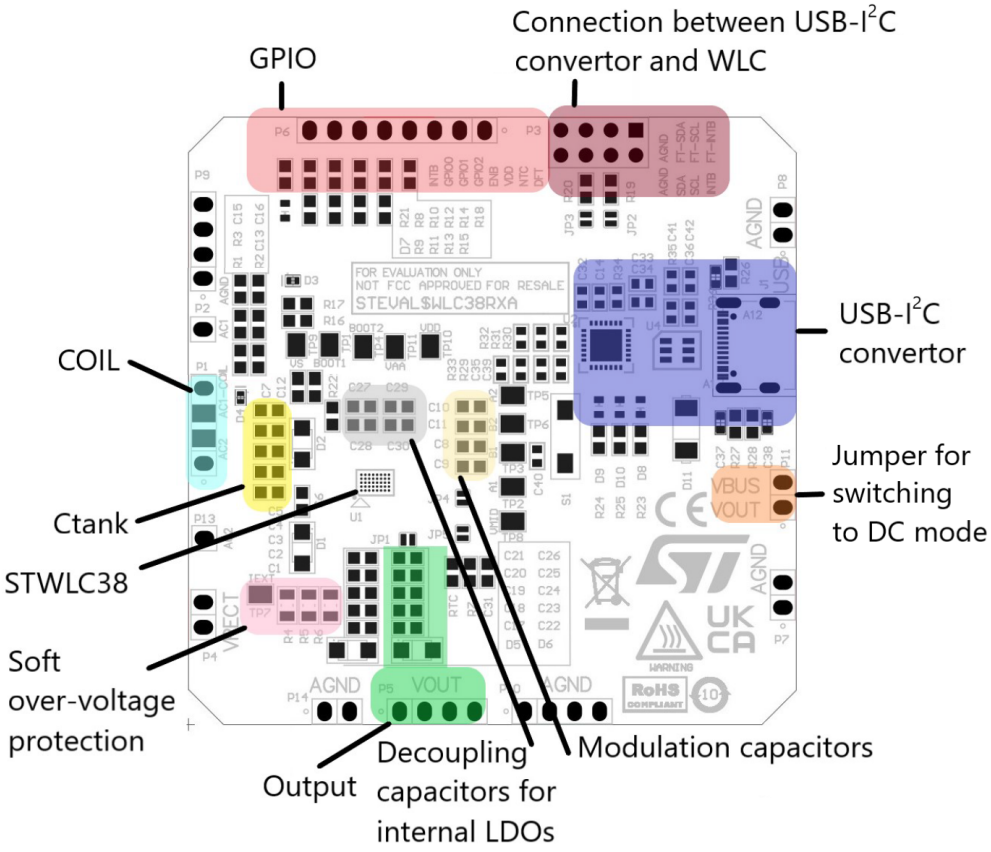
Parameter	Description
Qi compatibility	Qi 1.3 BPP and EPP protocol
Rx application PCB area	24 mm x 34 mm
Rx coil specifications	Inductance 8 uH, DCR 190 mΩ, ACR 225 mΩ Dimensions 57 mm x 57 mm x 0.42 mm (LD81FP008-1H)
Output voltage (Vout, continuous operation)	5 V Baseline Power Profile / 9 V Extended Power Profile
Output current (Iout, Load)	Recommended current load range is 0.1 – 1 A
Host MCU	STM32 used as a reference, the reference I^2C driver can be ported to any other MCU family
USB-to-I^2C converter	Embedded in the evaluation board
Efficiency	77.6% (5 W operation) with STEVAL-WBC86TX 81% (peak efficiency) with STEVAL-WBC86TX at 3 W
Applicable charging gap between Tx and Rx coils(z-distance)	3 – 13 mm (5 W output) with STEVAL-WBC86TX transmitter, maximum 16 mm – stable communication without output enabled
Misalignment between Tx and Rx coils (x-y offset from the center)	12 mm for 5 W output, 18 mm for stable communication without output enabled

3 Overview of the Board

The STEVAL-WLC38RX evaluation board is optimized for performance. The board features:

- STWLC38 wireless power receiver chip with BPP and EPP compliant firmware
- Very few external components, optimized BOM and PCB space
- On-chip high efficiency rectifier
- Support for external NTC for thermal monitoring
- On-chip thermal management and protections
- WPC Qi extended power profile (EPP, 15 W) compatible receiver chip
- On board USB-to-I^{2}C converter
- USB Type-C® connector for PC GUI connection

Figure 2. Main board elements



- Series resonant capacitors and the receiving coil together form a resonant circuit. This circuit is in charge of receiving the power signal, so any components/tracks involved should be rated accordingly
- USB/I^{2}C converter–provides a communication channel between a PC and STWLC38. LED D8 (Red) indicates that the I^{2}C converter is powered, D9 (yellow) indicates that STWLC38 is connected to the GUI. LED D10 (Green) indicates that I^{2}C communication was initialized and is ready. Switch S1 resets the converter. Note that header P3 connects the converter's I^{2}C signals to the STWLC38 I^{2}C signals. Short the corresponding pins with a jumper to establish a connection between the two ICs.
- Red LED (D7)–indicates STWLC38 core power status (LED is on when the core is powered).

3.1 Test points

STEVAL-WLC38RX features several headers and test points to provide easy access to key signals.

Figure 3. Headers and test points

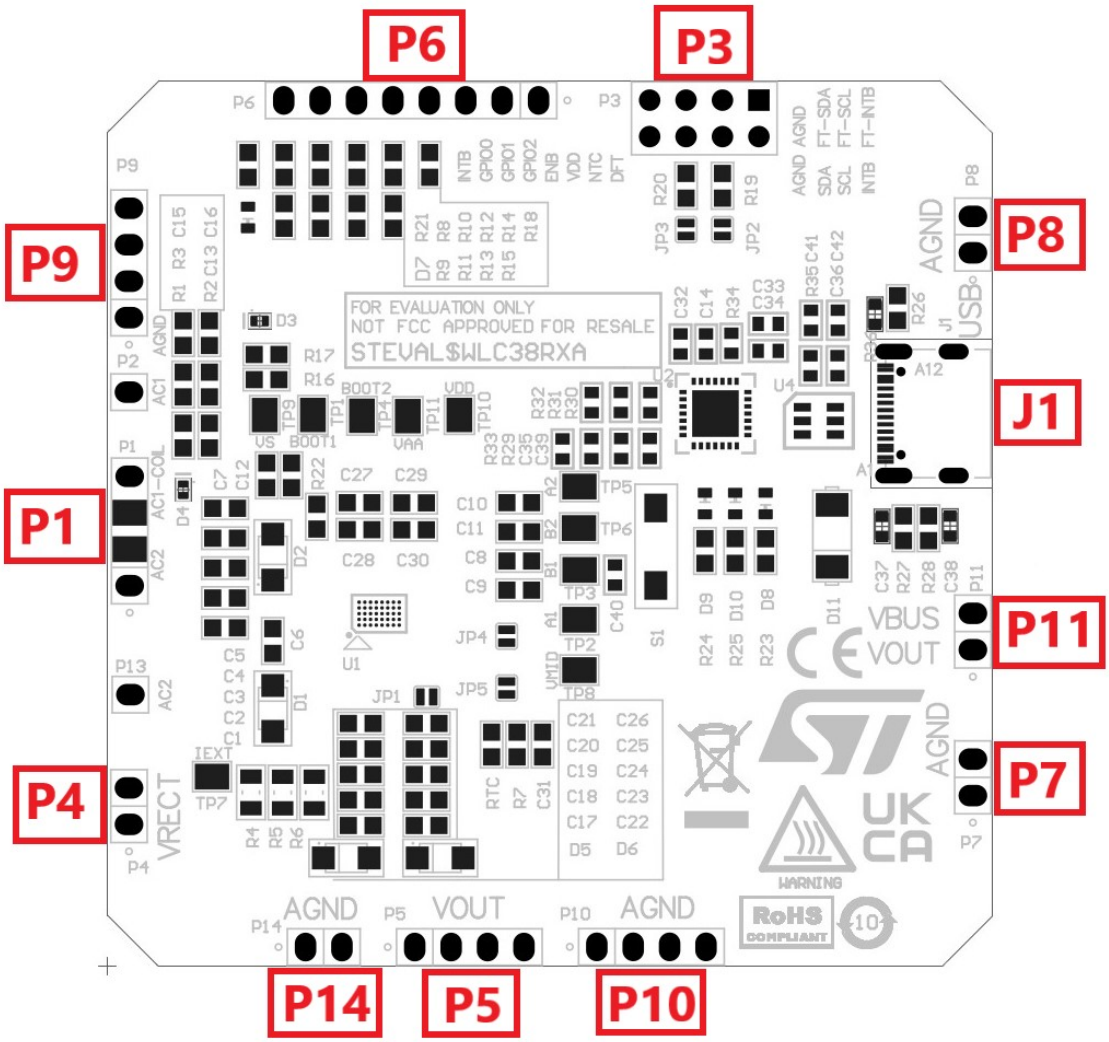


Table 2. header and test point descriptions

Connector	Name	Description
P1	Rx Coil	4-pin Header: for mounting External RX Coil: Pin 1,2 AC2; Pin3,4 AC1_COIL
P3	I^2C	2x4 pin Header: AGND, SDA, SCL, INTB
P6	Control signals	8 pin Header: Programmable GPIOs, ENB, INTB, NTC
P4	VRECT	2-pin Header: VRECT output
P5	VOUT	4-pin Header: VOUT output
P7, P8, P14	GROUND	2-pin Header: Ground connection
P9, P10	GROUND	4-pin Header: Ground connection
P11	VBUS-VOUT	2 Pin Header: VBUS to VOUT Wire Jumper

Connector	Name	Description
J1	USB-C	USB-C Connector

3.2 Basic operating modes

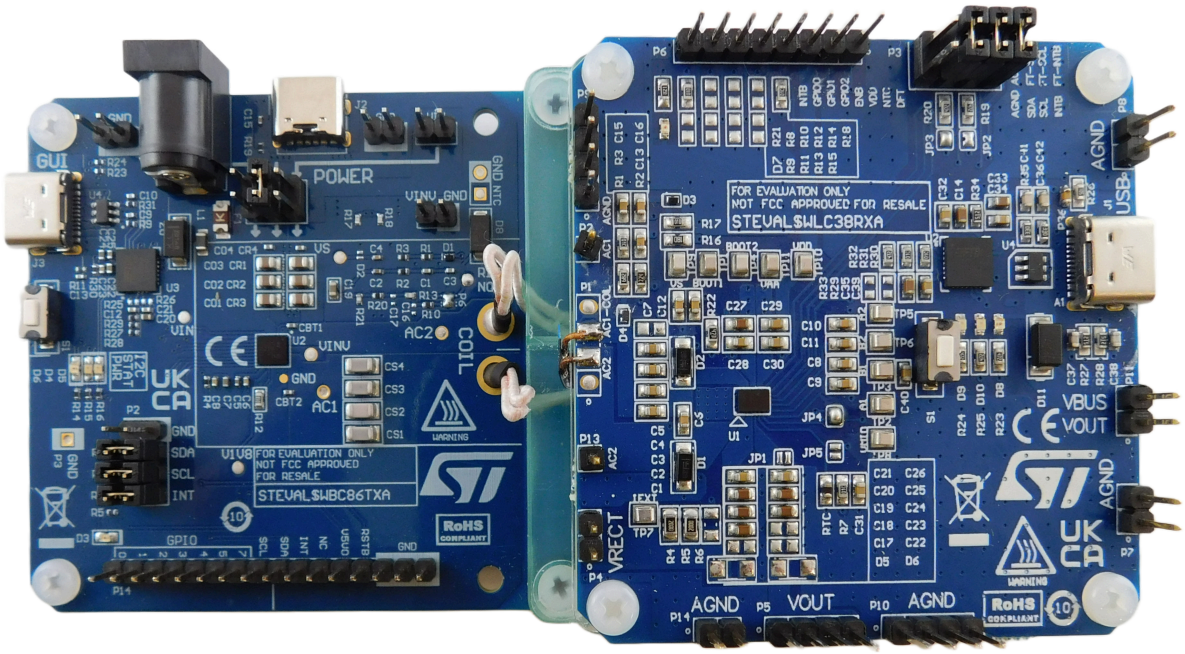
The receiver generally works in two modes - DC (also called Standalone) mode and AC mode.

DC mode is characterized by the wireless receiver being powered from a DC power supply such as external power supply or USB. This mode is used for Patch and Configuration loading (described later). Before entering this mode, device shall not be powered from power transmitter (device is put away from power transmitter or power transmitter is being powered off).

AC mode is used when the wireless receiver is powered from a wireless power transmitter. A stable and ongoing power transfer between the transmitter and the receiver is indicated by LED (D7). Make sure that Rx is not powered by external DC power before entering this mode.

3.3 Functional check

Figure 4. **Transmitter and receiver boards setup**



The first sign of an ongoing power transfer is the D7 LED, as this LED indicates the internal power supply of the device is ready. A continuously shining LED indicates a stable power supply, while a blinking or inactive LED indicates an unstable and/or insufficient power supply.

The LED D7 is permanently OFF means there is no power transfer between the transmitter and the receiver. Check if the power supply of the transmitter is connected and working properly. Power transfer termination may also be caused by a large enough misalignment (from an ideal center-to-center position) between receiver and transmitter coils. This issue is, to a certain degree, mitigated by an [ARC mode] feature, which improves spatial freedom of the charging.

The LED D7 is blinking means some power is transferred, but the amount is insufficient, or the transfer is unstable. One of the reasons for such instability may be communication issue between the power transmitter and the power receiver. The instability may as well be caused by one of the protections (OVTP, OCP, OTP) being triggered.

The built-in USB-I^{2}C converter features three LEDs. **LED D8 (RED)** indicates presence of power from the USB connector (J1). **LED D9 (YELLOW)** indicates that STWLC38 is connected to the GUI. **LED D10 (GREEN)** indicates that I^{2}C communication was initialized and is ready.

4 Graphical user interface (GUI)

STWLC38 (and other ST wireless charging devices) can be configured using the STCHARGE Wireless Power Studio GUI (STSW-WPSSTUDIO). The GUI can also be used to control, monitor and program the device.

For more information, please see STCHARGE Wireless Power Studio User Manual.

4.1 Connecting STWLC38 to PC GUI

Step 1. Connect the board to a PC by plugging a USB-C cable into the connector J1.

Make sure the STWLC38 I²C pins are connected to the USB-C connector. This can be done by shorting the appropriate signals (SDA, SCL, INT) on header P3.

Step 2. Power on the STWLC38 before connecting to the GUI, this can be done in one of the following ways:

- **Step 2a.** Place it onto a power transmitter. The device operates in Rx mode.
- **Step 2b.** Switch it into DC mode by connecting a DC power supply to the STWLC38 VOUT pin. The voltage must be no higher than 3 V.
- **Step 2c.** Switch it into DC mode by connecting the STWLC38 vout signal to the VBUS pin of the converter board.

Note: Please note that the Rx registers will be available only if the device is operating in Rx mode, DC mode is mainly used for updating patch and configuration file.

Step 3. Power up the board and open the GUI on your PC. Click the [**Connection**] button in the top menu.

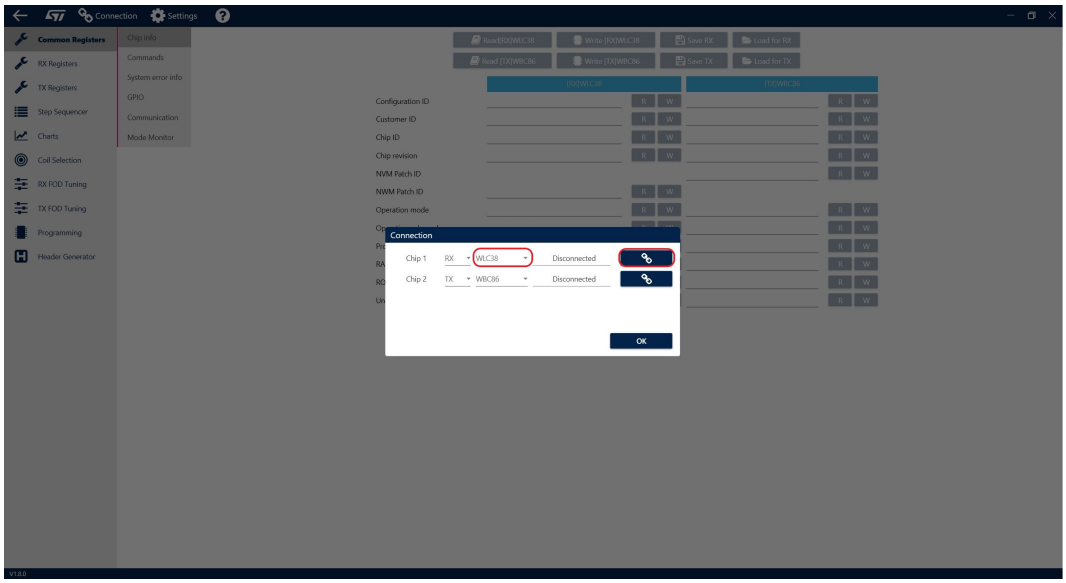
Up to two devices can be connected at a time. This allows the user to control both Rx and Tx at the same time).

Figure 5. PC GUI main screen



Step 4. Select [**WLC38**] as the Rx and click the [**Connect**] button on the right side of the window.

Figure 6. Device selection and connection



4.2 Patch and configuration files

Firmware of the device can be updated using a Patch file (a binary file in .memh format). The latest version of the Patch can be found at STWLC38. Updating the firmware is not required but may improve the performance of the board.

The device can be configured using a Configuration file, a binary file containing settings of all registers which can be found in the GUI. The GUI can also be used to generate a custom Configuration file, making it easier to quickly change configuration of the board and/or transfer the configuration to another board.

In GUI, registers are divided into two groups. The value changes of registers colored in RED will take effect only if a configuration file is generated and written to the device. Changes of registers colored in BLACK will take effect immediately and can be saved to the configuration file. During power-up, the values of those registers are set to the values defined in the configuration file written on the device. Registers that cannot be saved to the configuration file and whose value is lost upon reboot have this fact noted in the description.

4.3 Configuration file generation

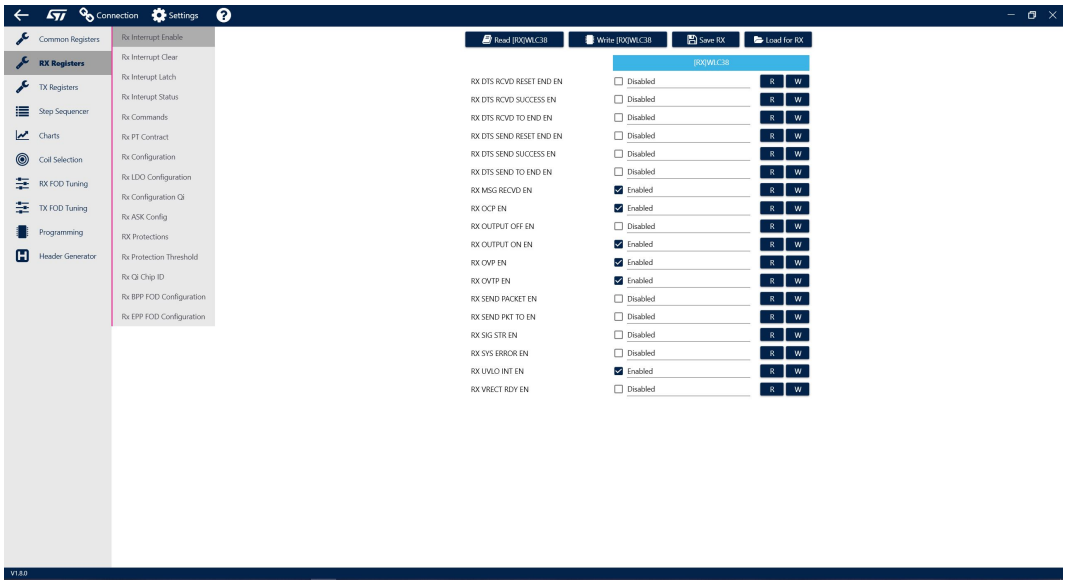
Using the STSW-WPSTUDIO makes generating the Configuration file quite simple.

Step 1. To make sure that the latest values are shown in the GUI, it is recommended to use the [**Read [Rx]WLC38**] button to dump values from the device to the GUI. Then, the required changes of the register may be made. This should be done, especially when going from DC mode to Rx mode.

Step 2. Click the [**Write [RX]WLC38**] button to save current configuration to temp memory.

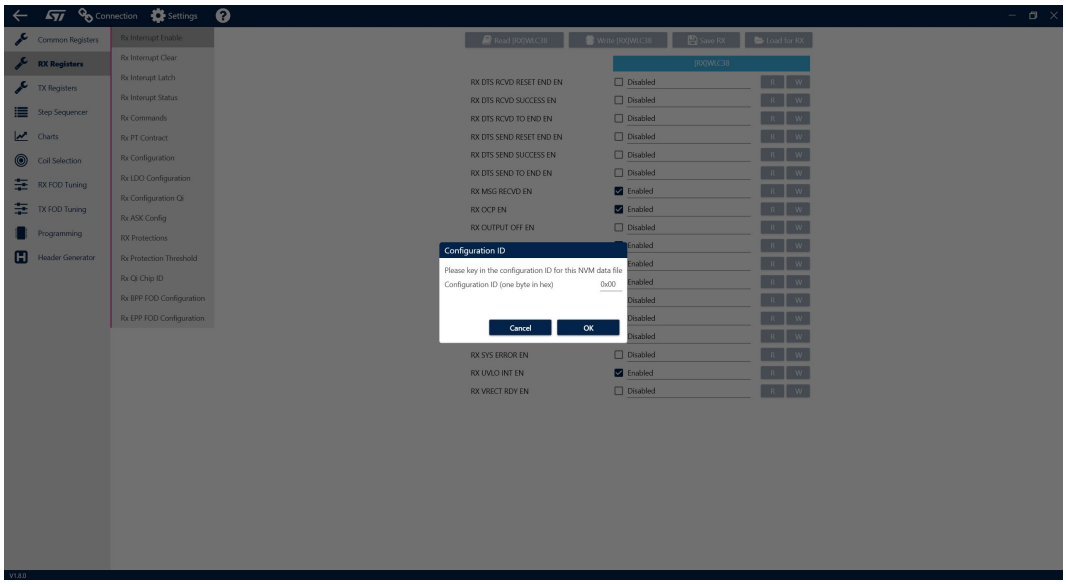
Step 3. Click the [**Save RX**] button in the RX Registers tab or the Common registers tab.

Figure 7. Save RX button



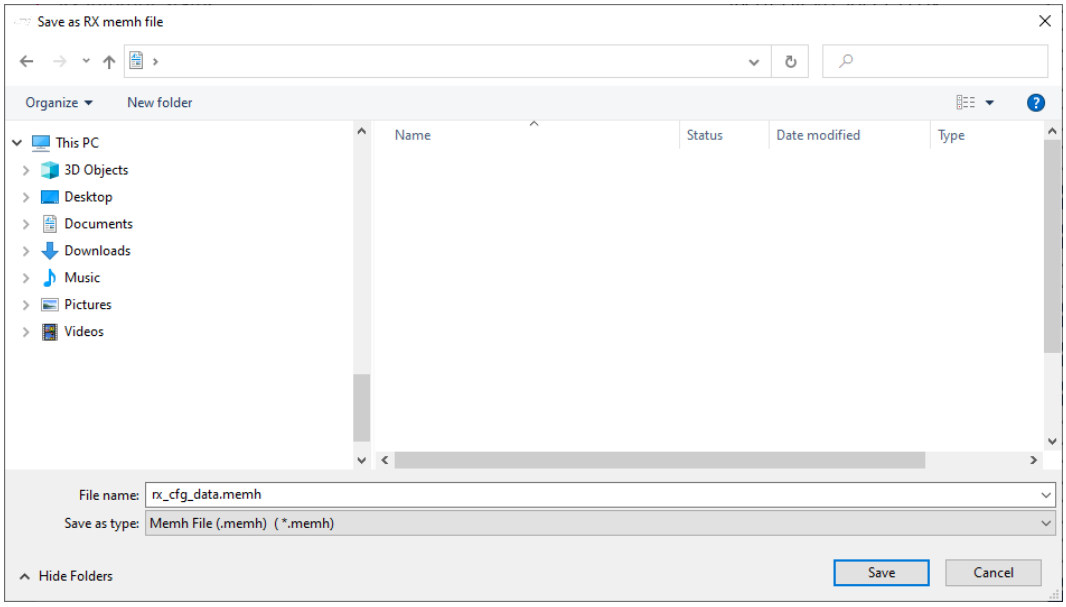
Step 4. Enter a Configuration ID number (used for version control) and press [**OK**].

Figure 8. Configuration ID field



Step 5. Choose your location for the save file.
After choosing a location, the configuration will be saved as a .memh file in the selected folder.

Figure 9. Save folder selection



4.4 Header file

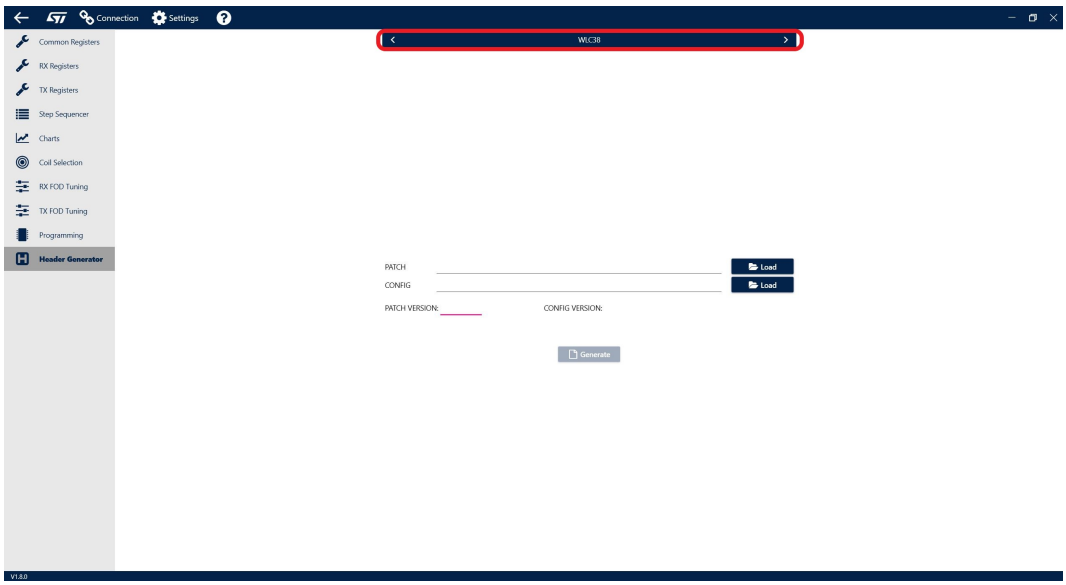
The GUI can also be used to generate a Header file, a binary .h file containing both Configuration and Patch files. The Header file makes programming the device using a host MCU easier, as both Configuration and Patch can be loaded at once by simply including the Header file in the host code.

4.5 Header file generation

A custom Header file can be generated in the Header Generator tab.

Step 1. Start by selecting [**WLC38**] in the top menu.

Figure 10. Device selection



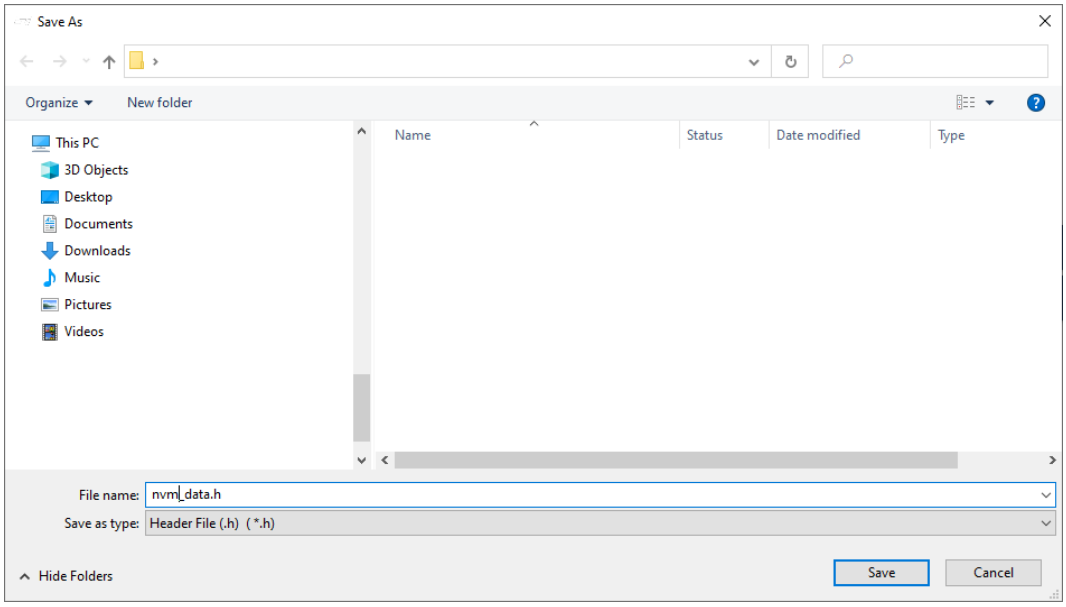
Step 2. Continue by selecting the patch and configuration files and press [**Generate**].
A pop-up window will appear, asking to confirm you have selected the correct Patch version.

Figure 11. Confirmation of patch version



Step 3. Choose a save destination.
After choosing a location, the Header file will be saved as a .h file in the selected folder.

Figure 12. Header file save



4.6 Programming the device

To program the device, the device must be switched into a so-called DC mode. Before switching to DC mode, first make sure power transfer is not active. The simplest way to achieve this is to either remove the receiver from the transmitter or to power down the transmitter. To enter DC mode, connect a DC power supply to the VOUT pins. This can be done by shorting the P11 jumper, which connects STWLC38's VOUT to the USB VBUS. Alternatively, an external power supply can be used. Voltage of the power supply should be higher than 3 V for this purpose.

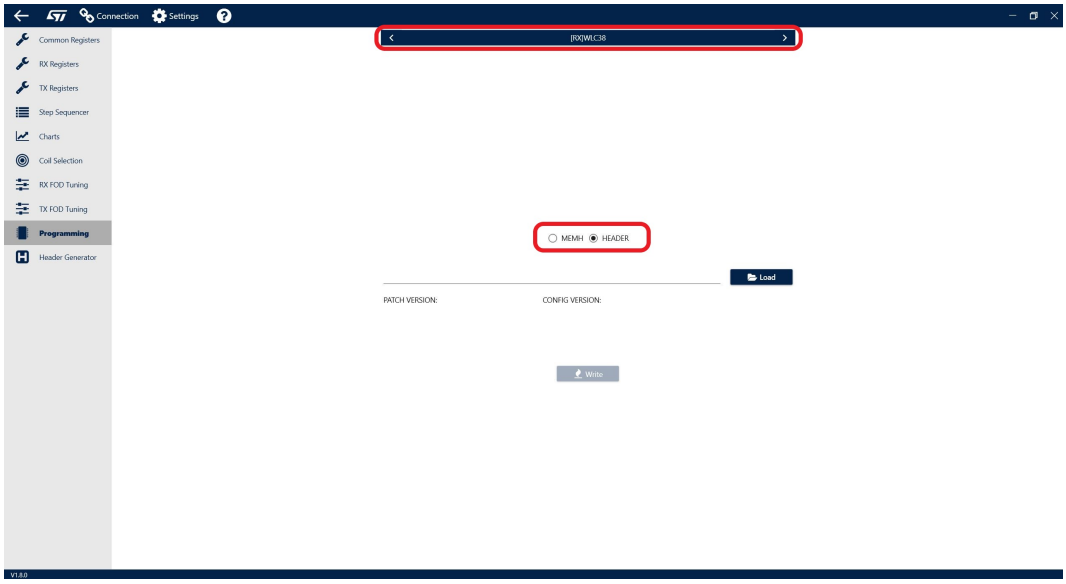
The device can be programmed in three ways – by changing the register values directly in the **GUI**, by using a **Header** file, which loads both Configuration and Patch files at once, or by loading the two **memh** files separately using the GUI.

Note: The values written into the registers via the GUI directly are only stored temporarily and will be lost upon chip reset.

Step 1. To load the **Header** file using the GUI, navigate to the Programming tab in the side menu.

Step 2. Select WLC38 in the top menu and [**HEADER**] in the toggle selector.

Figure 13. Loading the header file



Step 3. Select the .h file you want to write.
The GUI will automatically identify the Patch and Configuration files included in the Header file.

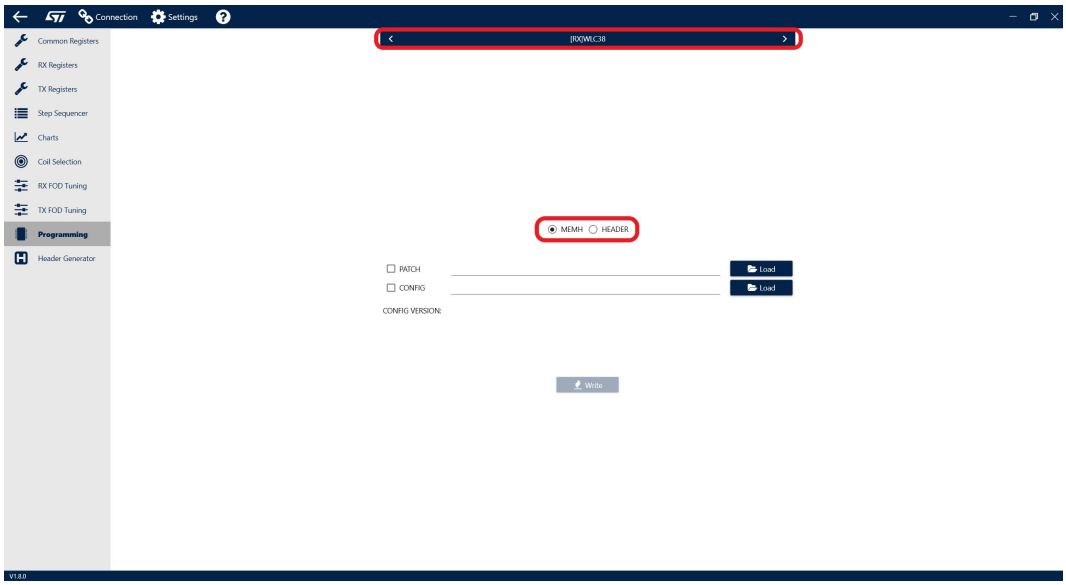
Step 4. Press the [**Write**] button to load the Header file into the device.

Step 5. To load the **memh** files (patch and configuration) using the GUI, navigate to the Programming tab in the side menu.

Note: STWLC38 has 32 kB of RRAM, which allows for multiple erase/rewrite cycles up to 1000 times.

Step 6. Select WLC38 in the top menu and [**MEMH**] in the toggle selector.

Figure 14. Loading the patch and configuration files



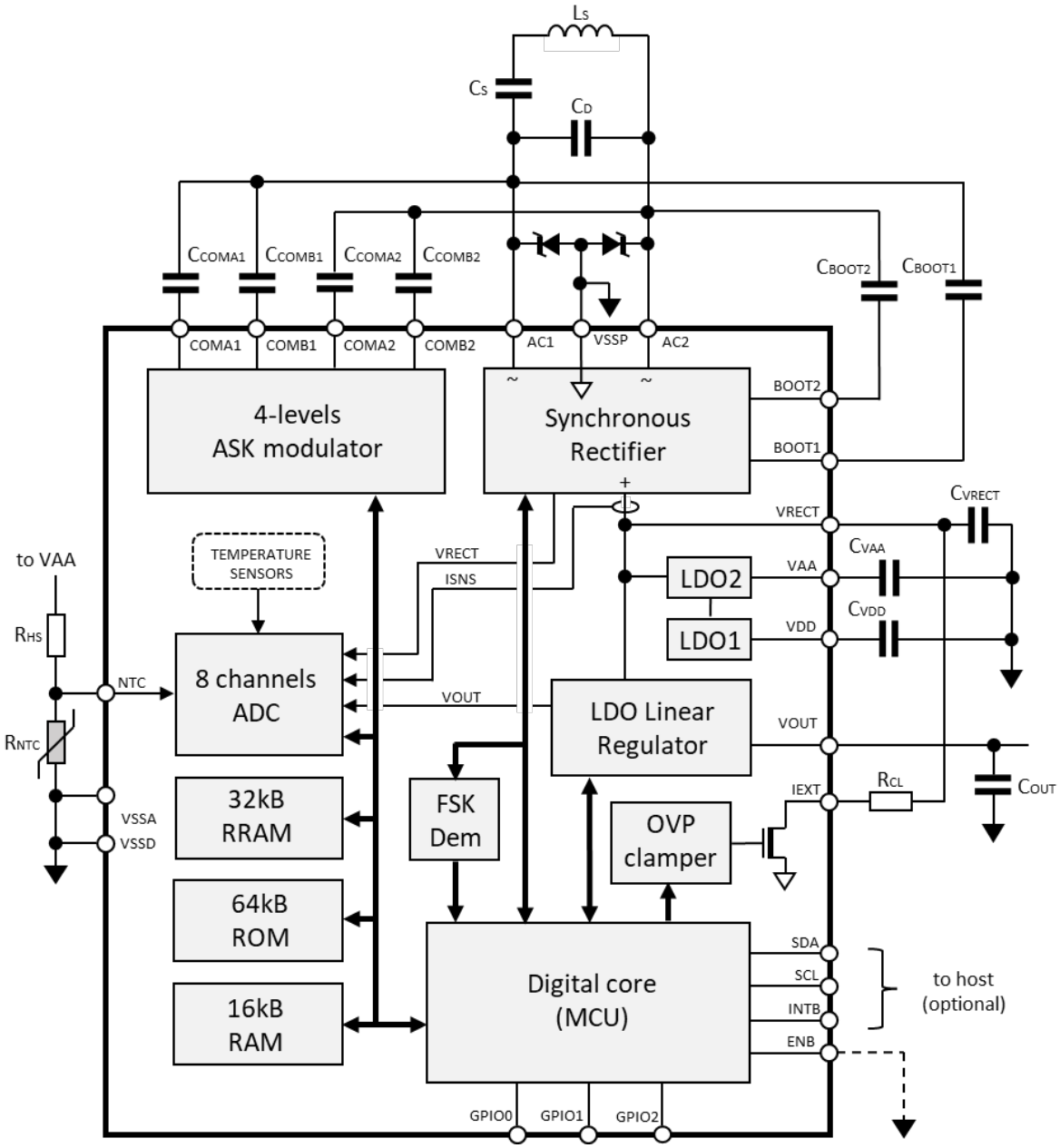
Step 7. Select the patch and configuration files you want to write.

Step 8. Press the [**Write**] button to load the .memh files into the device.

5 Device description and operation

5.1 System block diagram

Figure 15. **STWLC38 system block diagram**



5.2 Integrated power rectifier

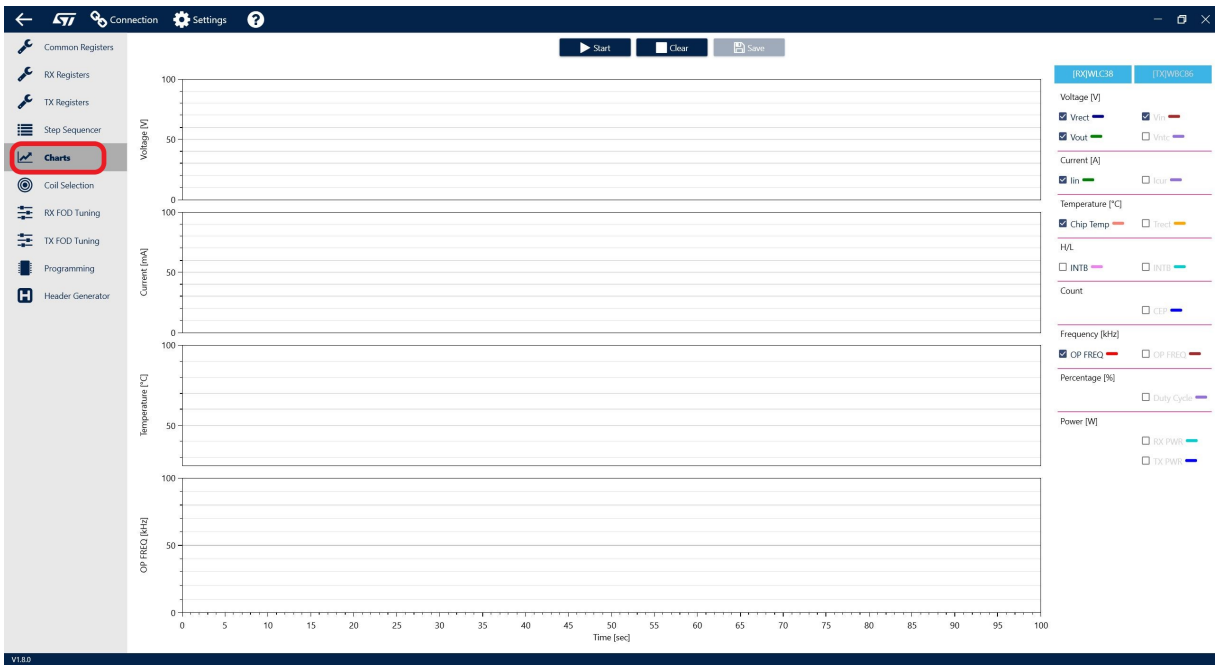
The synchronous rectifier is a key block in charge of converting the AC power signal from the receiving coil into a DC supply rail for the following linear regulator. The rectifier consists of four N-channel MOSFET transistors arranged into an H-bridge, conveniently driven by a control block that monitors the voltage at the AC1 and AC2 pins to optimize the commutations and to charge the external bootstrap capacitors for the high-side switches. Different driving schemes for the rectifier switches are possible. The MCU core dynamically selects the optimal one to maximize the overall efficiency as a function of the operating point.

When designing the filtering capacitor at the output of the synchronous rectifier, keep in mind that the capacitance value should be designed large enough to minimize the residual AC voltage ripple and to provide sufficient energy storage to sustain load transients and prevent the transients from impacting the ASK communication with the transmitter.

5.3 Internal ADC channels monitor

The GUI allows to monitor in real-time key parameters by reading the internal ADC channels of the STWLC38 and plotting voltages, current, and temperatures. For further details please refer to the GUI user manual UM3164.

Figure 16. PC GUI charts



5.4 Main LDO regulator

The main LDO regulates the rectified voltage to a target value specified by the user. The output voltage is configurable in 4 V – 12 V range with a 25 mV step. The default value of output voltage is 5 V for BPP (5W) mode and 9 V for EPP (15W) mode respectively. However, the target value can be easily changed in the GUI or by an external host controller.

The main LDO regulator is equipped with two protections:

1. IVL (input voltage loop): when the Vrect voltage falls under a set threshold, the Vout is temporarily disabled until the Vrect rises above the threshold again. The default threshold for this protection is 3 V but can be adjusted in 3.5 V to 10.5 V range with a 0.5 V step.
2. OCP (overcurrent protection): this protection is triggered when the rectifier current rises above a set threshold. The threshold can be adjusted to a value between 1.25 A and 1.93 A.
3. RX UVLO (undervoltage protection): when V_{RECT} falls under a set threshold, the set action may be triggered as send EPT or permanent disable of V_{OUT}.
4. Output short circuit protection is realized by a combination of overcurrent protection (OCP) and undervoltage protection (RX UVLO) mechanisms.

To maintain rectifier stability and ASK communication reliability , it is recommended to provide a certain minimum load to the rectifier. STWLC38 is equipped with Iload ballast function which draws sufficient current from the rectifier when the output load current is too small.

The rectifier current consists of the Iload ballast current, the output load current and the IC consumption current.

The Iload ballast current is automatically adjusted to maintain the minimal total load current of the rectifier, as set in the RX ILOAD BALLAST register.

Naturally when the sum of the output load current and IC consumption exceeds the RX ILOAD BALLAST threshold value there is no additional current sinked by ILOAD BALLAST block and it is disabled and will be automatically re-enabled when the total current would drop below the threshold.

5.5 Chip under voltage lockout

STWLC38 is equipped with an under voltage lockout, which triggers when the VAA voltage drops below 2 V, which results in the device being powered down. Normal operation is resumed as soon as the VAA voltage rises above 2.1 V.

5.6 Chip enable

The device can be disabled (forced into reset) by pulling the chip enable signal high. When the chip enable signal is released and left to be pulled down to ground, the device resumes normal operation. If the STWLC38 is in reset mode, then AC1 and AC2 are connected to GND to protect the device.

5.7 Power up sequence

When the STWLC38 receiver is placed onto a power transmitter, and the patch and configuration files are loaded into the device, it will initiate communication with the transmitter. After completing all the necessary steps described in the Qi specification, power transfer is established.

If Patch and/or Configuration files are not loaded, the device will not initialize the communication, therefore power transfer will not be established. Please refer to Section 4.6 Programming the device for device programming instructions. The communication will also not be initialized if either the Patch or Configuration or both files are corrupted.

During the initial phase of the power transfer STWLC38 operates in ARC mode, ST's proprietary mode, which makes the power-up sequence smoother and more reliable. For more information, see Section 5.12 Adaptive rectifier configuration (ARC) mode .

Once the VRECT voltage rises high enough, the internal power management system starts operating. The digital core is powered up and ready to control the internal circuitry. Default device settings are used until the digital core is woken up after which the firmware loads settings saved in the Cofiguration file saved in NVM. The device is now ready to operate.

5.8 LDOs

The device is equipped with two internal low drop-out regulators (LDOs) – a 2.5 V and a 1.1 V one, with the later deriving its power from the former.

The 1.1 V LDO powers the digital part of the IC, while the 2.5 V LDO powers the analog part of the IC but can also be used to power external low-power circuitry (such as LEDs). The maximum current externally drawn from this LDO shall not exceed 20 mA.

- External LDO capacitors should be placed as close to the IC as possible.
- Connect a 1 µF filtering capacitor between the VDD pin and ground.
- Connect a 4.7 µF filtering capacitor between the VAA pin and ground.

5.9 Protections overview

STEVAL-WLC38RX board uses both hardware and software protections to ensure safe operation of both the device and the board. The purpose of those protections is to avoid damage caused by unexpected operating conditions – over-voltage and/or over-current. Temperature is monitored as well – the device is equipped with both internal temperature sensor and pins for external NTC connection.

The majority of software protections can be configured in a GUI and/or by a host controller.

Each software protection can also be configured to disable the device's output and force the receiver to send an EPT (end power transfer) packet when triggered. If the transmitter receives this packet, it should terminate power transfer immediately.

Overvoltage protections (OVP)

- Ping overvoltage protection (POVP)
  - Protects the device against excessive rectifier voltage (VRECT) during the ping phase (first few milliseconds after the power-up)
  - The threshold is set to 14 V and is released as soon as the rectifier voltage drops below 11 V.
  - Gets disabled after the digital part of the device is powered up (the internal 2.5 V LDO reaches the target value).
  - When triggered, AC1 and AC2 pins of the device are shorted to ground
  - Cannot be disabled, and the threshold cannot be adjusted
- Soft overvoltage protection (SOVP)
  - Connects external resistors between the VRECT node and ground. This causes the VRECT voltage to drop, without dropping low enough to lose communication with the transmitter.
  - Can be disabled in the GUI
  - Can be configured to disable VOUT when triggered
  - Can also be configured to force the receiver to send EPT packet when triggered
- Hard overvoltage protection (HOVP)
  - Shorts AC1 and AC2 pins to ground when triggered
  - Can be disabled in the GUI
  - This protection can be configured to disable VOUT when triggered
  - The threshold of this protection can be set from 6 to 18 V with a 2 V step
- Firmware OVP
  - This protection can be configured to disable VOUT when triggered
  - The protection can also be configured to force the receiver to send EPT packet when triggered
  - The threshold of this protection can be set in range of 0 to 18 V

Undervoltage protections (UVLO)

- Input voltage loop (IVL)
  - It monitors the VRECT voltage and automatically reduces the device's output to maintain VRECT at the set threshold.
  - The threshold can be set from 3 to 10.5 V with a 0.5 V step. The default value is 4 V.
- RX UVLO
  - It monitors VRECT voltage and disables the device’s output when VRECT drops below a set threshold.
  - It can be configured to disable VOUT when VRECT drops below a set threshold.
  - Can also be configured to force the receiver to send EPT packet when triggered
  - The threshold can be set from 3 to 10.5 V with a 0.5 V step
  - Disabled by default

Overcurrent protections (OCP)

- Hardware OCP
  - Monitors the output current
  - Can be configured to disable VOUT when triggered
  - Can also be configured to force the receiver to send EPT packet when triggered
  - The threshold can be set to either 1.25, 1.5, 1.75 or 1.93 A
- Firmware OCP
  - Monitors the output current
  - Can be configured to disable VOUT when triggered
  - Can also be configured to force the receiver to send EPT packet when triggered
  - The treshold can be set from 0 to 1.93 A with 1 mA step

Overtemperature protections (OVTP)

- Hardware thermal shutdown (TSHUT)
  - Can be configured to disable VOUT when triggered
  - The threshold can be set from 105 to 135 °C with a 10 °C step.
- Firmware OVTP
  - Can be configured to disable VOUT when triggered
  - Can also be configured to force the receiver to send EPT packet when triggered
  - The threshold can be set from 105 to 135 °C with 0.1 °C step.
- NTC OVTP
  - Senses NTC voltage, usually used for coil temperature monitoring
  - Can be configured to disable VOUT when triggered
  - Can also be configured to force the receiver to send EPT packet when triggered
  - The threshold can be set from 0 to 2.5 V.

5.9.1 Soft overvoltage protection (SOVP)

SOVP protects the device against excessive rectifier voltage. When VRECT rises above a set threshold, the device will try to lower its value by connecting external power resistors (connected to IEXT pin) from VRECT to ground through internal transistors. This approach should achieve lowering the rectifier voltage without losing communication with the transmitter. Therefore, triggering this protection does not necessarily lead to regulation failure or power transfer termination. The protection can also be configured to disable the output and/or force the receiver to send EPT packet when triggered.

The SOVP can be set to either dynamic or static operation. In dynamic mode , when VRECT > VOUT SET + SOVP threshold, IEXT switch will be turned-on.

In static mode , when VRECT > 12 V + SOVP threshold, IEXT switch will be turned-on.

The SOVP threshold is programmable from 2 V to 5 V in step sizes of 0.2 V. It can be set by STSW-WPSTUDIO PC GUI tool or host controller via I^2C. SOVP is released when VRECT < SOVP threshold - 1 V.

In dynamic mode, the SOVP threshold is output setting dependent:

- Output enabled: the threshold is set as output voltage + margin
- Output disabled: the threshold is set to 12 V + margin

In dynamic mode the user may accidentally trigger protection. As previously mentioned, the threshold is set relative to the output voltage. When changing the output voltage setting, the SOVP threshold is adjusted automatically. However, the regulation loop is quite slow and may not be fast enough to accommodate a large voltage change. This might cause the rectifier voltage to temporarily exceed the SOVP threshold. Therefore, it is advised to change the output voltage slowly (in multiple smaller steps).

Figure 17. SOVP Rx protection threshold setting

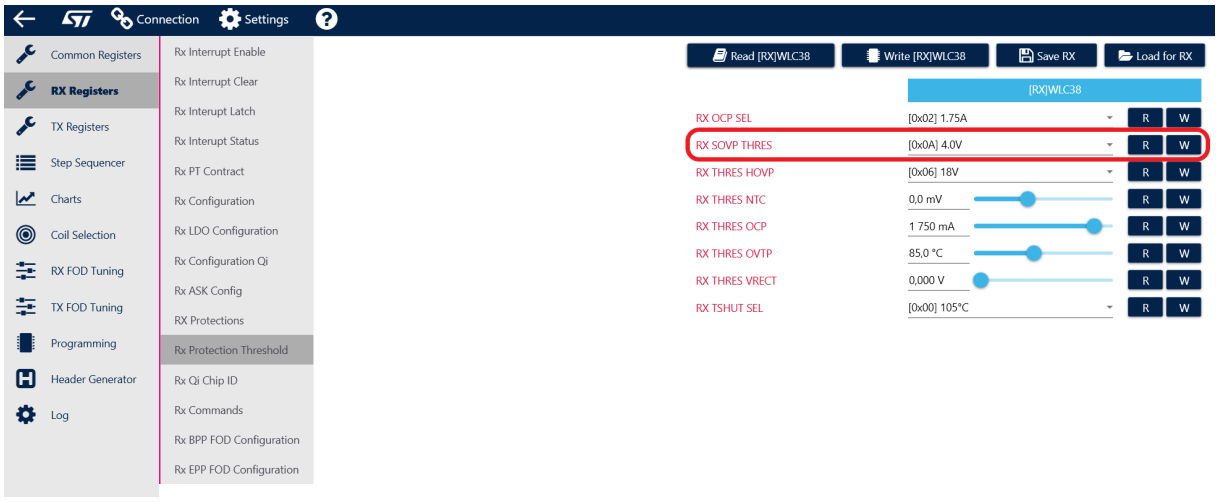
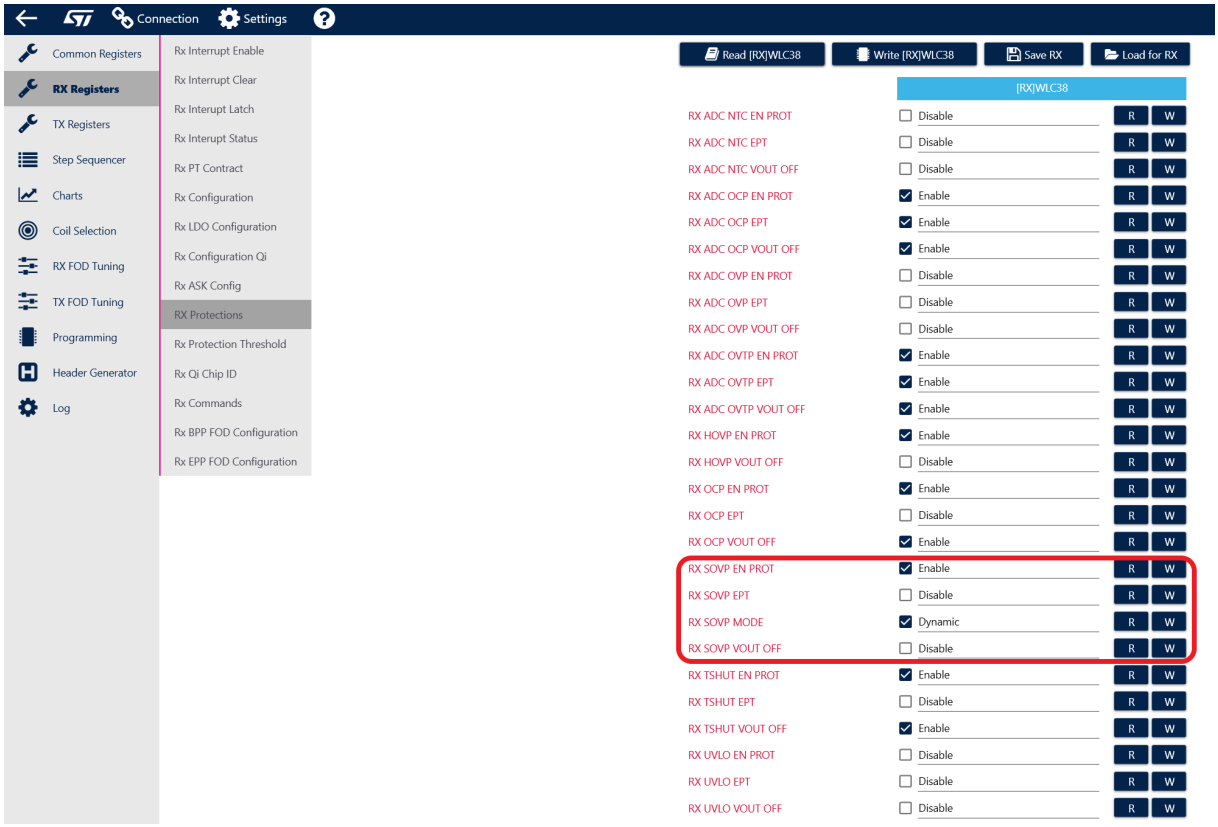


Figure 18. **SOVP Rx protection settings**



5.9.2 Hard overvoltage protection (HOVP)

HOVP is the fastest (hardware) over-voltage protection, which prevents the rectifier voltage from rising above a safe level. The default threshold value is 18 V. Triggering the protection causes STWLC38 to short AC1 and AC2 to ground, effectively shorting the receiving coil. This protection should be used as the last measure, as shorting the receiving coil will cause a significant rise in the transmitter current. Therefore, the HOVP threshold should be set higher than any other STWLC38 over-voltage protection (but still lower than AMR listed in the datasheet).

HOVP release value depends on SOVP setting.

- If SOVP is enabled, HOVP is released at SOVP threshold
- If SOVP is disabled, HOVP is released at 12 V + SOVP margin (refer to SOVP)

Figure 19. **HOVP Rx protection threshold setting**

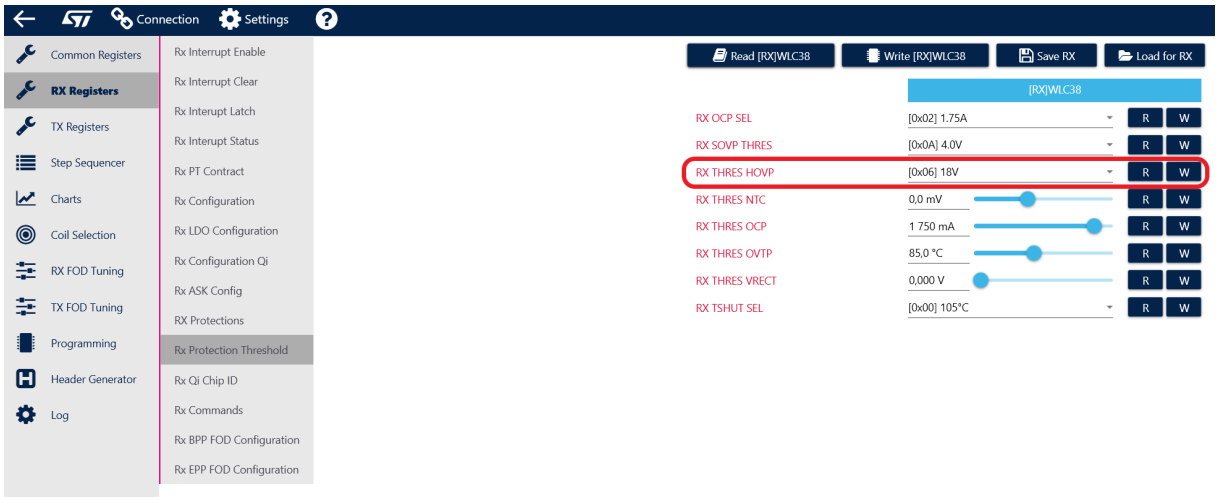
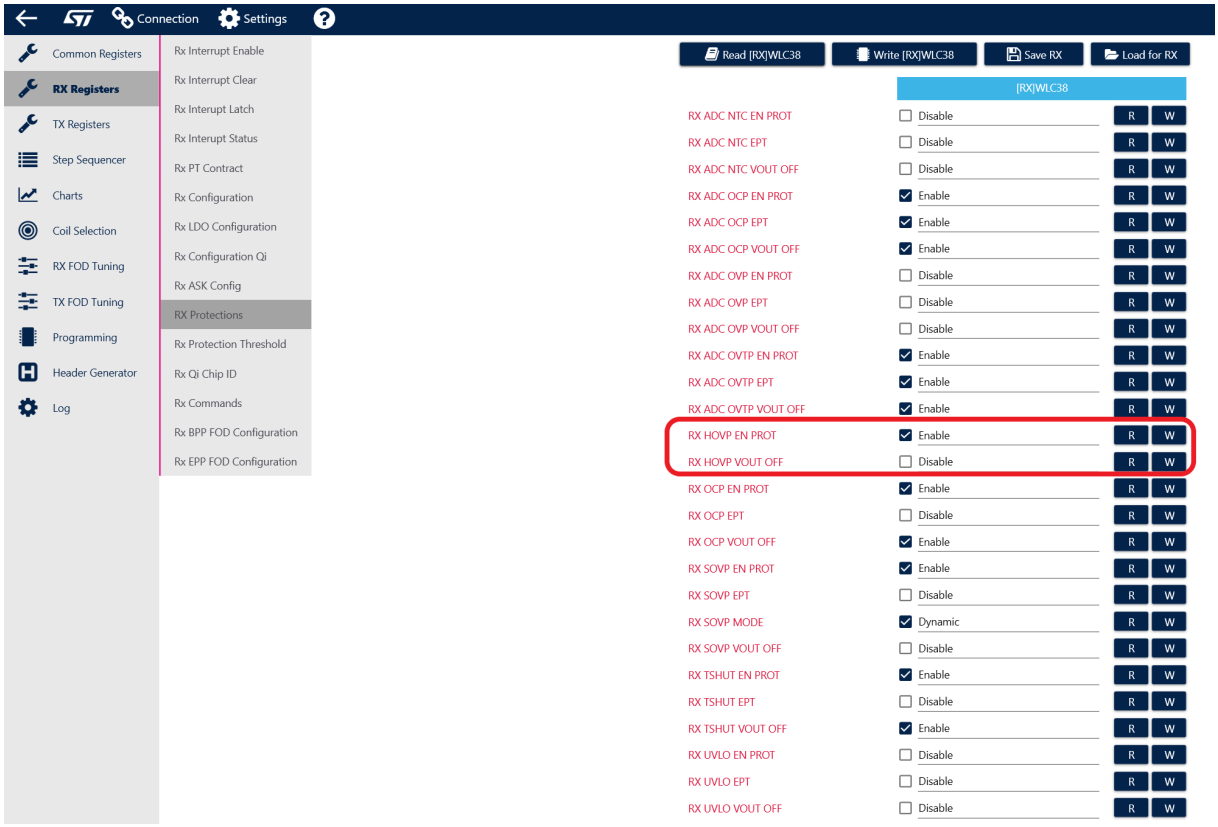
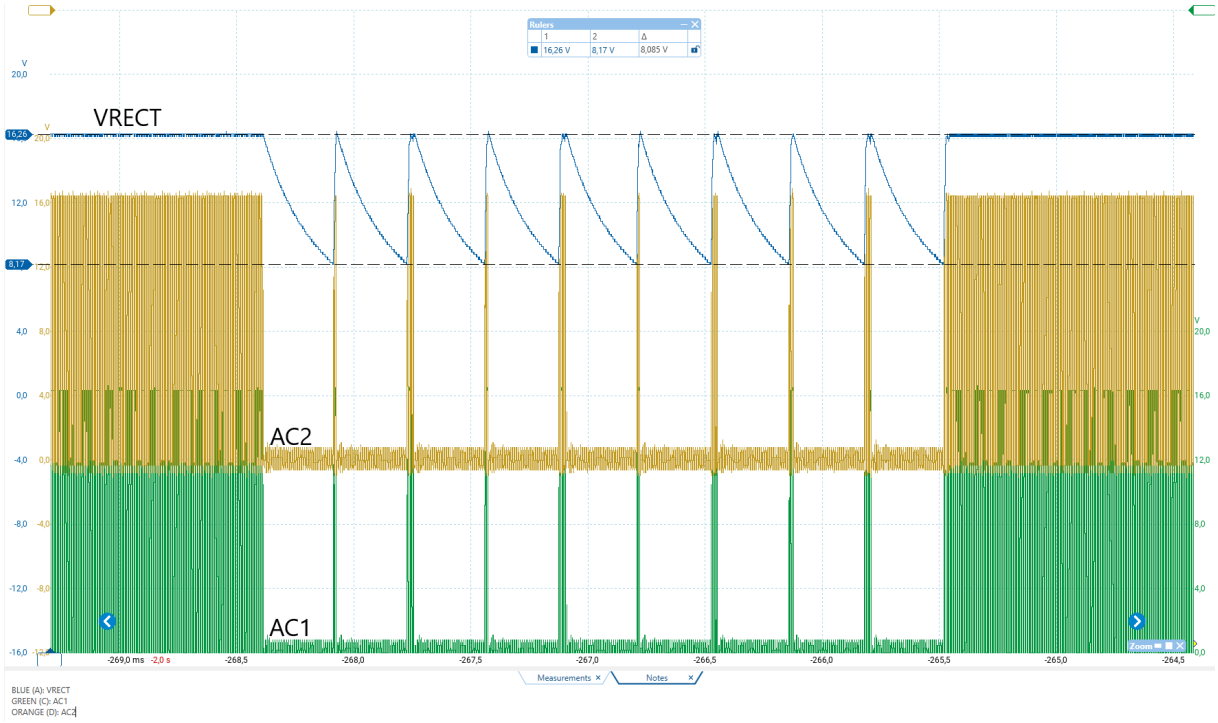


Figure 20. HOVP Rx protection settings



SOVP enabled, the VRECT voltage oscillates between the HOVP trigger threshold (18 V) and SOVP threshold.

Figure 21. VRECT and OVP thresholds



5.9.3 Firmware OVP

Firmware OVP is the slowest overvoltage protection. Therefore, it is recommended to set its threshold as the lowest among the available over-voltage protections. It monitors the rectifier voltage and can be configured to disable the device's output and/or force the receiver to send an EPT packet when triggered. The threshold can be adjusted from 0 to 16 V.

Figure 22. Firmware Rx protection threshold setting

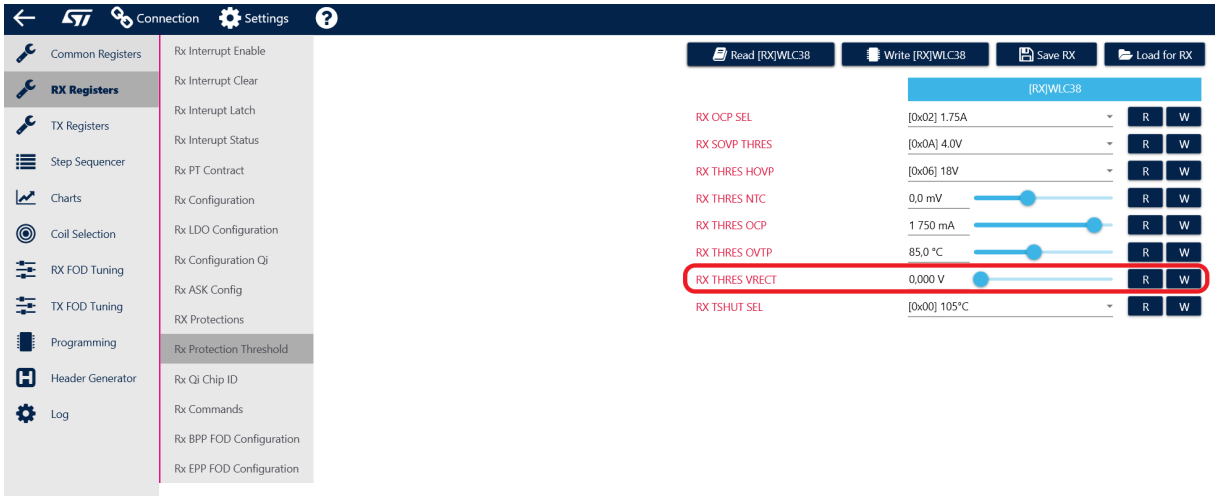
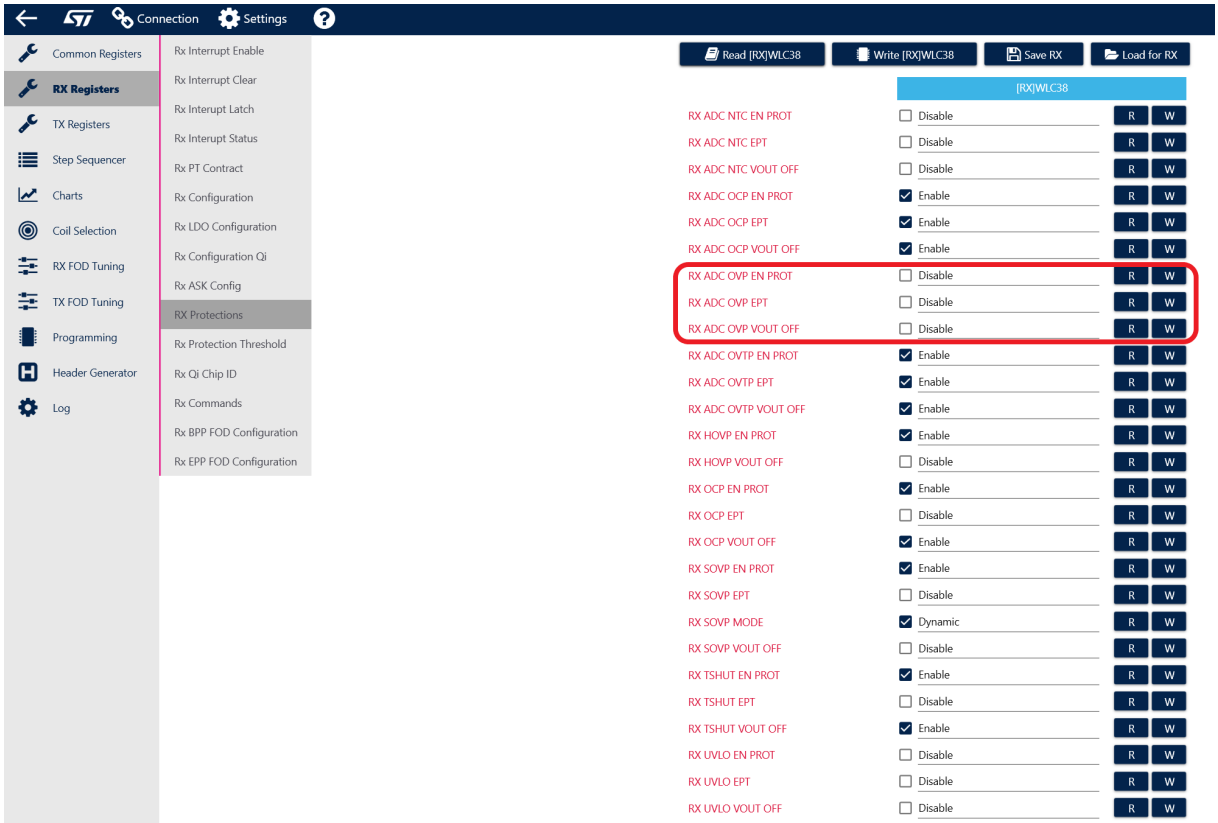


Figure 23. Firmware Rx protection settings



5.9.4 Overcurrent protections (OCP)

The STWLC38 IC is equipped with 2 separate overcurrent protections – one operated by hardware and one managed by the firmware.

The hardware OCP causes the set action as send EPT or disable Vout. This protection is triggered by HW. The threshold can be set to either 1.25, 1.5, 1.75 or 1.93 A.

Figure 24. Over current protection threshold setting

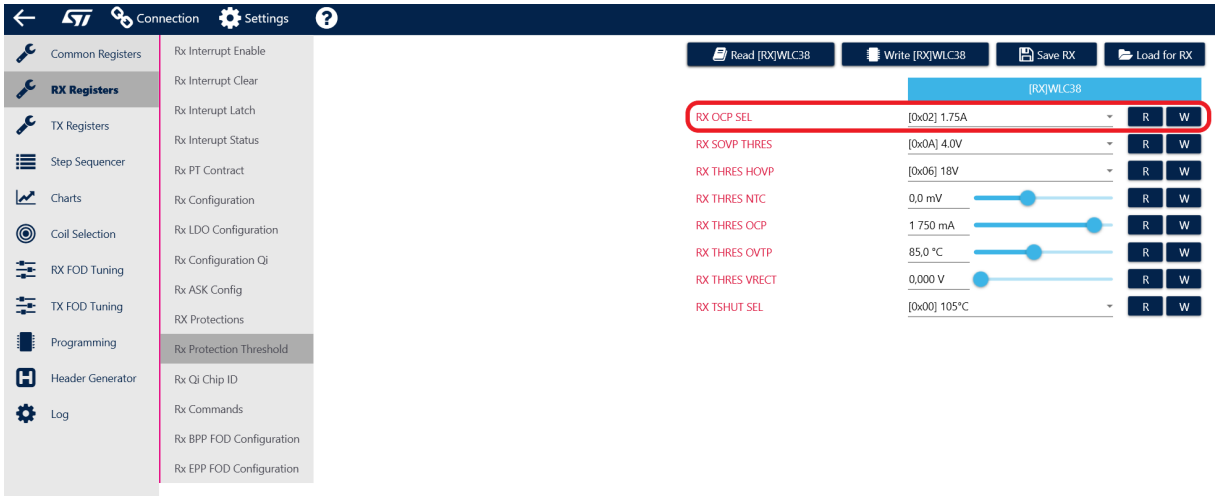
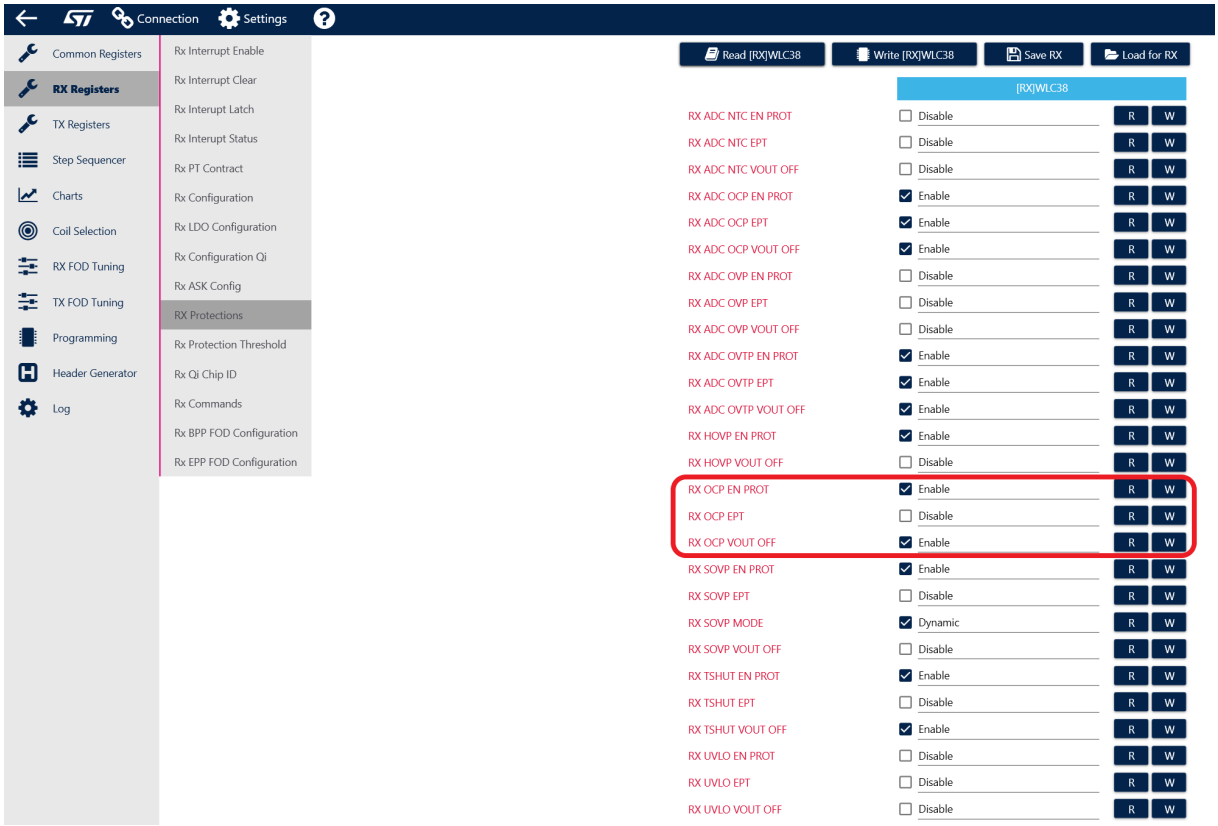


Figure 25. Hardware overcurrent Rx protection selection



The firmware OCP threshold can be adjusted from 0 to 1.75 A. It can also be configured to disable the device's output and/or force the receiver to send an EPT packet when triggered.

Figure 26. **Firmware overcurrent Rx protection threshold setting**

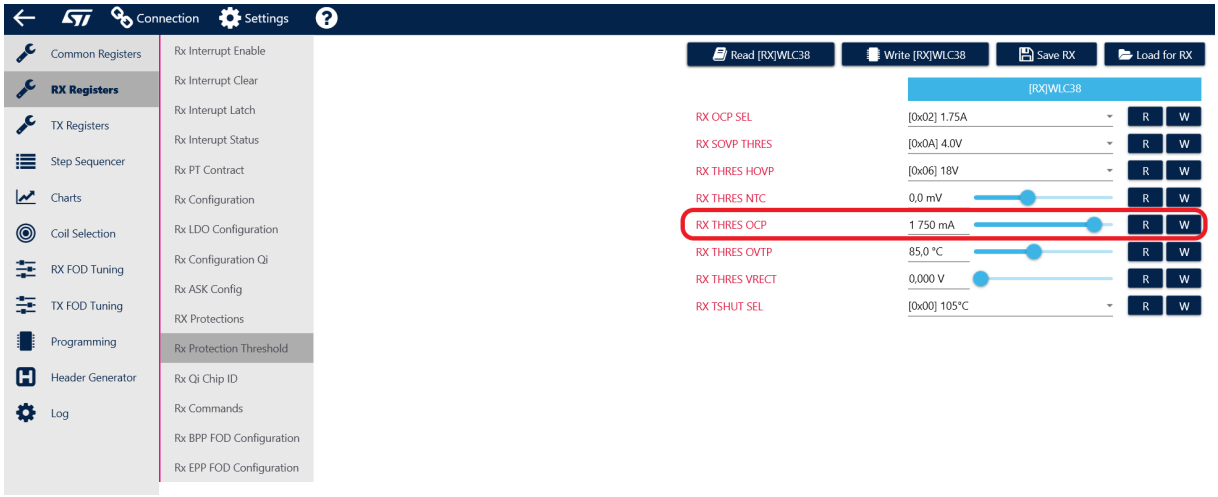
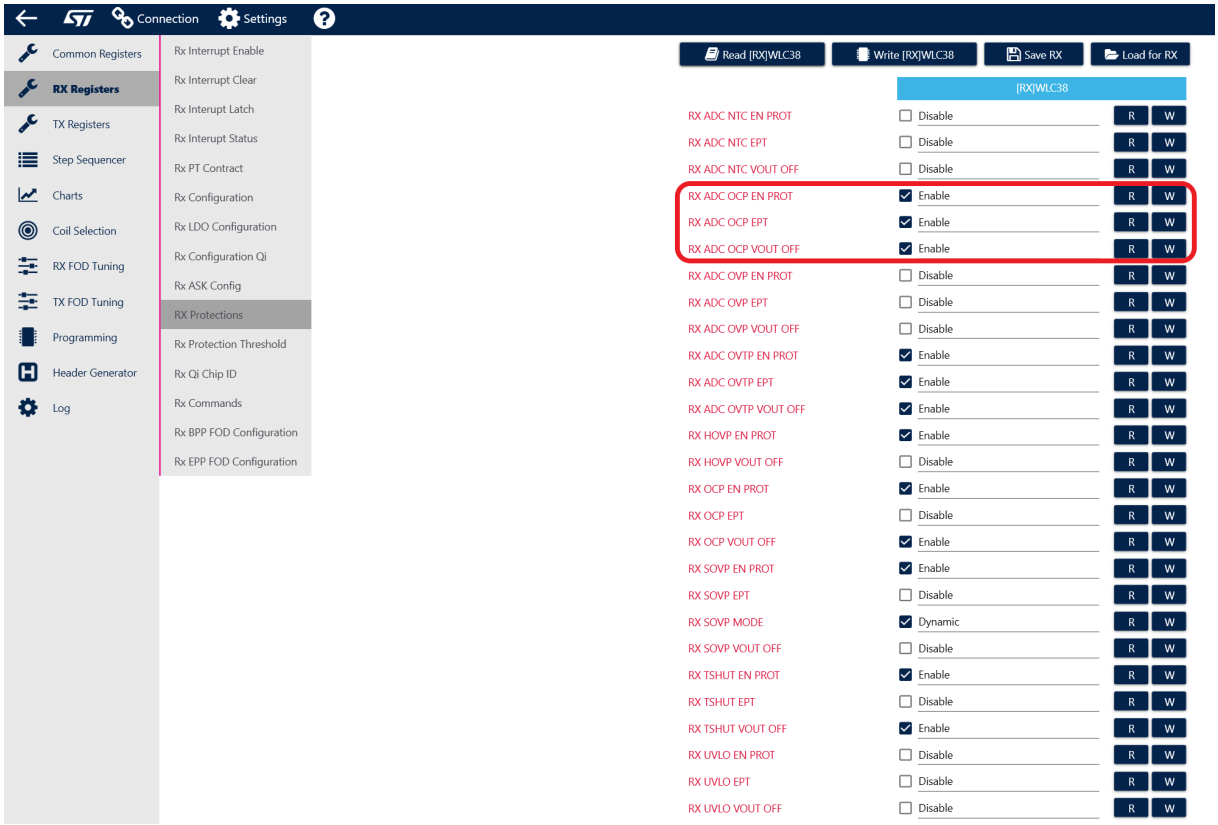


Figure 27. **Firmware overcurrent Rx protection settings**

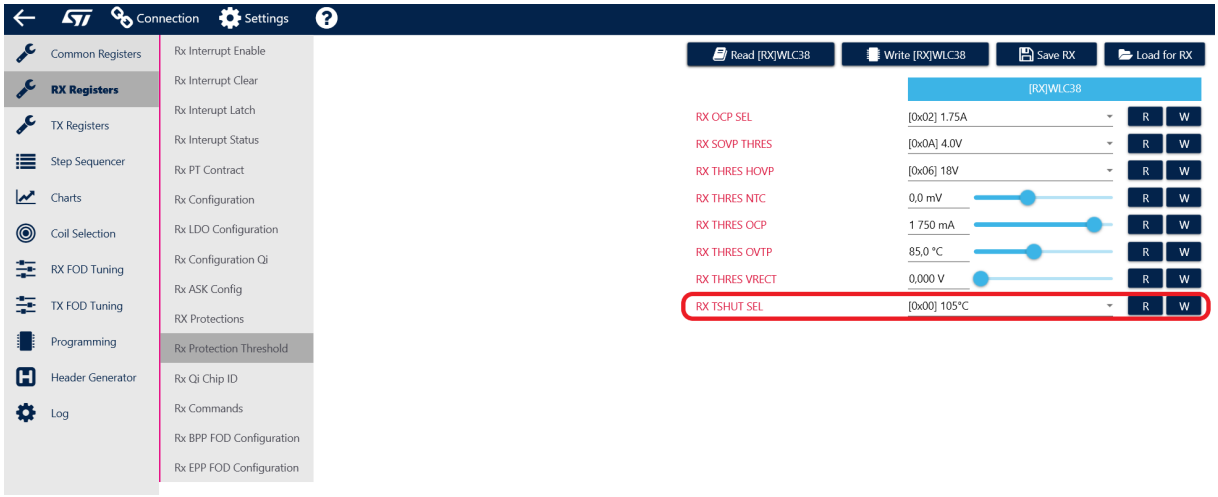


5.9.5 Overtemperature protections (OVTP)

The STWLC38 IC is equipped with one overtemperature protection operated by hardware and one managed by the firmware.

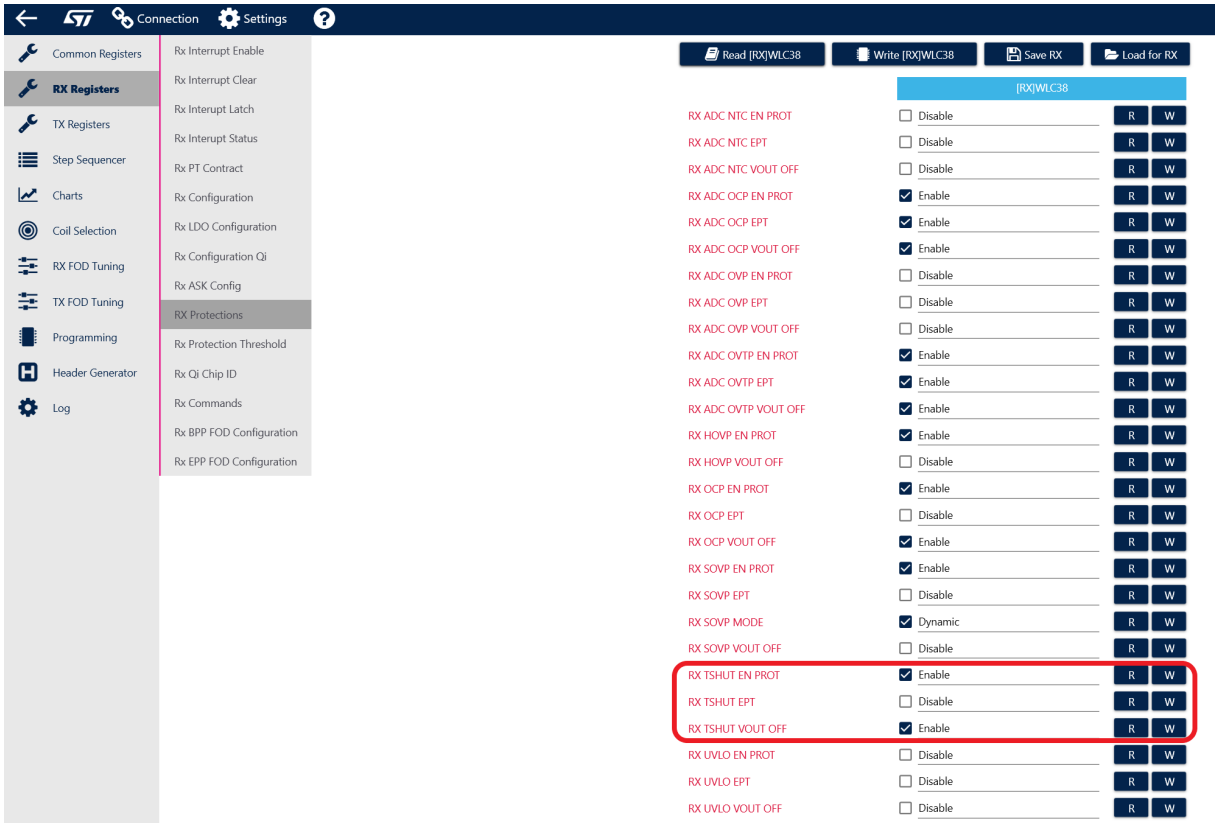
The hardware OVTP (called TSHUT) causes the AC1 and AC2 pins to be shorted to ground when triggered. The default threshold value is set to 105 °C (released at 95 °C), however the threshold can be adjusted in the GUI (10 °C step).

Figure 28. Hardware overtemperature Rx protection threshold setting



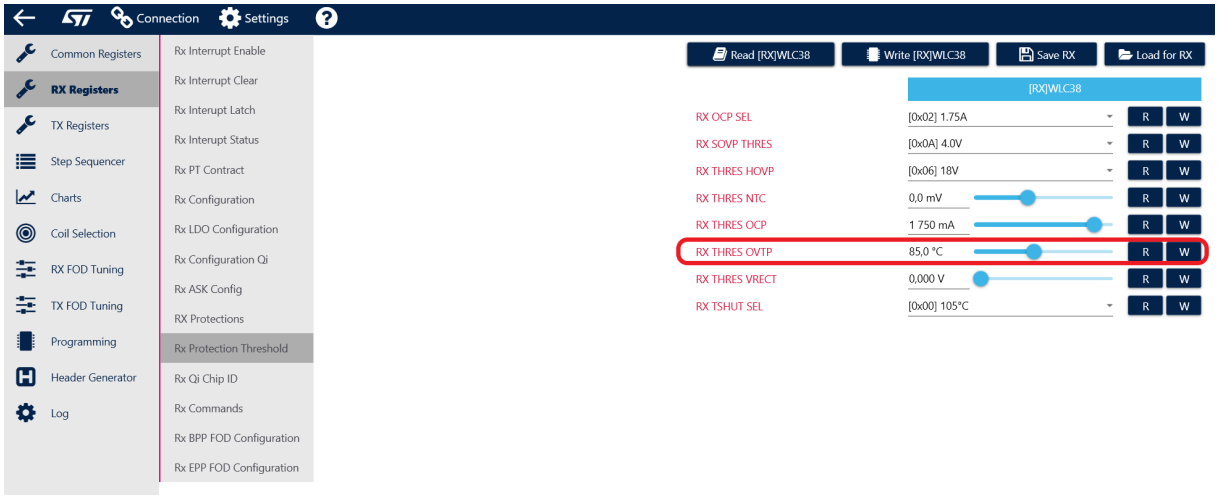
Hardware OVTP can also be configured to disable the device output and/or force the receiver to send an EPT packet when triggered.

Figure 29. Hardware overtemperature Rx protection settings



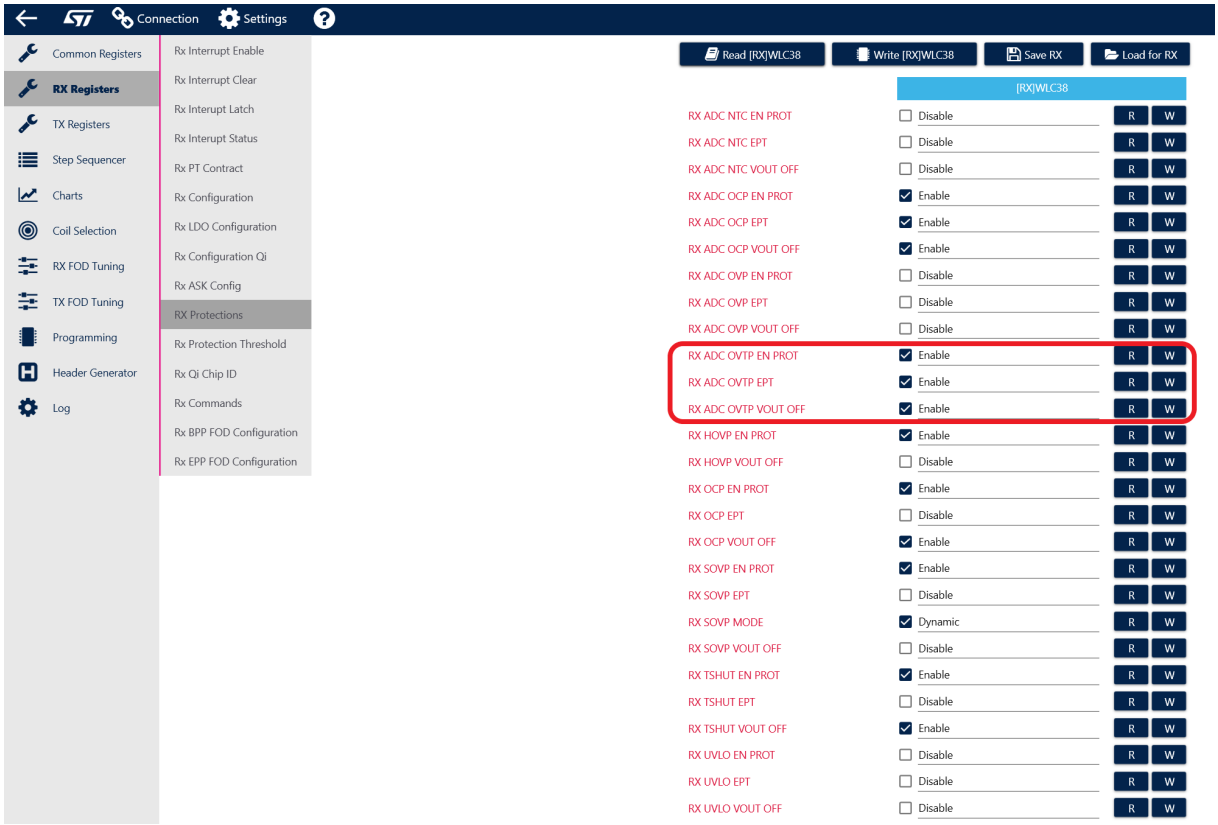
The firmware OVTP can be adjusted from 0 °C to 200 °C (with a 0.1 °C step) and can be accessed in the GUI or by a host controller.

Figure 30. Firmware overtemperature Rx protection threshold setting



Firmware OVTP can also be configured to disable the device output and/or force the receiver to send an EPT packet when triggered.

Figure 31. Firmware overtemperature Rx protection settings



5.9.6 NTC protection

An external NTC can be used to further monitor the operational temperature of the board. The user may choose a component or board region to be monitored by placing the NTC on or around it, although monitoring the receiving coil is the most common practice.

Together with a pull-up resistor, the NTC forms a voltage divider, which is connected to the STWLC38 NTC pin. It is recommended to design the voltage divider to reach the (low) NTC threshold at the highest allowed temperature.

The protection can also be configured to disable the device output and/or force the receiver to send an EPT packet when triggered.

Figure 32. NTC Rx protection threshold setting

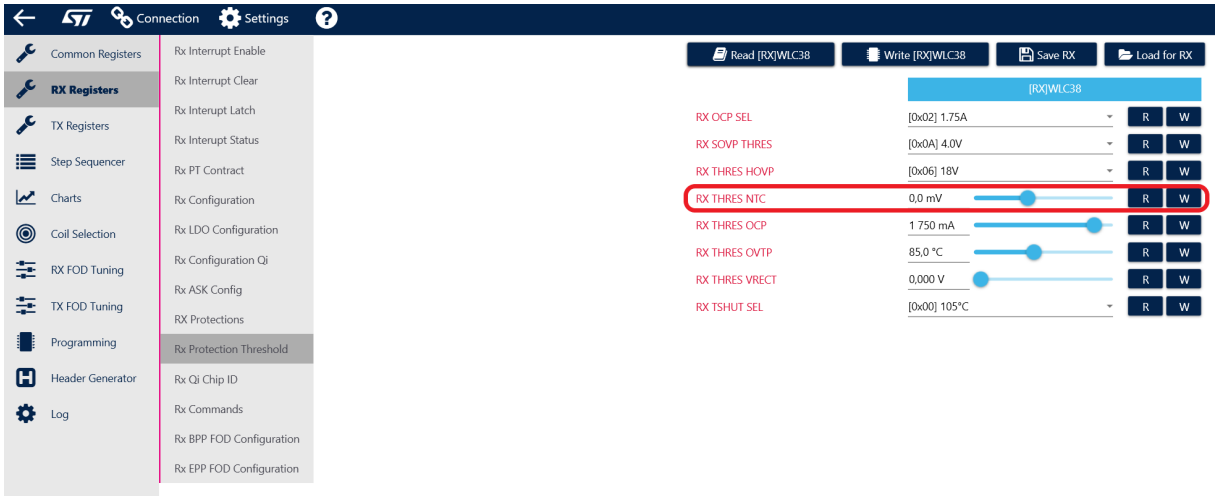
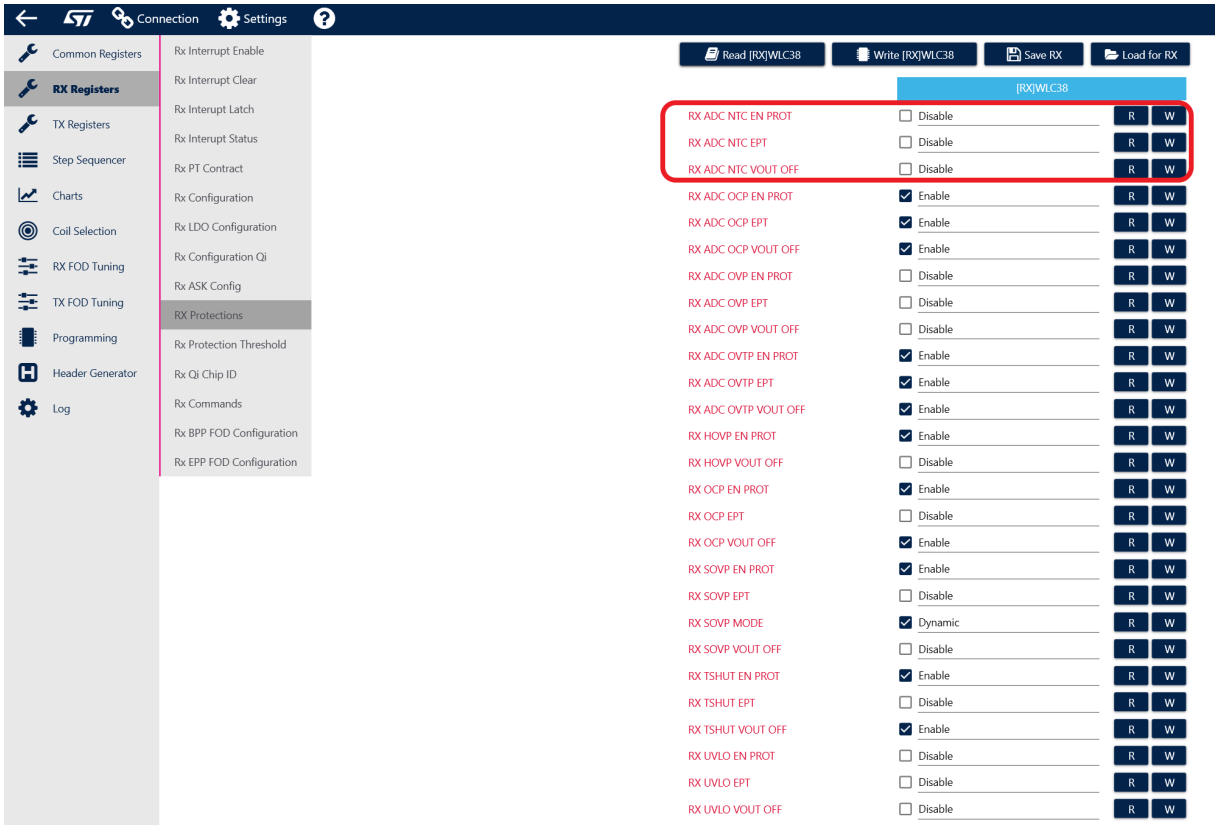


Figure 33. NTC Rx protection settings NTC ADC



5.9.7 Foreign object detection (FOD)

Foreign object is any object placed either on or near the transmitting coil, which is not considered a valid wireless power receiver and is magnetically active. Presence of the magnetic field generated by the transmitting coil may cause eddy currents to form in the foreign object (such as coins, keys etc.), which in turn would heat the object to potentially dangerous temperatures. To avoid possible damage to the device or even injury of the user, the power transmitter must be able to detect presence of a foreign object. This detection can be implemented in several different ways.

One of the most common ways is estimating the amount of power lost in the system. The power receiver indicates the total amount of received power from the power transmitter. This power consists of power available to the load and power loss which occurs on the supporting circuitry. Power loss on the power receiver side may occur in the following locations:

- The receiving coil and series resonant capacitors
- The rectifier
- The post-regulation stage
- In tracks and cables
- Other locations

A high power loss value indicates the probable presence of a foreign object.

STWLC38 uses precise internal AD converters to perform voltage and current measurements and a 32-bit arithmetic to keep sufficient accuracy. STWLC38 then uses the measured values to create a mathematical model to estimate the received power, which are then sent to the power transmitter for evaluation.

For more information regarding FOD please refer to the Qi specification.

5.10 WPC Qi wireless power transfer

5.10.1 Wireless power interface

Wireless power interface is the area in which power transfer takes place. It consists of the transmitter (primary) power interface and the receiver (secondary) power interface. The main component of the interfaces is the transmitting/receiving coil. The user should avoid placing any objects which are magnetically active and not required for power transfer on or near the wireless power interface in order to avoid possible damage or injury (see Section 5.9.7 Foreign object detection (FOD)). Detection of a valid receiver on the primary wireless power interface is performed by analog and digital pings.

Analog ping is a short burst generated by the power transmitter. The power transmitter then evaluates if the resonant frequency and/or resonant circuit capacitance changed, as such change might indicate presence of a potential power receiver. However, the burst does not carry sufficient energy to wake the power receiver.

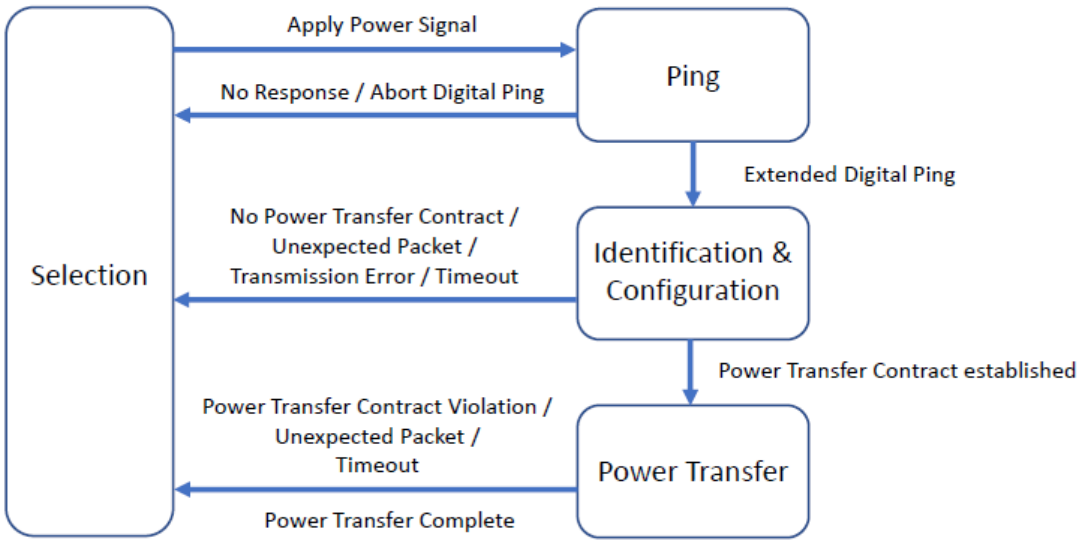
Digital ping is a more energetic burst of power generated by the transmitter; its parameters are defined by the Qi specification. The target of this burst is to generate enough power for the receiver to establish communication between the transmitter and the receiver.

The Qi specification defines a specific way the receiver is expected to answer the digital ping. The reason is that only a proper answer can inform the transmitter that a valid power receiver was placed on the wireless power interface.

5.10.2 Qi Baseline Power Profile

The flowchart in the figure below shows steps required to achieve power transfer in the Baseline Power Profile (BPP) according to Qi 1.2.4.

Figure 34. Qi Baseline Power Profile flowchart

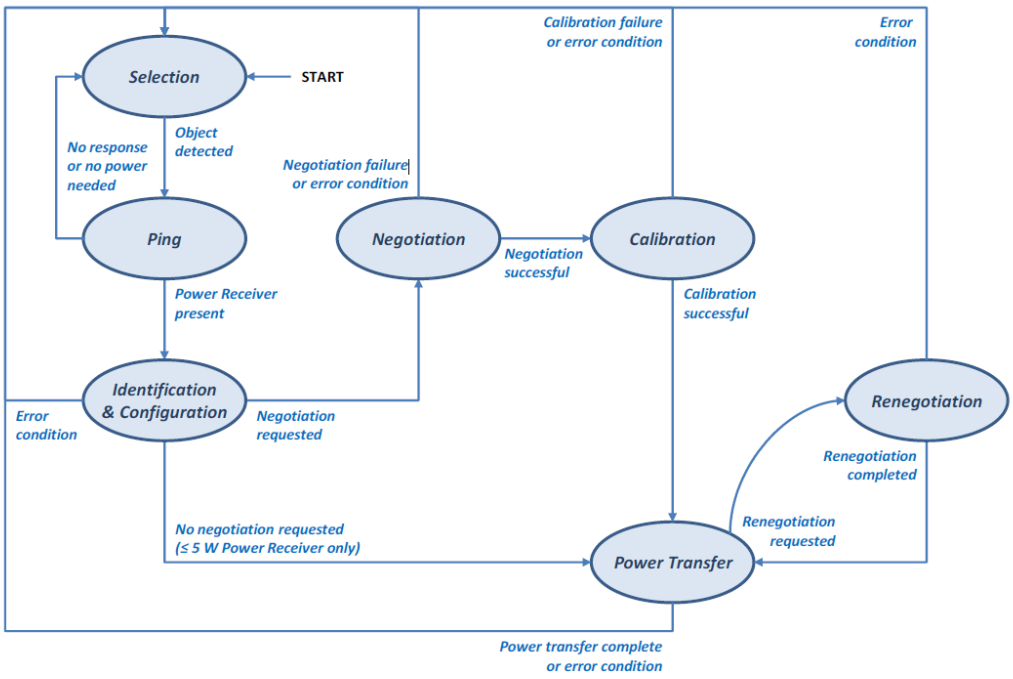


- **Selection:** is the default state in which the power transmitter periodically scans its wireless power interface for a potential power receiver. When a suitable power receiver is detected, the power transmitter proceeds to the (digital) ping phase.
- **Digital ping:** this phase is an interrogation session during which is the potential power receiver expected to reply through amplitude shift-keying (ASK) modulation as defined by the Qi specification. After a valid power receiver is detected, the transmitter proceeds to the Identification & Configuration phase.
- **Identification & Configuration:** this phase is aiming to identify the receiver and to gather information necessary for a stable and reliable power transfer, such as the maximum power or FSK communication parameters. After receiving the information, the power transmitter creates a Power transfer contract, basically a summary of the operational parameters. After the power receiver is identified and the power transfer is created, the power transmitter proceeds to the power transfer itself.
- **Power transfer:** this is the final step. The power transmitter initially increases and subsequently modulates the transmitted power in response to control (feedback) data from the receiver. The receiver periodically sends information to the power transmitter, such as required power (CEP), received power (RPP) etc. which are used to maintain a closed control loop. If a critical event (e. g. over-voltage, over-current or over-temperature) occurs, the power transmitter shall terminate the power transfer immediately.

5.10.3 Qi Extended Power Profile (EPP)

Extended Power Profile (EPP) is another wireless power transfer profile described by the Qi specification and is used to deliver up to 15 W of power. To enter EPP, both the transmitter and the receiver must comply with the Qi EPP specification. Negotiation and calibration steps are required before entering power transfer. A renegotiation phase might be used after initiating the power transfer. In the renegotiation phase, the power receiver may request adjustments of the power transfer contract.

Figure 35. Qi Extended Power Profile flowchart



- **Negotiation**: in this phase the power receiver negotiates with the power transmitter to finetune the power transfer contract.
- **Calibration**: in this phase the power transmitter can improve its ability to detect foreign objects during power transfer by receiving additional information at two different loads from the power receiver.

Important notes:

- Changing the output voltage must respect the overall system design (selected coil, transmitter type etc.).
- Output load transient response strongly depends on correct output capacitors design. Severe load transients may lead to temporary output voltage collapse due to the overall TX-RX response time.
- The initial load at power-up should not exceed 2.5 W, instead it should be ramped up smoothly to full power after power transfer is established.

5.11 Bidirectional communication

The amount of power transmitted is (from a control standpoint) fully dependent on the operating conditions of the power transmitter (i.e., its bridge voltage, operating frequency, and duty cycle).

Since there is no direct electrical feedback from the load (system output) to the power transmitter, the power receiver must establish communication with the power transmitter to provide this feedback instead.

To match load requirements and prevent excessive power transmission, the power receiver communicates to the power transmitter the required power level using amplitude shift keying (ASK). This is done by the receiver modulating the amount of power it draws from the power signal. The transmitter detects this as a modulation of the voltage across and/or current through the transmitting coil and adjusts its operating conditions accordingly.

The power transmitter can communicate to the receiver as well, this time using frequency shift keying (FSK). This is done by the power transmitter directly modulating the power signal. Communication in this direction enables proprietary communication.

5.11.1 ASK communication

A state (either HI or LO) is characterized by the amplitude being constant (with a certain variation Δ) for at least 150 ms.

If the power receiver and power transmitter coils are properly aligned, then for all appropriate loads, at least one of the following three conditions shall apply.

- The difference of the amplitude of the transmitter coil current in the high and low state is at least 15 mA.
- The difference of the amplitude of the transmitter coil voltage in the high and low state is at least 200 mV.
- The difference of the transmitter coil current in the high and low state is at least 15 mA. The transmitter coil current is measured at instants in time that correspond to one quarter of the cycle of the control signal driving the half-bridge inverter.

Figure 36. HI and LO states in ASK communication

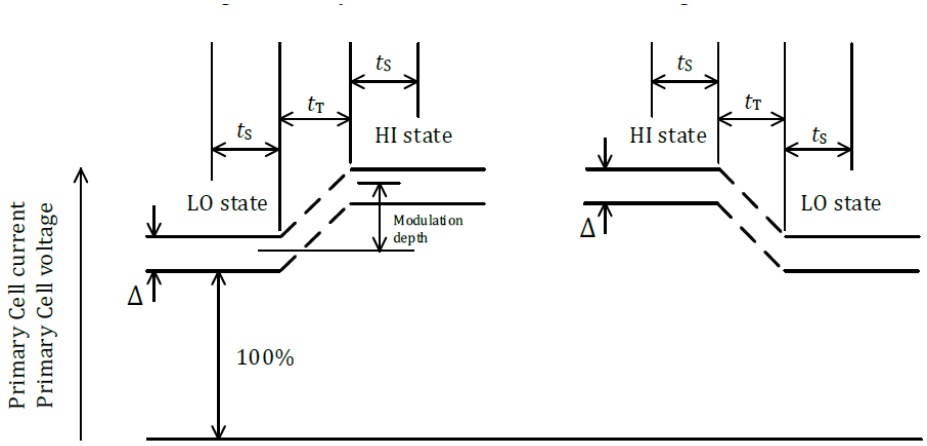


Table 3. HI and LO state parameters

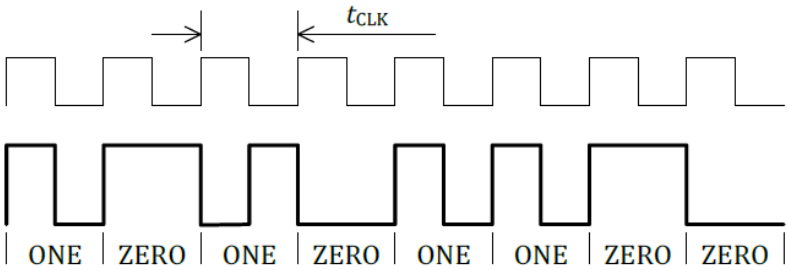
Parameter	Symbol	Value	Unit
Maximum transition time	t_T	100	µs
Minimum stable time	t_S	150	µs
Current amplitude variation	Δ	8	mA
Voltage amplitude variation	Δ	110	mV

The power receiver uses a differential bi-phase encoding scheme to modulate the data bits onto the power signal. The power signal is then demodulated by the power transmitter demodulation circuit and the demodulated signal is processed.

The communication itself is carried out via packets. Each packet consists of 4 parts and each of those parts consists of several bits/bytes.

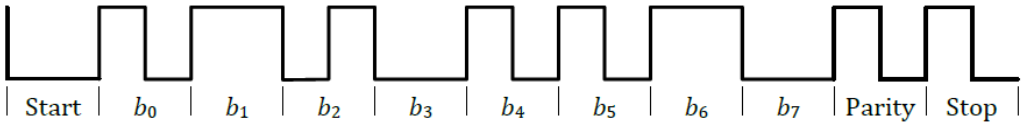
Each communication **bit** is synchronized to the power receiver's internal 2 kHz clock signal period. A ONE bit is encoded as two transitions of the power signal, while a ZERO bit is encoded as a single transition.

Figure 37. **Example of a differential bi-phase encoding scheme**



To transmit a single **byte** of data, the power receiver must send an 11-bit sequence. This sequence consists of a START bit (ZERO), the 8 data bits of the byte (LSB first), a parity bit and a STOP bit (ONE). The parity is odd, meaning an even number of ONE bits in the data byte results in the parity bit being equal to ONE, while an odd number of ONE bits in the data byte results in the parity bit being equal to ZERO. An example of such message can be seen in the picture below:

Figure 38. **Example of asynchronous serial format**



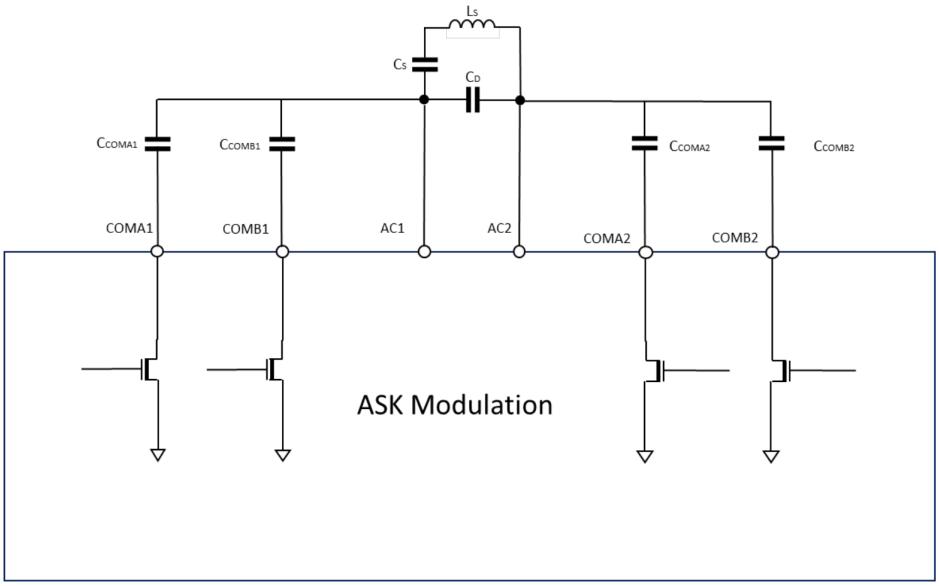
A **packet** consists of a preamble, header, message, and checksum. The preamble contains 11 to 25 ONE bits and enables the power transmitter to synchronize with the incoming data. The header, message and checksum are sequence of three or more bytes. The header indicates the packet type while also implicitly providing the size of the message. The checksum consists of a single byte. It is calculated as an exclusive-OR of both the header and message bytes and enables the power transmitter to check for transmission errors.

For a more detailed explanation please refer to the Qi Specification, Section *Power Receiver to Power Transmitter communications interface*.

STWLC38 uses integrated low resistance N-channel MOSFET transistors to modulate the power signal. The internal switches are connected to the COMA1, COMB1, COMA2 and COMB2 pins. As shown below, these pins are connected to the AC1 and AC2 nodes through external capacitors, therefore connecting those capacitors modulates the impedance seen by the coil, which is reflected onto the primary (transmitter) coil as a change in current and/or voltage.

The figure below shows the ASK modulation capacitors connection, as well as optional ASK demodulation circuit, which is required for a transmitter mode operation.

Figure 39. **ASK modulation capacitor connections**



5.11.1.1 *ASK communication example: sending a proprietary packet from STWLC38 to STWBC86*

The explanation of ASK communication is described in detail in Bi-directional communication application note on STWLC38 and STWBC86.

5.11.2 FSK communication

The power transmitter modulates the power signal by switching between its normal operating frequency f_{op} and its modulated frequency f_{mod}. The difference between f_{op} and f_{mod} can be described by two parameters: **polarity** and **depth**.

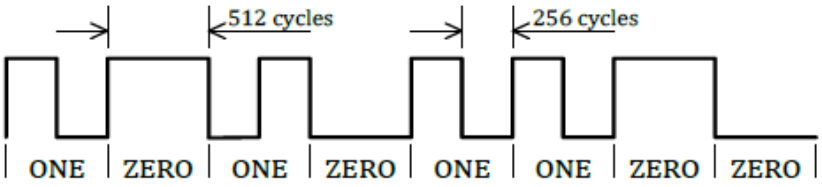
Polarity is given by the difference between f_{mod} and f_{op} – a positive polarity corresponds to a positive difference, while a negative polarity corresponds to a negative difference.

Depth describes the magnitude of difference between f_{op} and f_{mod}. Please note that a negative polarity results in a higher induced voltage on the power receiver coil and should therefore be used with care.

The FSK communication is executed via packets. Each packet consists of 3 parts – a header, a message and a checksum. The header is a single byte which indicates the packet type, while also implicitly providing the size of the message part. The message size can range from 1 up to 27 bytes. The checksum is a single byte which provides a way to check for transmission errors. It is calculated as an exclusive-OR of the header bytes and the message bytes.

Each FSK **bit** is aligned to 512 periods of the power signal. A ONE bit is represented by two transitions over the 512-period slot - the frequency change occurs at the very start of the slot and after 256 power signal periods pass. A ZERO bit is represented by a single transition (at the start of the 512-period slot).

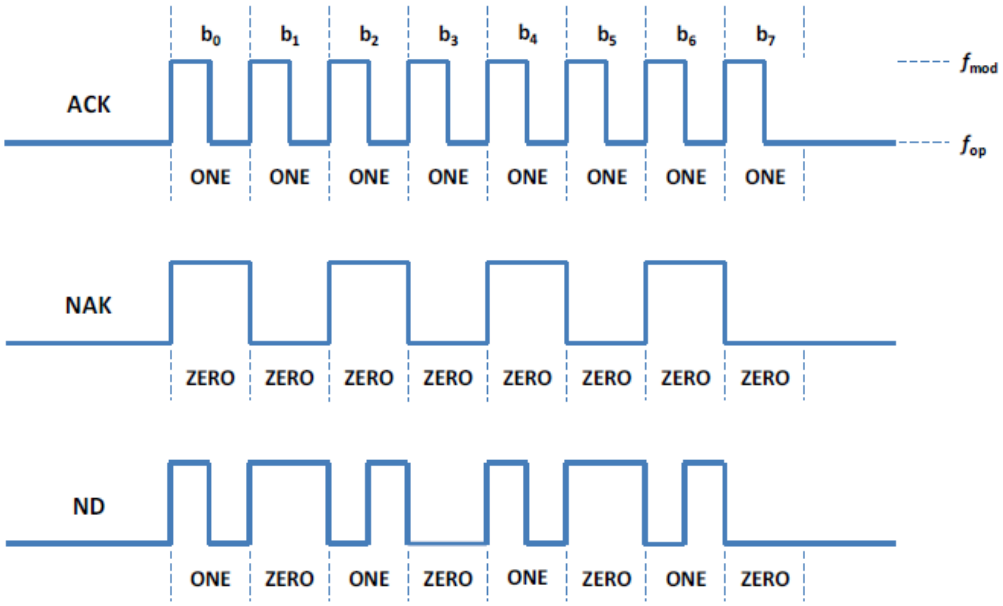
Figure 40. **Example of differential bi-phase encoding**



Each data **byte** is transferred as a sequence of 11-bits. The sequence consists of a start bit (ZERO), the data byte itself, a parity bit and a stop bit (ONE). The parity used for byte encoding is even – the parity bit set to a ONE if the data byte contains an odd number of ONE bits. If the data byte contains an even number of ONE bits, the parity bit is set to a ZERO instead. This encoding scheme is used for all 3 parts of the FSK packet - the header, message and checksum.

There is also an additional type of message which can be sent from the transmitter to the receiver – a response to a receiver's message. A response consists of 8 bits, so the receiver can use quite a simple logic to decode it. The response can be either Acknowledge (ACK), Not-Acknowledge (NACK) or Not-Defined (ND). ACK is encoded as a series of 8 ONE bits, NACK is encoded as 8 zero bits and ND is encoded as a series of 8 bit of alternating ZERO and ONE bits ('01010101').

Figure 41. Format of three defined responses



For a more detailed explanation please refer to the Qi Specification, Section *Power Transmitter to Power Receiver communications interface*.

5.11.3 Most common Qi communication packets

- **Control Error Packet (CEP)** provides feedback from the power receiver to the power transmitter about the amount of power required by the load. CEP is a two's complement signed integer; its maximum range is -128 to +127. Its value is calculated by the receiver as a portion of the currently transmitted power, by which the current power level should be increased/decreased. A negative CEP indicates less power is required, while a positive CEP indicates more power is required. CEP equal to zero indicates no changes of the operation point are required.
- **Received power packet (RPP)** value is calculated by the power receiver as a portion of the maximum power value contained in the configuration packet and is used to inform the transmitter about the amount of power received by the receiver. This information is mainly used for foreign object detection.
- **Signal strength packet (SS)** indicates the degree of coupling between the transmitting and receiving coils. Its value is an unsigned integer, and it is calculated from the current power/voltage level as a portion of the maximum expected power/voltage.
- **Proprietary packet (PP)** enables sending custom messages between the transmitter and the receiver. Qi standard defines different proprietary packet types, which can range in length from 1 to 20 bytes.
- **Charge status packet (CSP)** contains an unsigned integer which states the current charge status of the battery in the receiving device. CSP equal to 100 means the battery is fully charged, while CSP equal to 0 means the battery is completely discharged. If the receiving device does not contain a battery or it cannot provide charge status information, the CSP value should be set to 0xFF.

- **End power transfer (EPT)** is used by the power receiver to request power transfer termination. The power transmitter shall terminate the power transfer immediately after receiving an EPT packet. The EPT packet should contain a message indicating the reason for the termination. The following table lists the codes used for various reasons.

Table 4. EPT codes

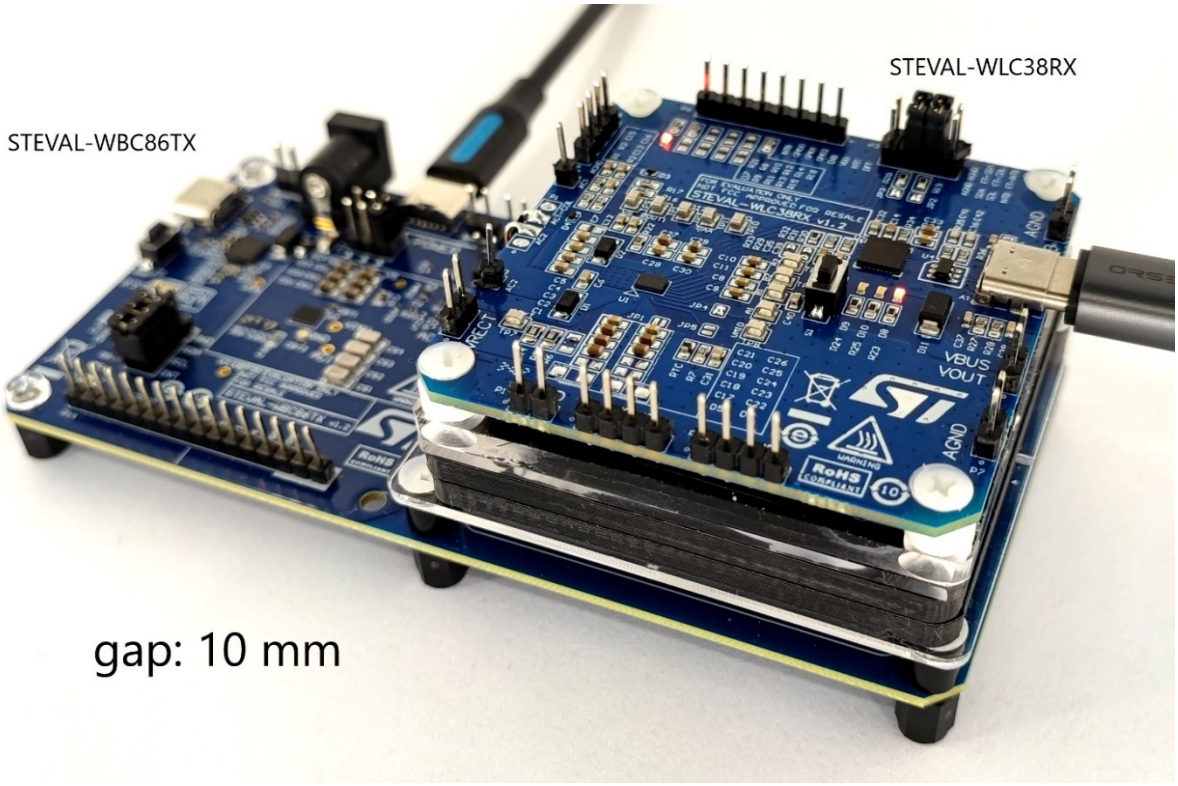
Reason	Value	Recommended usage of the values (Informative)
EPT/nul	0x00	use if none of the other codes is appropriate
EPT/cc	0x01	charge complete; use to indicate that the battery is full
EPT/if	0x02	internal fault; use if an internal logic error has been encountered
EPT/ot	0x03	Over-temperature, use if (e.g.) the battery temperature exceeds a limit.
EPT/ov	0x04	Over-voltage; use if a voltage exceeds a limit
EPT/oc	0x05	Over-current; use if the current exceeds a limit
EPT/bf	0x06	battery failure: use if the battery cannot be charged
EPT/nr	0x08	no response: use if the target operating point cannot be reached
EPT/an	0x0A	aborted negotiation: use if Power Transfer Contract cannot be negotiated
EPT/rst	0x0B	restart; use to restart the power transfer

5.12 Adaptive rectifier configuration (ARC) mode

ARC mode improves the ping-up and spatial freedom of the system in both X and Y directions during power transfer, without any change in hardware or optimization of the coil. The feature mode can provide up to 3 additional millimeters of spatial freedom in the 4 basic spatial directions (up/down/left/right). The ARC mode function can be configured to be disabled by the firmware after VRECT voltage rises above a set threshold, the threshold can be configured in the GUI.

The 3 mm improvement is depending on the coils used in Wireless charging system. The result was validated with the following RX and TX setup.

Figure 42. **STEVAL-WBC86TX and STEVAL-WLC38RX setup**



The ARC mode is dedicated rectifier setting what allows to increase the spatial freedom of position. The ARC mode is automatically enabled and the IC stays in ARC mode until the Vrect exceeds the set threshold (RX ARC THRES). Then the ARC mode is disabled, the rectifier switches to a different operating mode where output can be enabled.

The table below shows how ARC mode improves ping up spatial freedom.

Table 5. **Possible separation without and with ARC mode**

Separation (mm)	Ping without ARC mode	Ping with ARC mode
Centre	OK	OK
4	OK	OK
8	OK	OK
12	OK	OK
16	OK	OK
18	Cannot ping up	OK
19	Cannot ping up	OK
20	Cannot ping up	Cannot ping up

5.13 I²C Interface

STWLC38 can operate fully independently i.e., without being interfaced with a host system. In applications in which the STWLC38 is a part of peripherals managed by the host system, the two SDA and SCL pins could be connected to the existing I^2C bus. The device works as an I^2C slave and supports standard (100 kbps), fast (400 kbps). STWLC38 has been assigned 0x61 7-bit hardware address.

The pins are up to 3.3 V tolerant, and the pull-up resistors should be selected as a trade-off between communication speed (lower resistors lead to faster edges) and data integrity (the input logic levels must be guaranteed to preserve communication reliability). When the bus is idle, both SDA and SCL lines are pulled HIGH.

5.13.1 Data validity

The data on the SDA line shall remain stable during the high state of every SCL clock pulse. The high and low states of the SDA line shall only change when the SCL clock signal is low.

5.13.2 Start and stop conditions

Both the SDA and the SCL lines remain high when the I^2C bus is not busy. A START condition is indicated by a high-to-low transition of the SDA line when SCL is HIGH, while the STOP condition is indicated by a low-to-high transition of the SDA line when SCL is HIGH. A STOP condition must be sent before each START condition.

5.13.3 Byte format

Every byte transferred over the SDA line contains 8 bits. Each byte received by STWLC38 is generally followed by an acknowledge (ACK) bit. The most significant bit (MSB) is transferred first. A single data bit is transferred during each clock pulse.

The device generates an ACK pulse (by pulling the SDA line low during the acknowledge clock pulse) to confirm a correct device address or data bytes reception.

5.13.4 Interface protocol

The interface protocol consists of:

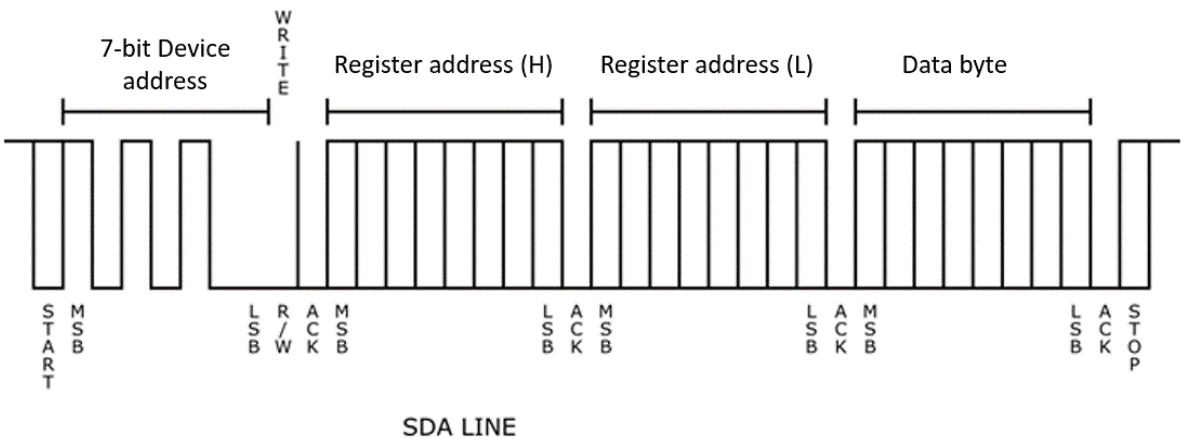
- A start condition (START)
- A device address + R/W bit (read =1 / write =0)
- A register H address byte
- A Register L address byte
- A sequence of n data bytes (each data byte must be acknowledged by the receiver)
- A stop condition (STOP)

The register address byte determines the first register in which the read or write operation takes place. When a read or write operation is finished, the register address is automatically incremented.

5.13.5 Writing to a single register

Writing to a single register begins with a START condition followed by the device address 0xC2 (7-bit device address plus R/W bit cleared), two bytes of the register pointer and the data byte to be written in the destination register. Each transmitted byte is acknowledged by the STWLC38 through an ACK pulse.

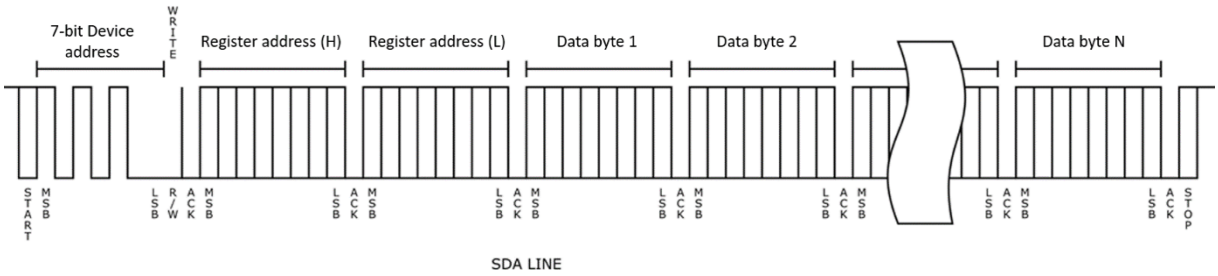
Figure 43. Writing to a single I²C register



5.13.6 Writing to multiple registers with incremental addressing

STWLC38 supports writing to multiple registers with auto-incremental addressing. When data is written into a register, the register pointer is automatically incremented, therefore transferring data to a set of subsequent registers (also known as page write) is a straightforward operation.

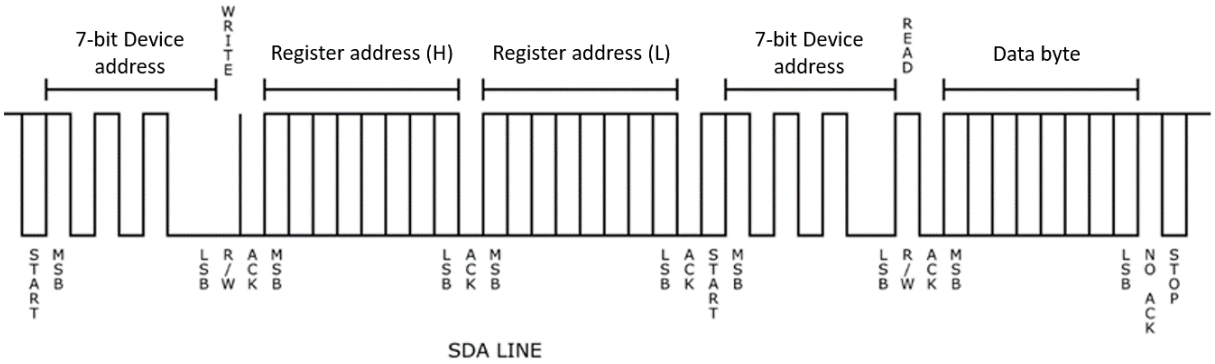
Figure 44. Writing to multiple I²C registers



5.13.7 Reading from a single register

Reading from a single register begins with a START condition followed by the device address byte 0xC2 (7-bit device address plus R/W bit cleared) and two bytes of register pointer, then a restart condition is generated and the device address 0xC3 (7-bit device address plus R/W bit asserted) is sent, followed by data reading. ACK pulse is generated by the STWLC38 at the end of each byte, but not for data bytes retrieved from the register. A STOP condition is finally generated to terminate the operation.

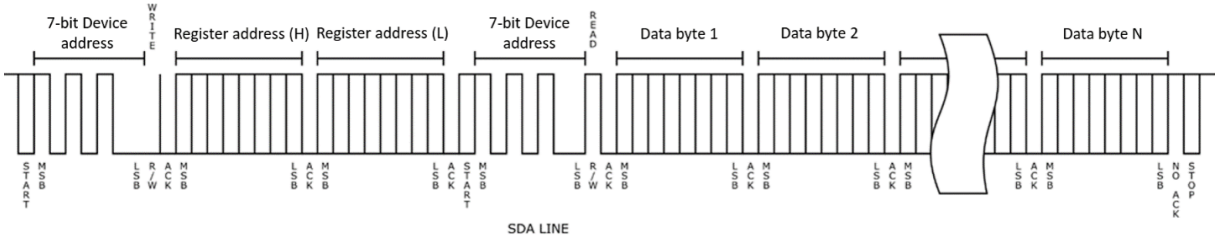
Figure 45. Reading from a single I²C register



5.13.8 Reading from multiple registers with incremental addressing

Similarly, to multiple bytes (page) writing, reading from subsequent registers relies on an auto-increment of the register: The master can extend data reading to the following registers by generating and an ACK pulse at the end of each byte. Data reading starts immediately, and the stream is terminated by a NMAK at the end of the last data byte, followed by a STOP condition.

Figure 46. Reading from multiple I²C registers



5.14 GPIOx and INTB pins

GPIO0 through GPIO2 are programmable general-purpose I/O pins. These pins can be configured as inputs or outputs (push-pull or open-drain) and assigned various functions.

Table 6. GPIO pins

Code	I/O	Function
0x01	I	Pull-up
0x02	I	Pull-down
0x03	O	Interrupt (Open drain)
0x04	O	Interrupt (Push pull)
0x05	O	FW ready
0x06	I	Disable the main LDO
0x07	I	Disable ASK communication
0x0B	O	Indication of successful Negotation. (Push-pull, active high)
0x0C	O	Indication of successful Negotation. (Push-pull, active low)
0x0D	O	Indication of successful Negotation. (Open drain)
0x29	O	Open drain, active low
0x2A	O	Open drain, active high
0x2B	O	Push-pull, active low
0x2C	O	Push-pull, active high

The INTB pin is an interrupt output line that can be assigned to any internal interrupt condition and used to inform the host system about a specific event.

5.15 Interrupt registers

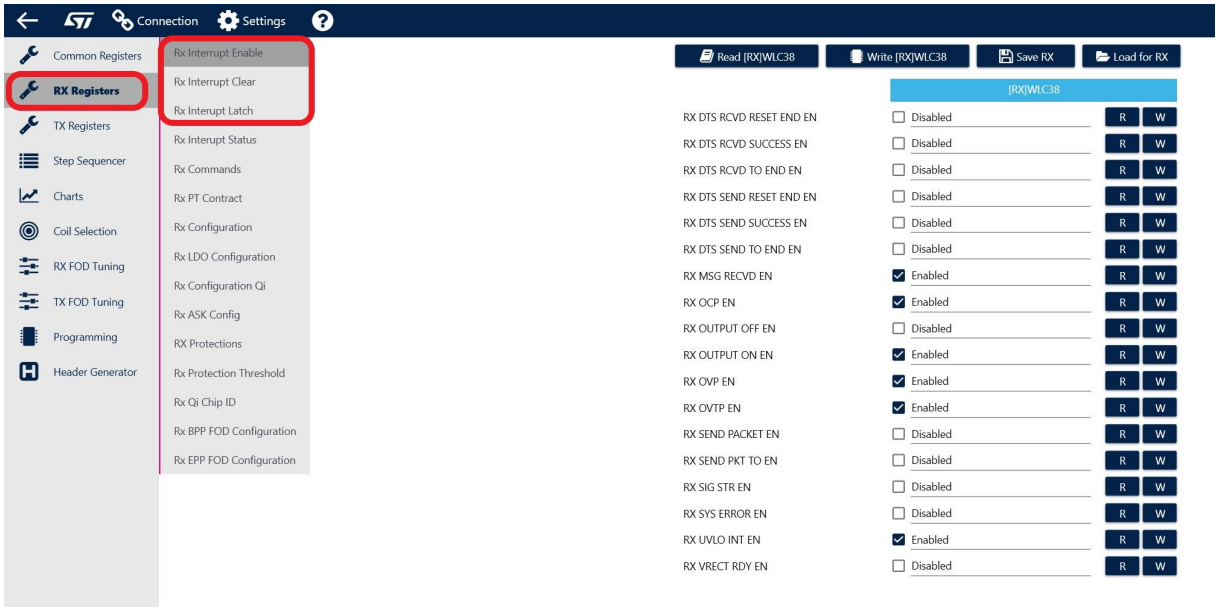
There are 4 bits (enable, clear, latch and status) assigned to each interrupt sorted into separate tabs.

The enable tab can be used to either enable the corresponding interrupt (write), or to check whether the interrupt is already enabled (read).

The latch tab can be used to determine which interrupts have been triggered (read only). After being triggered, the bit remains set to 1 until cleared by writing a 1 into the corresponding clear register (write only).

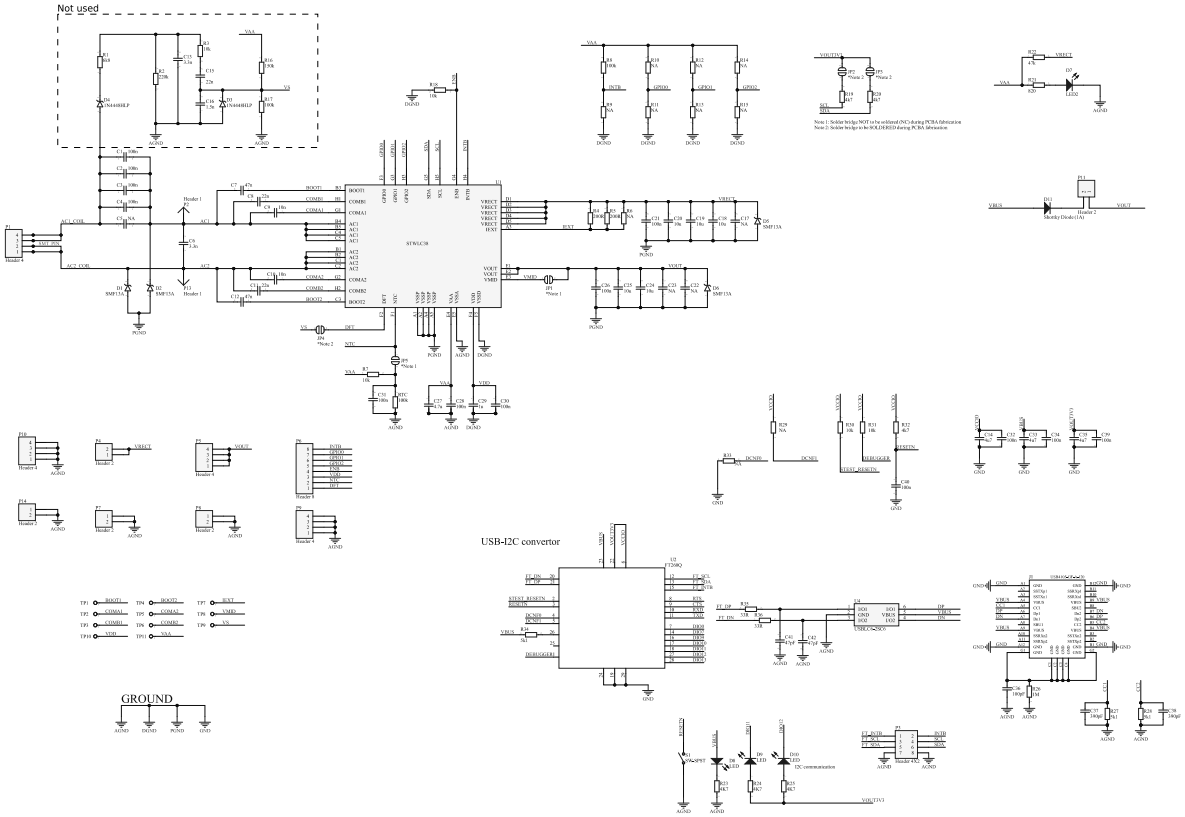
The status tab can be used to determine which interrupt is being triggered at a given moment (read only). The status bit goes back to zero after the triggering condition is removed.

Figure 47. **Interrupt register settings**



6 Schematic diagrams

Figure 48. **STEVAL-WLC38RX circuit schematic**



7 Bill of material (BOM)

Table 7. **STEVAL-WLC38RX BOM**

Item	Q.ty	Ref.	Value	Description	Manufacturer	Order code
1	11	TP1, TP2, TP3, TP4, TP5, TP6, TP7, TP8, TP9, TP10, TP11		TEST POINT	Harwin	S2751-46R
2	9	C1, C2, C3, C4, C21, C26, C28, C30, C31	100n	SMD 0603 CAP	Wurth	885012206095
3	7	R9, R10, R11, R12, R13, R14, R15	N.M.	SMD 0603 RES	N.M.	N.M.
4	5	C18, C19, C20, C24, C25	10u	SMD 0603 CAP	Murata	GRM188R6YA106MA73D
5	5	JP1, JP2, JP3, JP4, JP5		SMD 0603 JUMPER	N.M.	N.M.
6	5	P4, P7, P8, P11, P14		Header 2	Harwin	M20-9770242
7	5	R19, R20, R23, R24, R25	4k7	SMD 0603 RES	Panasonic	ERJ-3EKF4701V
8	4	C5, C17, C22, C23	N.M.	SMD 0603 CAP	N.M.	N.M.
9	4	C32, C34, C39, C40	100n	SMD 0603 CAP	Wurth	885012206095
10	4	D1, D2		TVS diode	Littelfuse	SMF13A
11	4	C14, C27, C33, C35	4u7	SMD 0603 CAP	Murata	GRM188R61E475KE11D
12	3	P5, P9, P10		Header 4	Harwin	M20-9730445
13	5	R3, R7, R18, R30, R31	10k	SMD 0603 RES	Panasonic	ERJ-3GEYJ103V
14	2	R8, R17	100k	SMD 0603 RES	Panasonic	ERJ-3GEYJ104V
15	2	C41, C42	47pF	SMD 0603 CAP	Wurth	885012006055
16	2	C6, C13	3.3n	SMD 0603 CAP	Wurth	885012206086
17	2	C7, C12	47n	SMD 0603 CAP	Wurth	885012206093
18	3	C8, C11, C15	22n	SMD 0603 CAP	Wurth	885012206091
19	2	C9, C10	10n	SMD 0603 CAP	Wurth	885012206089
20	2	C37, C38	390pF	SMD 0603 CAP	Samsung	CL10B391KB8NNNC
21	2	D3, D4		DFN-2	Diodes	1N4448HLP-7
22	2	P2, P13		Header 1	Adam Tech	PH1-01-UA
23	2	R4, R5	200R	SMD 0805 RES	Panasonic	ERJ-P06F2000V
24	2	R27, R28	5k1	SMD 0603 RES	Panasonic	ERJ-3GEYJ512V
25	2	D5, D6	N.M.	TVS diode	N.M.	N.M.
26	2	R35, R36	33R	SMD 0603 RES	Panasonic	ERJ-3GEYJ330V
27	1	C16	1.5n	SMD 0603 CAP	Wurth	885012206084
28	1	RTC		SMD 0603 RES	Vishay	NTCS0603E3104FXT
29	1	C29	1u	SMD 0603 CAP	Wurth	885012206052
30	1	C36	100pF	SMD 0603 CAP	Wurth	885012206077
31	1	D7		LED	Wurth	150060RS75000
32	1	D8		LED	Wurth	150060RS75000
33	1	D9		LED	Wurth	150060YS55040
34	1	D10		LED	Wurth	150060VS55040
35	1	D11		Shottky Diode (1A)	Diodes	B220A-13-F
36	1	J1		USB-C	Wurth	629722000214

Item	Q.ty	Ref.	Value	Description	Manufacturer	Order code
37	1	P1		Header 4	Luxshare	LD81FP008-1H
38	1	P3		Header 4X2	Harwin	M20-9980445
39	1	P6		Header 8	Harwin	M20-9730845
40	1	R1	6k8	SMD 0603 RES	Panasonic	ERJ-3GEYJ682V
41	1	R2	220k	SMD 0603 RES	Panasonic	ERJ-3GEYJ224V
42	1	R6	N.M.	SMD 0805 RES	N.M.	N.M.
43	1	R16	150k	SMD 0603 RES	Panasonic	ERJ-3GEYJ154V
44	1	R21	820	SMD 0603 RES	Panasonic	ERJ-3GEYJ821V
45	1	R22	47k	SMD 0603 RES	Panasonic	ERJ-U03J473V
46	1	R26	1M	SMD 0603 RES	Panasonic	ERJ-3GEYJ105V
47	1	R32	4k7	SMD 0603 RES	Panasonic	ERJ-3EKF4701V
48	2	R33, R29	N.M.	SMD 0603 RES	N.M.	N.M.
49	1	R34	5k1	SMD 0603 RES	Panasonic	ERJ-3GEYJ512V
50	1	S1		SW-SPST	Bright	P-DT2112C
51	1	U1		STWLC38	STMicroelectronics	STWLC38JRM
52	1	U2		FT260Q	FTDI	FT260Q
53	1	U4		SOT23-6	STMicroelectronics	USBLC6-2SC6
54	1	COIL			Luxshare	LD81FP008-1H
55	4	Standoff			Harwin	R30-6700394
56	4	Screw			Wurth	97790803111
57	1	Pair of wires between COIL and PCB			CNC Tech	610224
58	1	Glue			Loctite	LOCTITE 454
59	3	2.54 jumper			Wurth	60900213421

8 STEVAL-WLC38RX description and operation

8.1 STEVAL-WLC38RX coil description

Table 8. Electrical characteristics of the receiver coil

Parameter	Specification
Part number	LD81FP008-1H
Vendor	Luxshare-ICT
Inductance (@ 100 kHz)	8 +/- 0.2 µH
DC resistance (DCR)	190mΩ (±10%)
AC resistance (ACR)	225mΩ (±10%)

Figure 49. Mechanical characteristics of the receiver coil 1

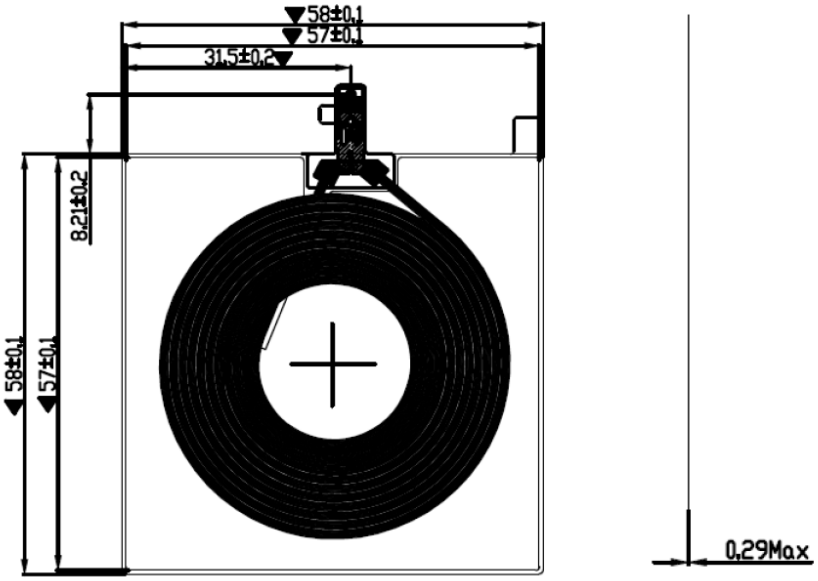


Figure 50. **Mechanical characteristics of the receiver coil 2**

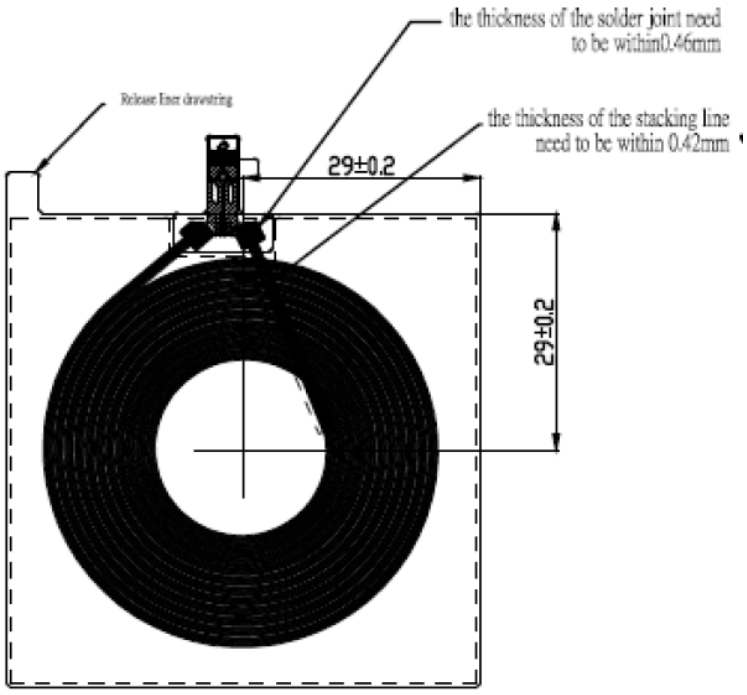
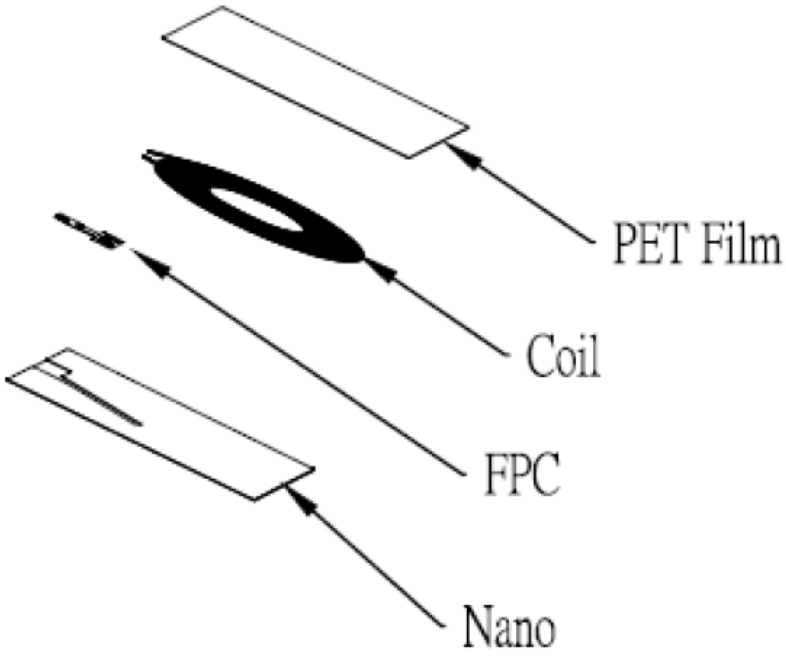


Figure 51. **Mechanical characteristics of the receiver coil 3**



8.2 STEVAL-WLC38RX PCB layout

Figure 52. **STEVAL-WLC38RX PCB layout top layer**

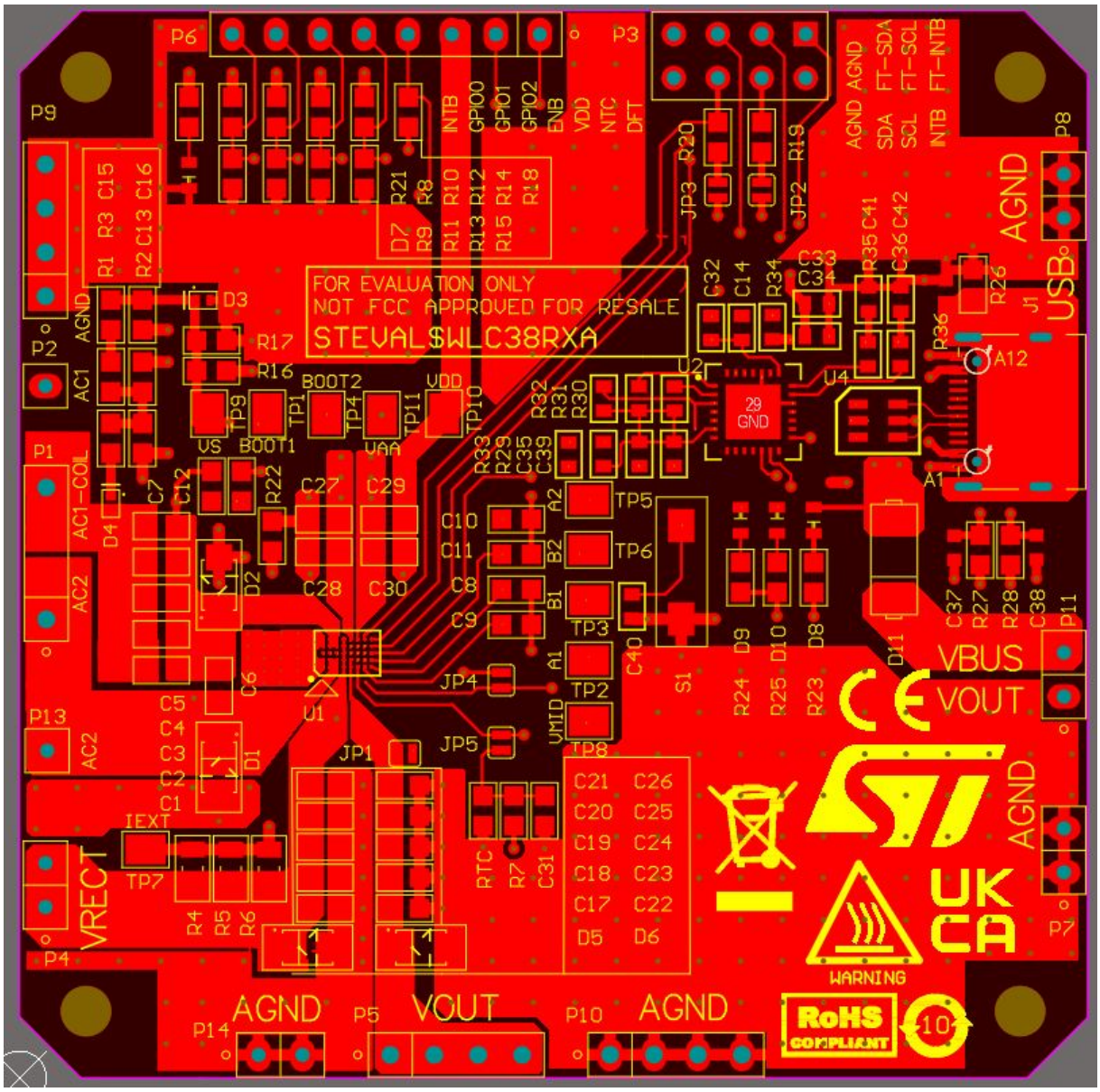


Figure 53. **STEVAL-WLC38RX PCB layout inner 1 layer**

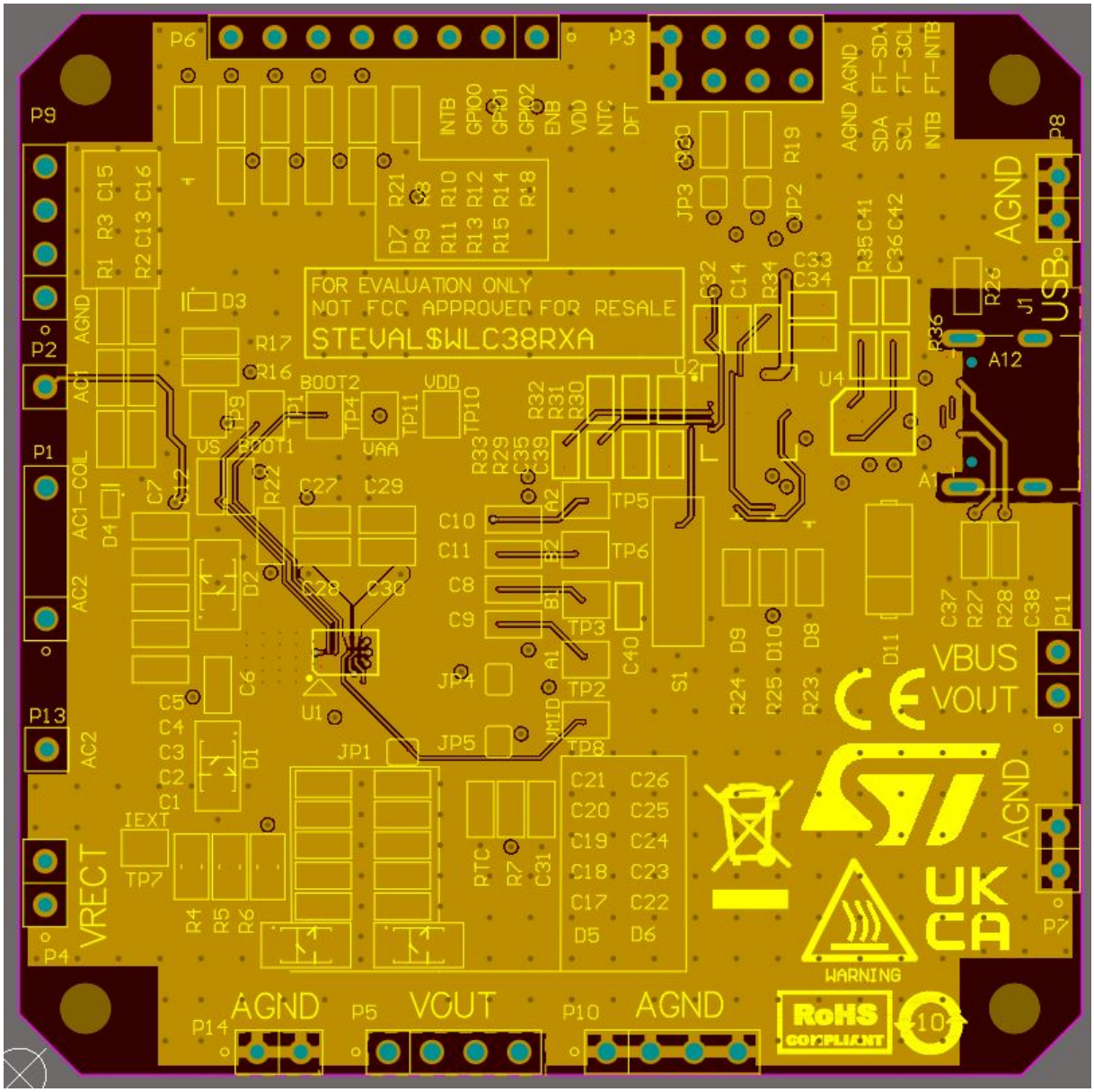


Figure 54. **STEVAL-WLC38RX PCB layout inner 2 layer**

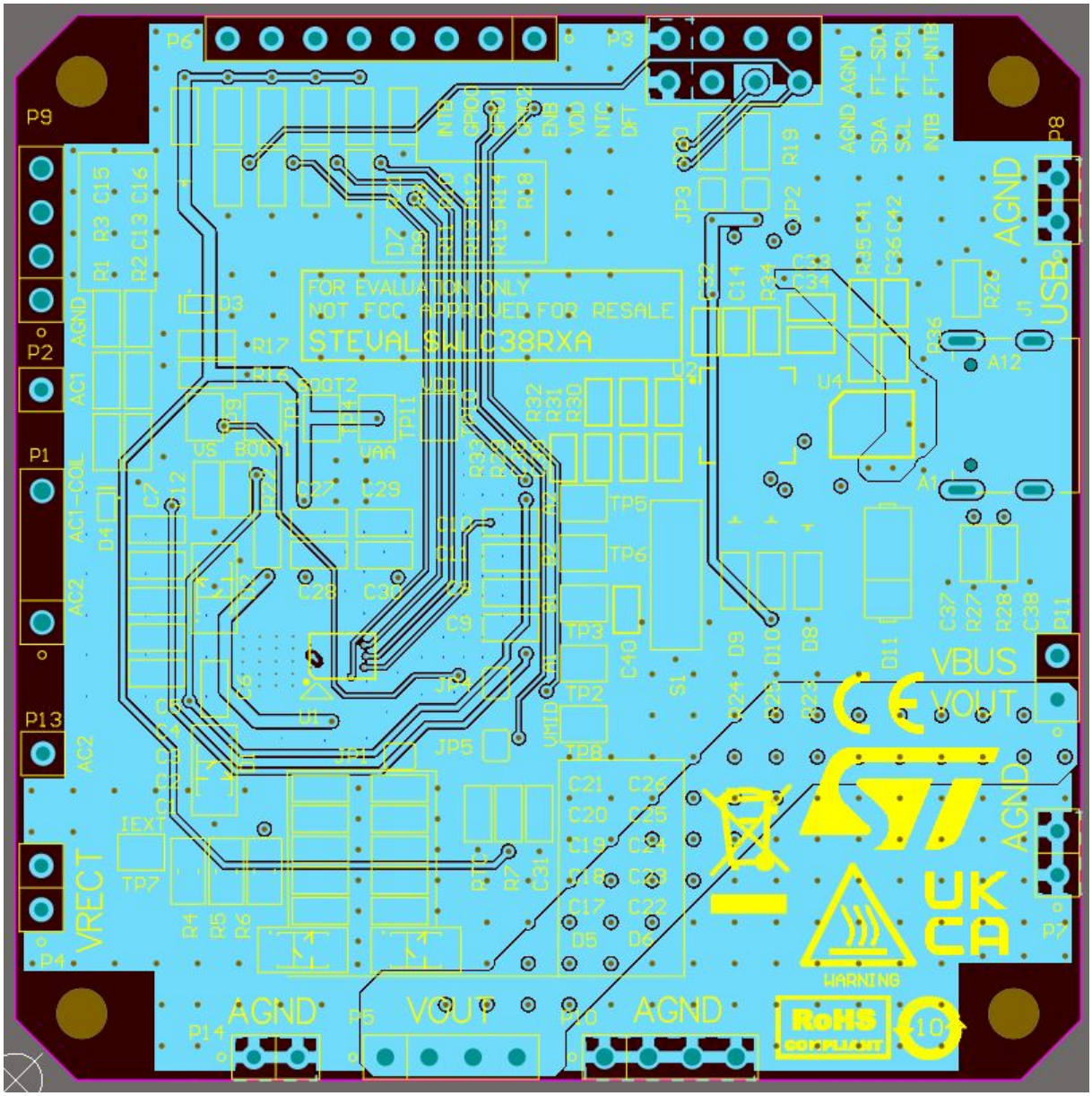
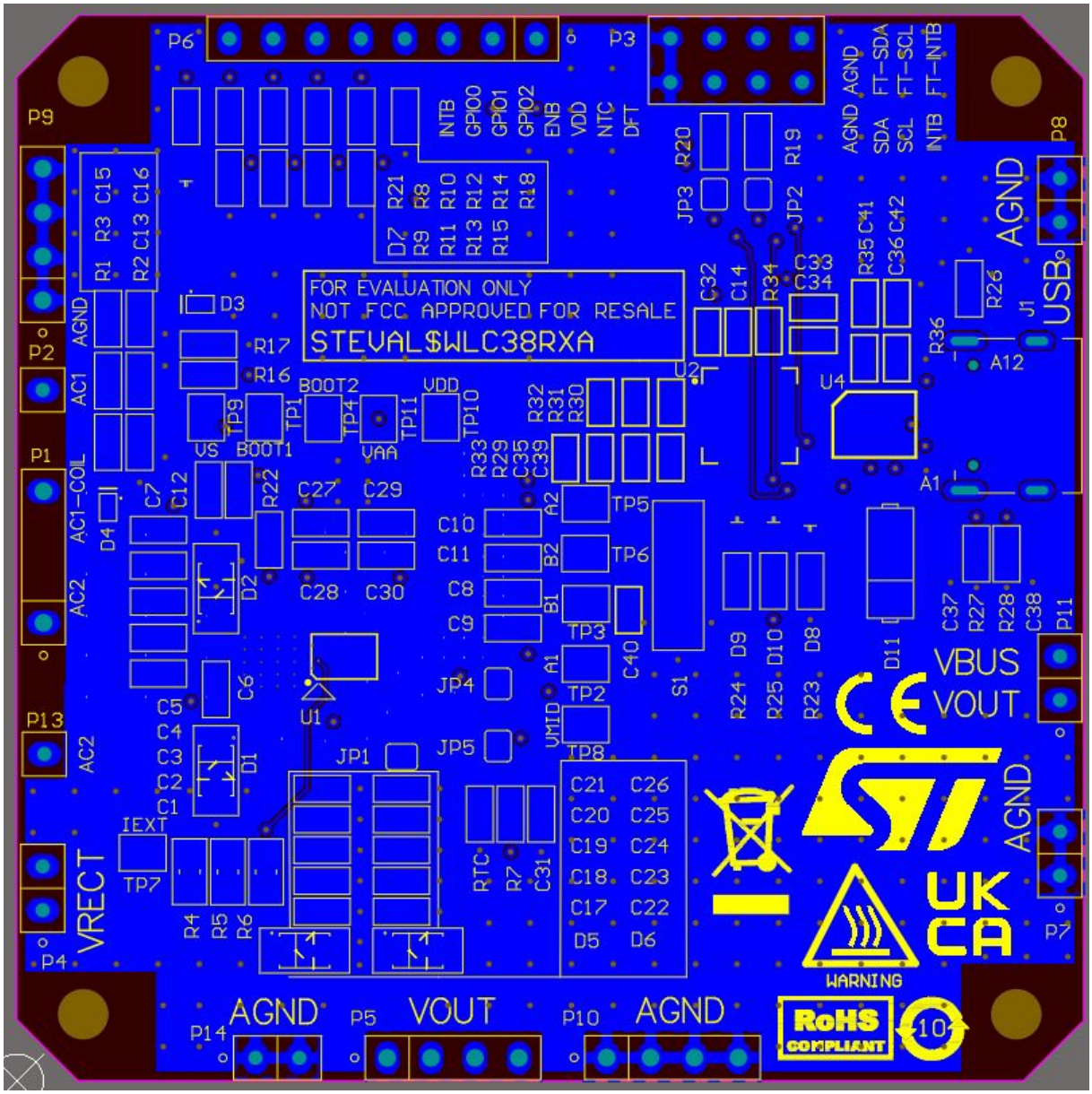


Figure 55. **STEVAL-WLC38RX PCB layout bottom layer**



8.3 STWLC38 default configuration

Table 9. **STWLC38 default configuration**

Parameter	Value
Output voltage (BPP mode, 5 W)	5 V
Output voltage (EPP mode, 15 W)	9 V
Rectifier mode	half-sync up to 160 mA full-sync above 160 mA
Communication capacitors	COMA and COMB enabled up to 200 mA Iout COMB enabled above 200 mA Iout
Input voltage loop (IVL)	4 V
Hardware over-current protection (HW OCP)	1.75 A

Parameter	Value
Software over-current protection (SW OCP)	1.75 A
Soft over-voltage protection (SOVP)	16 V
Software over-temperature protection (SW OVTP)	85 °C
Hardware over-temperature protection (TSHUT)	105 °C
Iload	32 mA
ARC mode disable threshold	6 V
Vrect ready threshold (Vout is enabled)	Vout setting + 240 mV
Manufacturer ID	0x0022
I^2C address	0x61 (7-bit address)
16bit register address	high-byte first low-byte second

- LDO setting (output current dependent)
  - Highest LDO drop: 320 mV
  - Lowest LDO drop: 128 mV
- Active interrupts
  - Message received from TX
  - OCP (over-current) protection
  - RX Output on
  - OVP (over-voltage) protection
  - OVTP (over-temperature) protection
  - Power Transfer Contract to EPP (15W). The Transmitter confirmed the EPP support.
- GPIO functions
  - GPIO3 as Interrupt
  - other GPIOs: no function

8.4 Typical performance characteristics

Board specifications as defined before (see Reference design specifications)

Table 10. Typical performance characteristics

Output current [mA]	Output voltage [V]	Rectifier voltage (Vrect) [V]	IC Temperature* [°C]
200	5.00	5.20	34
500	5.00	5.15	38
700	5.00	5.14	41
1000	4.97	5.10	48

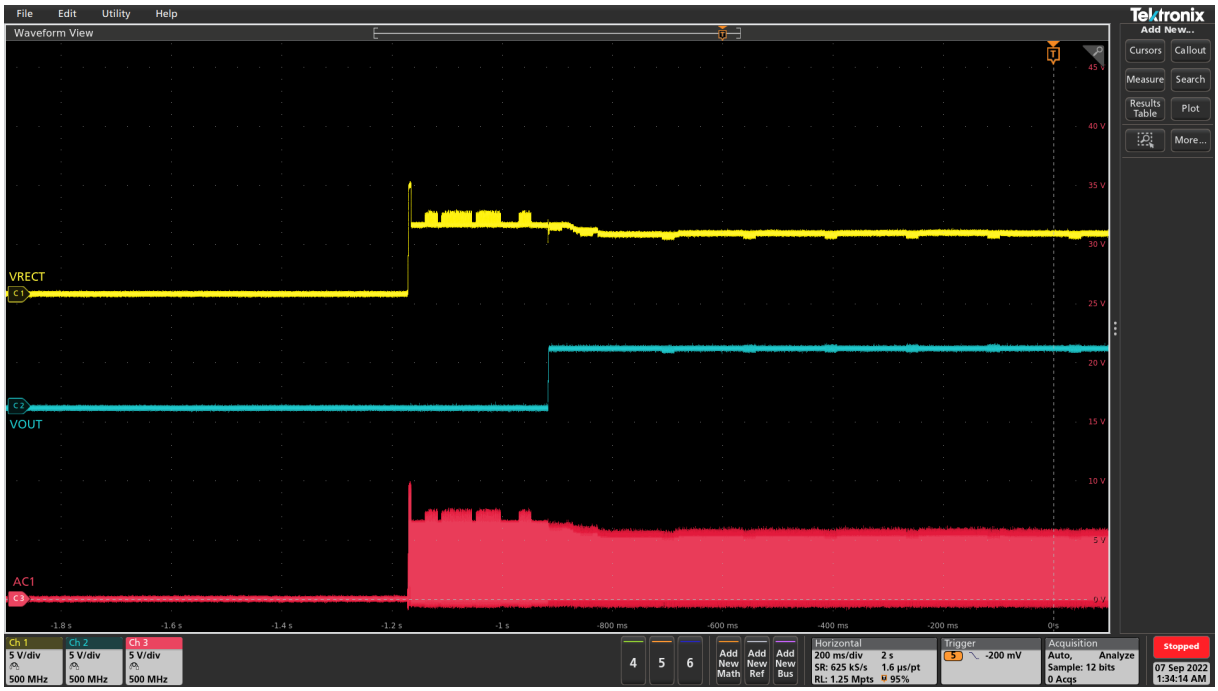
*5 minutes at each point, internal thermal sensor, ambient temperature was 23 °C

8.4.1 Power up waveforms

The following figure shows the power up waveforms (receiver side) of a STWLC38 (RX) and STWBC86 (TX) setup. Transmitter and Receiver coils are aligned; a 500-mA resistive load is connected to the receiver output.

Figure 56. Power up waveforms of a STEVAL-WLC38RX and STEVAL-WBC86TX setup

YELLOW→VRECT, BLUE→VOUT, RED→AC1



8.4.2 Efficiency and spatial freedom in the XY plane

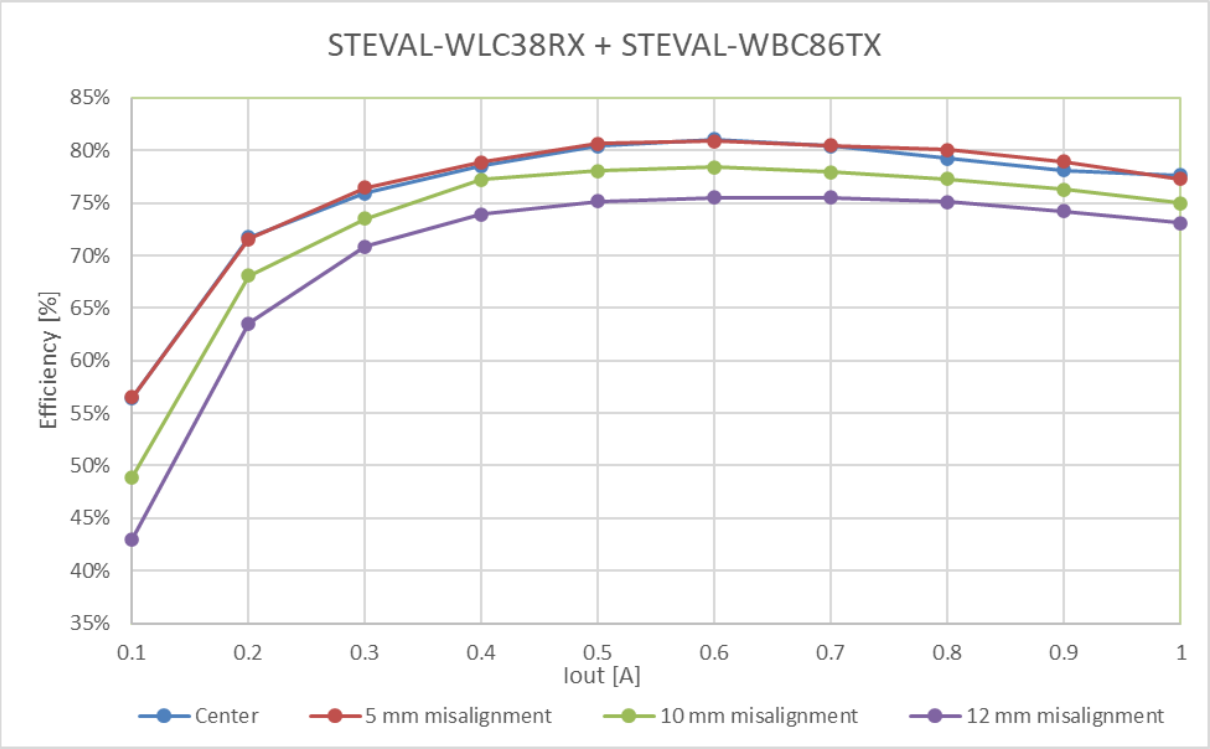
Efficiency is one of the most important metrics of wireless charging performance evaluation. Spatial freedom, size of the area in which a power receiver can be placed on the power transmitter which still allows sufficient power to be transmitted, is another important metric.

Efficiency and spatial freedom of STEVAL-WLC38RX were measured with STEVAL-WBC86TX as the transmitter. The efficiency was measured from the transmitter DC input to the receiver DC output. The measurement does not include any power losses in the wall adapter or the cable.

The test setup consisted of:

- 20 W PISEN wall adapter, model A829-120167C-EU1
- USB-C power cable from Vention, model COT
- STEVAL-WBC86TX as Transmitter
- STEVAL-WLC38RX as Receiver
- Electronic load in CC mode, model BK Precision 8500.

Figure 57. **Efficiency curves for various misalignments in the X-axis**



The data clearly demonstrates how the ARC mode feature improves spatial freedom. However, minimizing coil misalignment is recommended whenever possible, as a large misalignment leads to lower efficiency of power transfer. Since FOD is evaluated based on power loss in the system, a large misalignment may also trigger FOD protection.

8.4.3 Spatial freedom in the Z-axis

Z-axis distance between the coils, also known as charging gap, is an additional parameter which significantly affects charging performance. Therefore, STEVAL-WLC38RX was also tested at various charging gap distances. The measurements examined two key performance indicators – efficiency and maximum delivered power.

Figure 58. **Efficiency curves for various misalignments in the Z-axis**

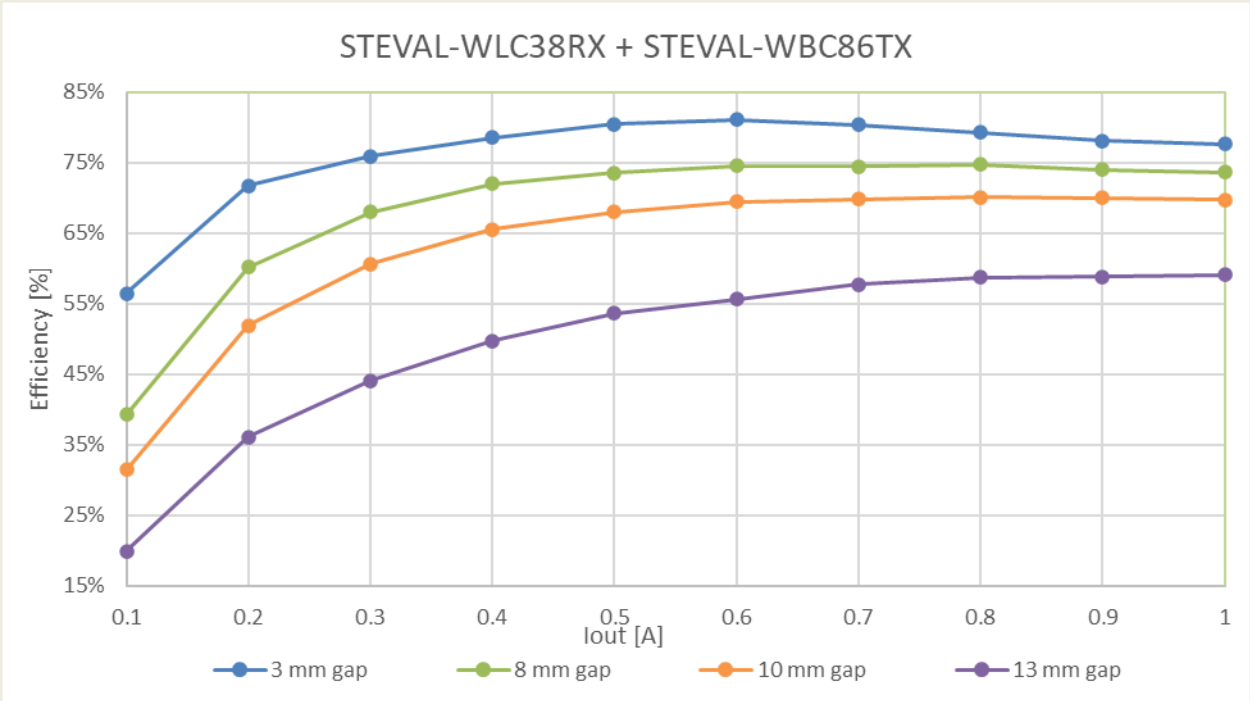


Table 11. **maximum power for Z-axis gaps**

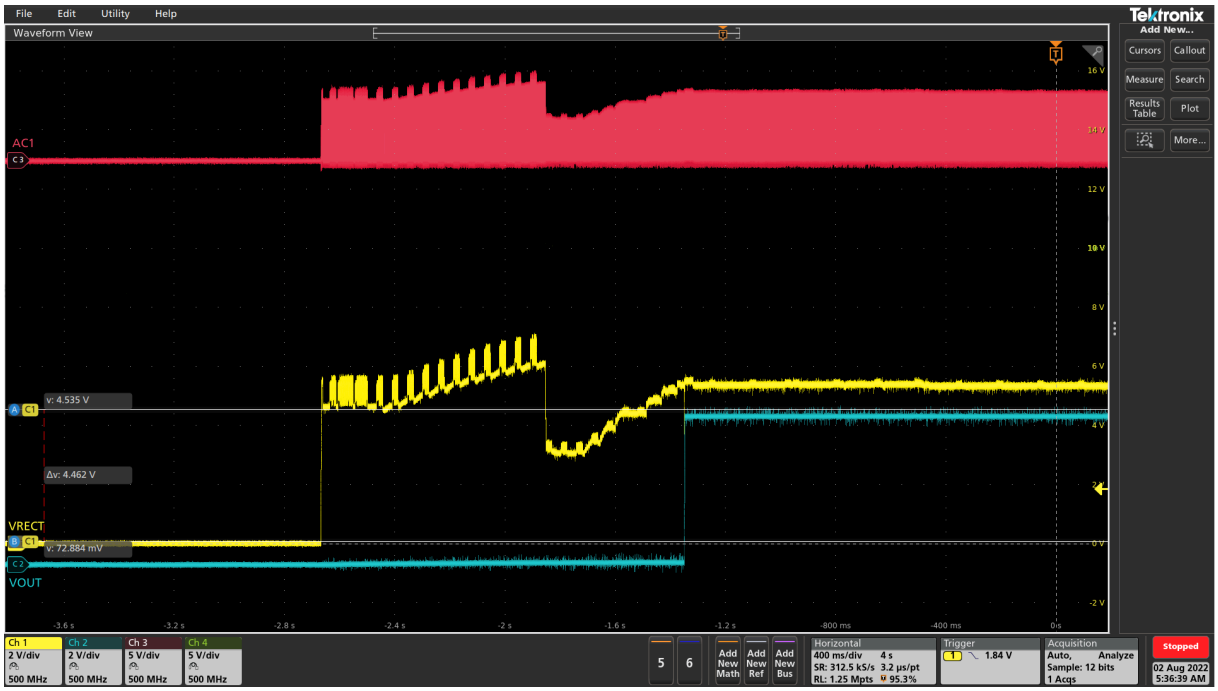
Z gap [mm]	Maximum power(1) [W]
4	5.002
6	5.000
8	5.001
11	5.000
13	5.000
16	3.093

(1) The maximum power is also limited by the transmitter, as STEVAL-WBC86TX is only BPP compatible

ARC mode also improves performance at higher Z-axis distances. The following waveform shows how STWLC38 is able to establish communication even at a high Z-distance (16 mm), and after reaching the default 6 V VRECT threshold, switches to a different operating mode where output can be enabled.

Figure 59. **STEVAL-WLC38RX communication performance at high Z distances**

YELLOW→VRECT, BLUE→VOUT, RED -> AC1



8.4.4 Thermal performance

Figure 60. **Thermal performance with a 5 W load (5 V/1 A) after 5 minutes**

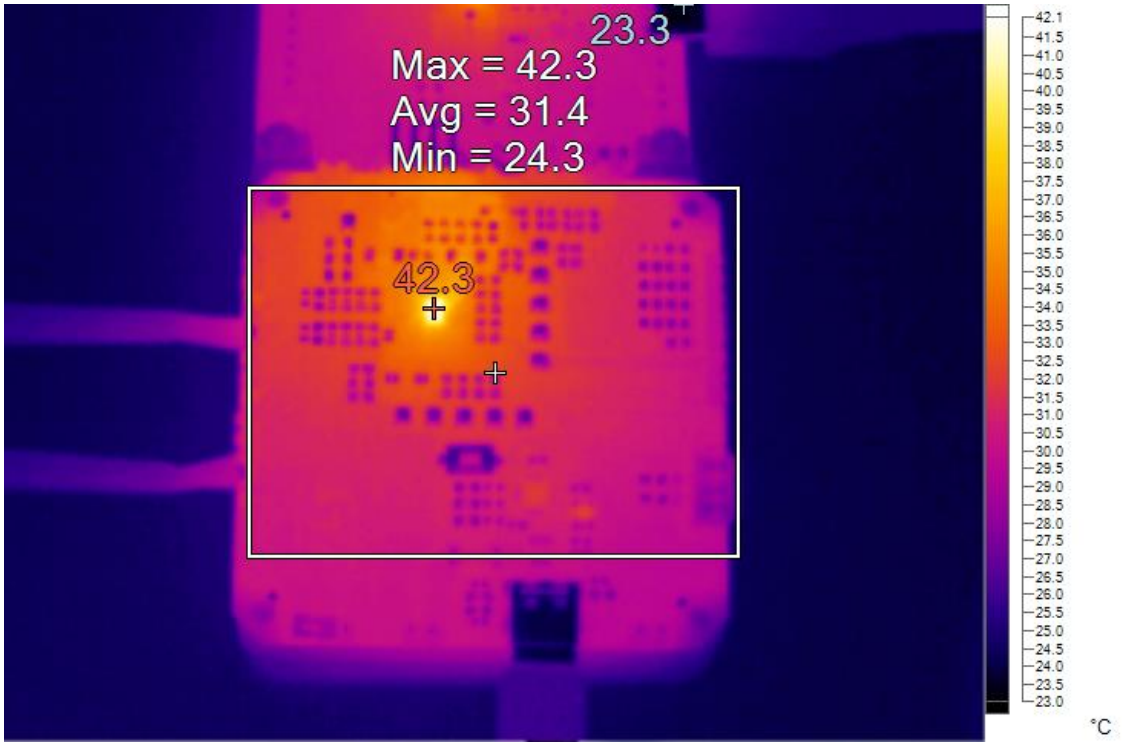
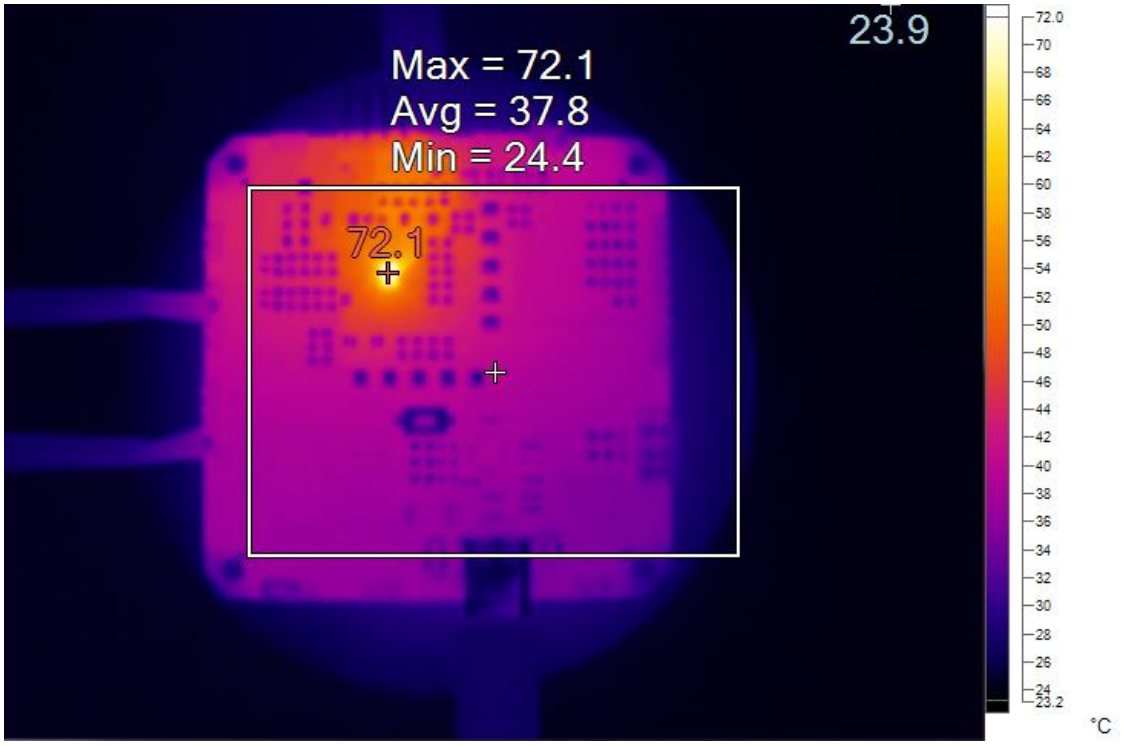


Figure 61. **Thermal performance with a 15 W load (9 V/1.7 A) after 5 minutes**



9 Designing a 5 W Qi BPP or 15 W Qi EPP wireless power receiver based on STEVAL-WLC38RX evaluation board

The design should begin with external component selection, as those components have a significant impact on the board performance. Modifications of the external components are also harder to implement after the board is manufactured and assembled.

9.1 External component design

STWLC38 requires only a few external components to work properly. The design should begin with coil and series resonant capacitor selection, as those are the crucial part of the design and affect most of the device's functions.

9.1.1 Coil selection

Coil selection is a complex problem, as there are several factors which influence the resulting performance. When selecting the coil, the designer should consider the coil's inductance, DCR, ACR, current rating, dimensions, maximum z-distance and layout.

DCR, ACR (DC and AC resistance) should be kept as low as possible to minimize power loss caused by the current flowing through the coil.

Dimensions, maximum z-distance and coil layout must be considered to achieve the best coupling possible, as better coupling results in higher efficiency. Similar sized coils might offer a higher coupling factor, on the other hand a different sized RX/TX coil might offer greater freedom of positioning.

Using the default coil combination found in ST evaluation kits should be a good starting point. For more information regarding coil selection, please refer to AN5961.

9.1.2 Series and parallel resonant capacitor selection

Selection of the series and parallel capacitors is greatly influenced by the transmitting coil design. The value should be selected so that the resulting circuit has its resonant peak located at, or very close to, 100 kHz. The following formula can be used to calculate the series capacitance value:

$$f = \frac{1}{2 \cdot \pi \cdot \sqrt{L \cdot C}} \rightarrow C = \frac{1}{L \cdot (2 \cdot \pi \cdot f)^2}$$

The parallel capacitance can be omitted, as it is only used for backwards compatibility with certain power transmitters (required during Qi certification).

Voltage rating of the capacitors should consider that during operation, voltage generated across the capacitors is usually much higher than the input/output voltage of the system. Also, a quick movement of the receiving coil from a misaligned position to the transmitting coil center may generate a voltage peak, as the regulation loop is quite slow. Therefore, the minimum recommended voltage rating of the series resonant capacitors is 50 V. X5R/X7R dielectric capacitors are recommended if price is the deciding factor, C0G dielectric capacitors are recommended for better performance (lower derating caused by voltage biasing and temperature shift).

For more information regarding capacitor value selection, please refer to Wireless Power Consortium documentation, which describes capacitors used in the Qi certified topologies.

9.1.3 VRECT and VOUT capacitor selection

VRECT and VOUT capacitor value selection is mostly a tradeoff between load transient response time and capacitor cost and/or size. A capacitance of 30 uF is recommended for a 15 W application. For lower power applications, the capacitance on VOUT and VRECT can be reduced depending on power scaling.

The VRECT capacitor voltage rating should respect the maximum input voltage. Voltage peaks generated by quick coil movements should also be a concern when determining the voltage rating. Leaving a greater margin is advisable. The VOUT capacitor voltage rating is determined by the selected output voltage. Ceramic X5R or X7R (preferred) dielectric capacitors are recommended for both VRECT and VOUT.

Capacitor derating caused by temperature shift and DC bias should also be considered, as the capacitance decrease caused by those effects may be quite significant. Please refer to the capacitor datasheet for more information.

9.1.4 ASK modulation capacitor selection and configuration

STEVAL-WLC38RX features two pairs of modulation capacitors, with each pair having a different capacitance value – 10 nF for COMA, 22nF for COMB.

The modulation capacitors used in STEVAL-WLC38RX were chosen to ensure good communication performance under all operating conditions. Using a custom board design may require changing the COMA and COMB capacitors.

The COMA and COMB capacitors are used for power signal modulation. Although the modulation period is short, the voltage drop it creates across the LDO must be considered, as this drop further increases overall losses of the system. This is particularly important when the device is delivering a high load current.

The GUI allows the customer to enable a particular pair to be active during low output current operation, while using the other and/or same during high current operation. The same can be set for low/high VRECT voltage condition. Therefore, four separate capacitor combinations can be selected in total.

Figure 62. Rx ASK configuration

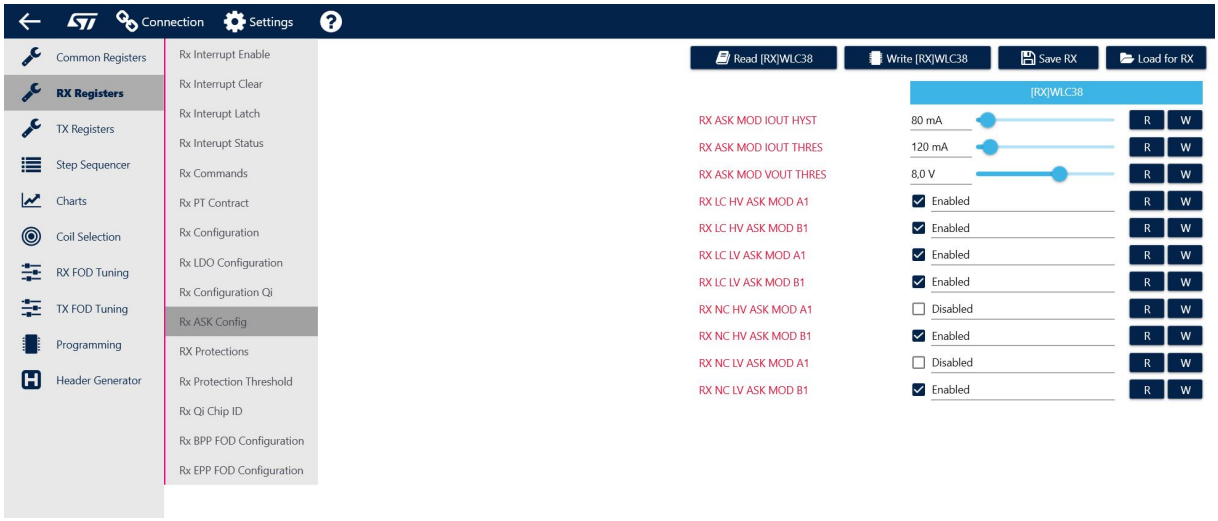


Table 12. Default STWLC38 ASK configuration

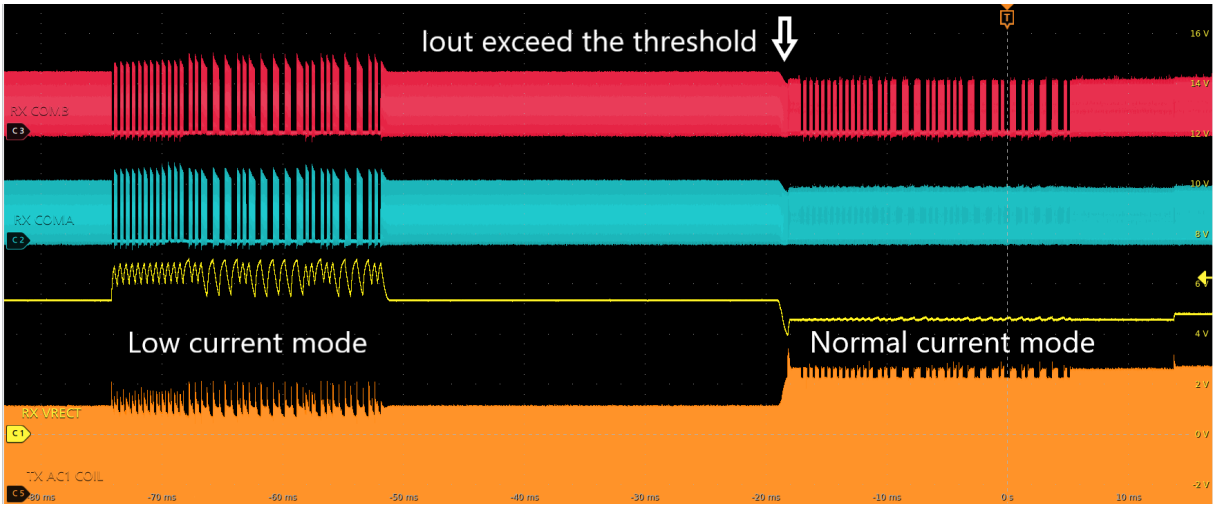
Setting	Low Current (LC)	Threshold	Normal Current (NC)
Low Voltage (LV)	COMA enabled	Low Current to Normal Current: 200 mA Normal Current to Low Current: 120 mA	COMA disabled
	COMB enabled		COMB enabled
Threshold	8V		8V
High Voltage (HV)	COMA enabled		COMA disabled
	COMB enabled		COMB enabled

The following screenshot shows an example of RX Vrect (yellow), RX COMA (blue), RX COMB (red) and TX COIL sensing (orange). In the picture you can see how the modulation capacitors affect the Vrect voltage and coil sense voltage of a coupled transmitter - used for ASK demodulation.

The modulation effect shall be considered in regards to the overall power loss, as the LDO voltage drop (Vrect - Vout) multiplied by the output current adds an addition power loss into the system.

Changing the modulation capacitor value may have an impact on the TX coil voltage. This voltage is used for ASK demodulation circuit. The impact depends on various variables such as coil, gap, voltage, transfered power, misalignment and another parameters.

Figure 63. **The transition from Low current to Normal current and the effect on modulation**



This should be considered when configuring the LDO drop curve (see LDO output voltage configuration).

9.1.5 SOVP clamp resistor selection

SOVP clamp resistors, also known as IEXT resistors, are used as a part of the SOVP protection. When triggered, STWLC38 pulls VRECT to ground through those resistors in an attempt to lower the VRECT voltage. Therefore, the resistance value must be selected according to the selected SOVP threshold and power dissipation capabilities of the resistors. IEXT pins' current capabilities must also be considered, as the internal switches can only withstand up to 300 mA of current.

9.1.6 ESD/TVS diodes

STEVAL-WLC38 also features two external TVS diodes which protect the STWLC38's input pins (AC1 and AC2) against excessive input voltage. The breakdown voltage of those diodes is 14.4 V.

9.1.7 NTC thermistor connection

The figure below shows external NTC connection. Together with the R7 pull-up resistor the NTC forms a voltage divider, which is connected to the STWLC38 NTC pin. It is recommended to design the voltage divider to reach the (low) NTC threshold at the highest allowed temperature.

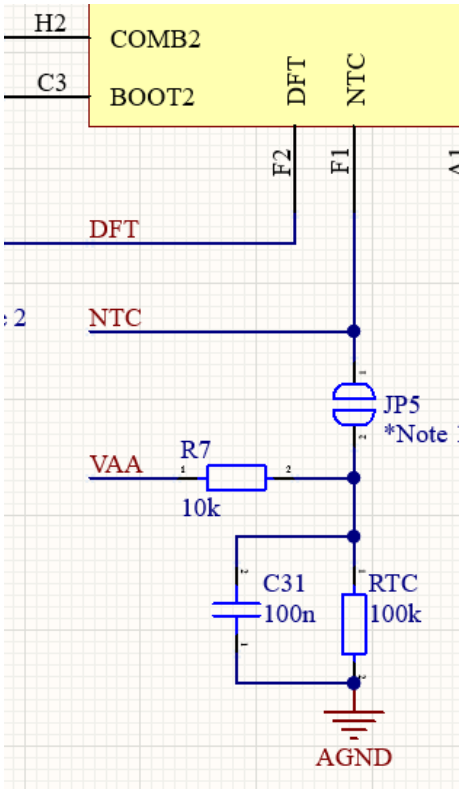
$$ntc = \left(\frac{accumulated_{ADC}}{2^{10} \times acc} \times 2 \times 2.25 \times 0.8 + 1.8\right)$$

ADC raw data two's complement (2 bytes)

acc constant default value 16

ntc calculated value in mV

Figure 64. **External NTC connection**



9.2 PCB layout guidelines

- Power tracks (AC1, AC2, VRECT, VOUT) and power ground tracks should be kept wide enough to sustain high current. Duplicating these tracks in inner layers and adding vias is advisable wherever possible to lower impedance as much as possible.
- AC1, AC2, BOOT1, BOOT2, COMAx, and COMBx generate noise. Using shielding near these traces (by placing ground planes below) is recommended.
- Power ground carries the sum of ripple current from VRECT/Crect and DC current from VOUT/Load. Current return paths from LDO capacitors should be routed separately from the high current paths.
- AC1 and AC2 tracks should be routed close together to minimize the area of the resulting loop.
- Communication (I^2C) and sensing signals should be routed far from noise generating nets (AC1, AC2, COMAx, COMBx, BOOT) to minimize the effects of interference induced from those high di/dt nets.
- COMM1, COMM2, BOOT1, BOOT2 capacitors should be placed as close to the device as possible.
- VRECT and VOUT capacitors should be placed close to the device to minimize the area of high current loops.
- Auxiliary LDO capacitors should be placed as close to the device as possible.
- Thermal performance and grounding can be enhanced by dedicating one layer as a ground plane. No signal/power tracks should be routed on this layer to ensure ground integrity.

9.3 LDO output voltage configuration

Configuration of the LDO has a major impact on efficiency, as it directly affects the magnitude of the power loss created by the passing current. Therefore, minimizing the voltage drop is desired when aiming for higher efficiency.

However, a higher LDO drop improves communication stability and helps prevent negative modulation, which might leak into the output voltage. Lastly, different rectifier operating modes also affect the VRECT waveform.

Therefore, LDO drop configuration is a tradeoff between efficiency and communication/output voltage stability.

Figure 65. **Default LDO configuration**

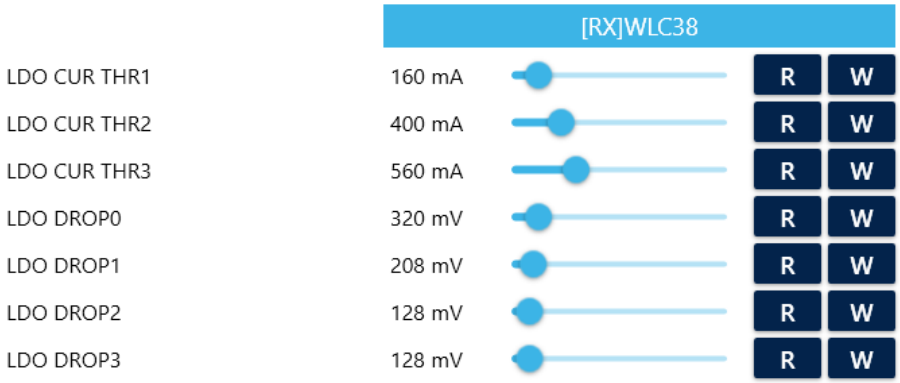
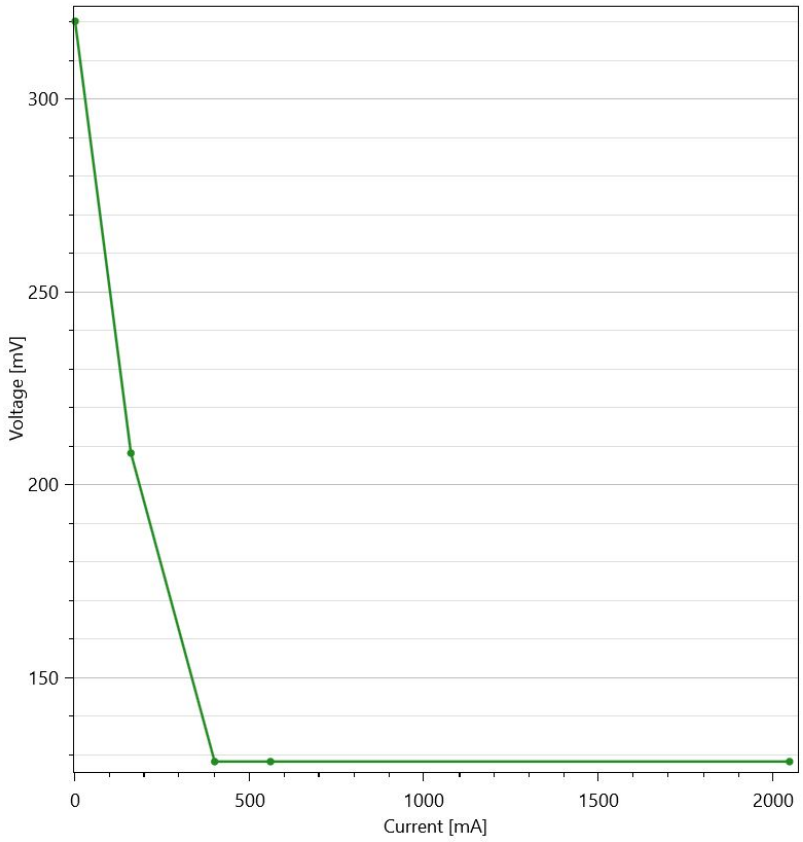


Figure 66. **Default LDO drop curve**



9.4 FOD tuning

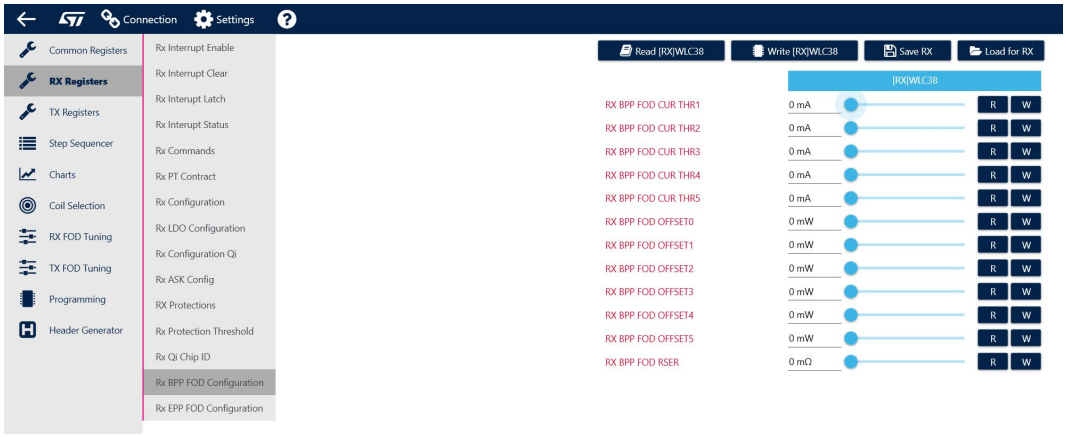
The aim of this tuning is to provide instructions and a step-by-step example of how to fine-tune STWLC38 parameters to comply with Qi requirements regarding foreign object detection.

Note: In the final product FOD tuning procedure requires a calibrated, Qi-compliant transmitter (like Nok9 system, GRL, etc.) capable of reporting both the transmitted power and received power (sent back from the receiver under test) without terminating the power transfer in case a FOD condition is detected.

9.4.1 BPP

Step 1. Begin by setting all FOD tuning registers to zero and generating a new Configuration file for STWLC38.

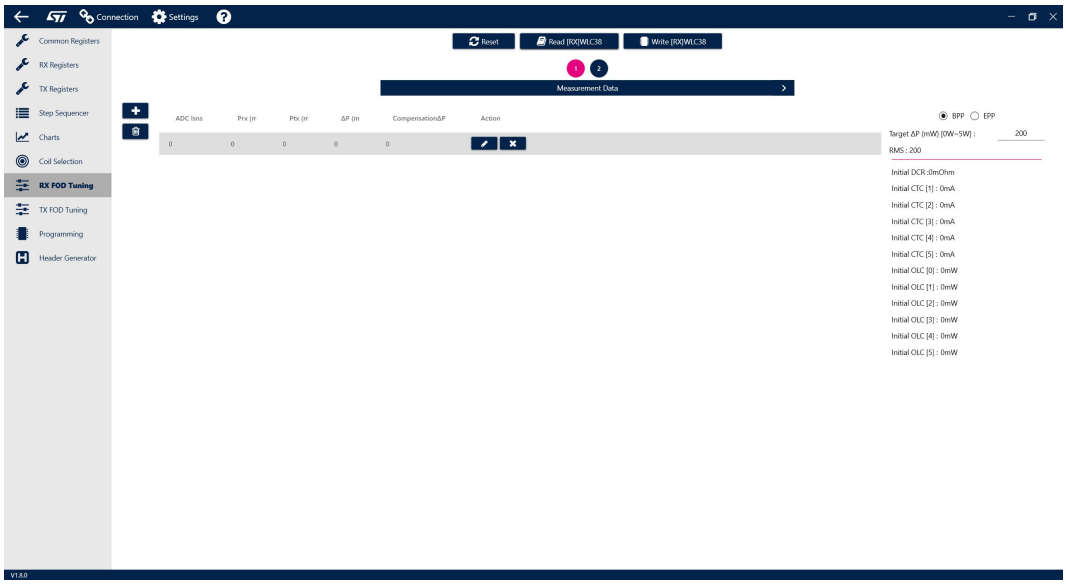
Figure 67. Cleared Rx FOD tuning registers



Step 2. Load the new (cleared) Configuration file into STWLC38.

Step 3. Open the FOD tuning tool by navigating to the FOD Tuning tab.

Figure 68. FOD Tuning tab



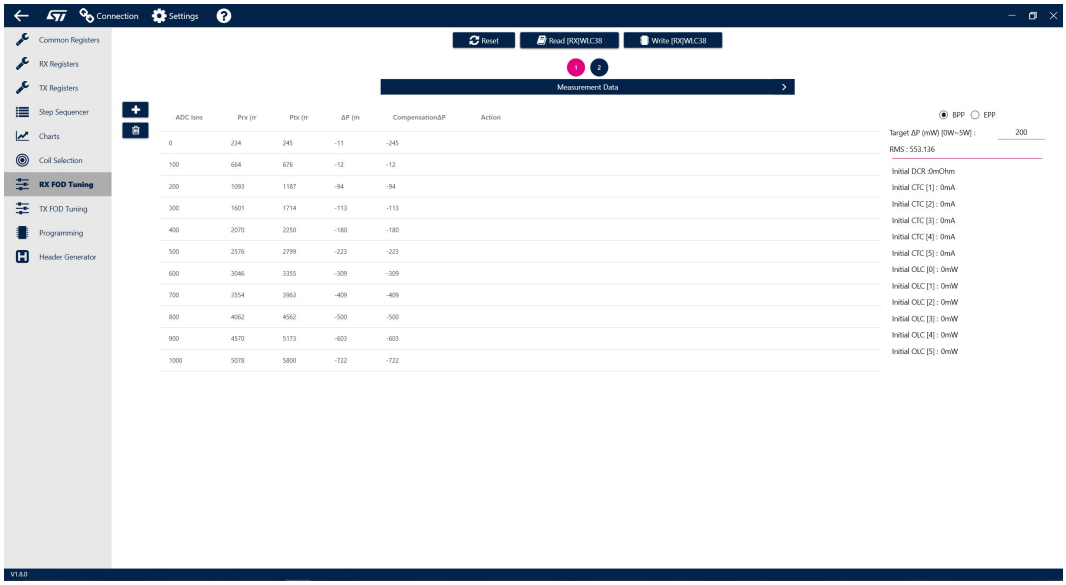
Step 4. Place STWLC38 onto a Qi-compliant power transmitter.

Step 5. Connect an electronic load to STWLC38 output (VOUT) and read Prx (RX Received power) and Ptx (Tx Transmitted power) from the transmitter registers. If the Transmitter is STEVAL-WBC86TX, the reading of Ptx and Prx is automatically done click on [+] and [Read From RAM]. As transmitter may be used reference Qi compliant transmitter and then the Prx and Ptx shall be manually written to the table.

Step 6. Increase the load from 0 to 1000 mA in 100 mA steps (in case of a 5 W setup) and record the Prx and Ptx values (use the [+] button to add more lines).

The Qi standard allows up to 350 mW difference between Prx and Ptx. The default target of the tool is set to 200 mW (can be adjusted on the right-hand side). The lower value provides a margin which helps prevent a false FOD trigger.

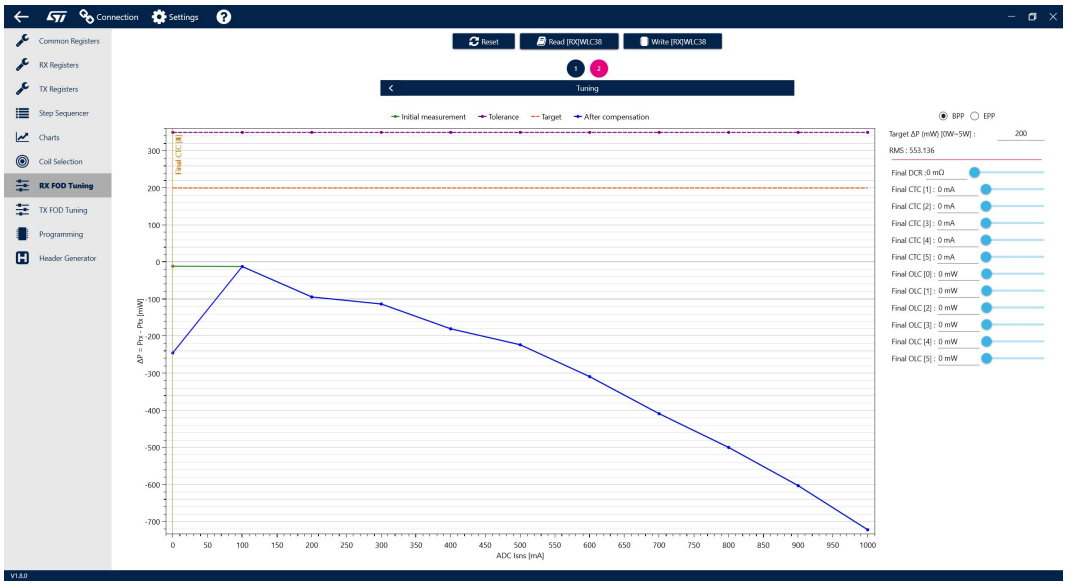
Figure 69. Load adjustment



Step 7. Navigate to the second page of the tool by clicking the 2 button above.

The initial Rx/Tx power difference, labeled ΔP, is shown in green. The target of the tuning is to find compensation values, which would move the blue (compensated ΔP) curve as close to the 200-mW (orange dashed) line as possible.

Figure 70. Rx/Tx power difference curve



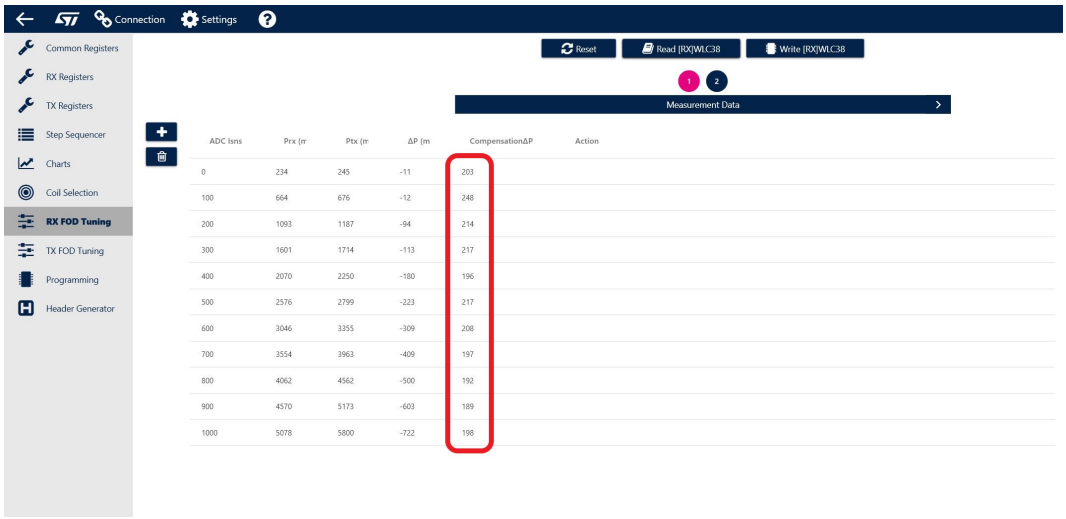
Step 8. Move the sliders on right-hand side to achieve a flat line close the target ΔP.

Figure 71. Tuning to obtain a flat BPP FOD curve



Step 9. Check the compensated values by navigating back to the first page of the tool.

Figure 72. Compensated values



Step 10. Copy the compensation values located on the right to the Rx Registers (RX BPP FOD Configuration) and generate a new Configuration file.

Step 11. Load the Configuration file into STWLC38.

Step 12. Repeat the previous measurement under the same conditions and check if the results are the same.

Step 13.

The results will not be exactly the same as there are a lot of variables affecting the power difference (X/Y missalignment, Z gap, transmitter measurement accuracy, STWLC38 measurement accuracy etc.), but they should be very similar.

Step 14. Repeat the test with a 5 mm misalignment in the X and Y axis.

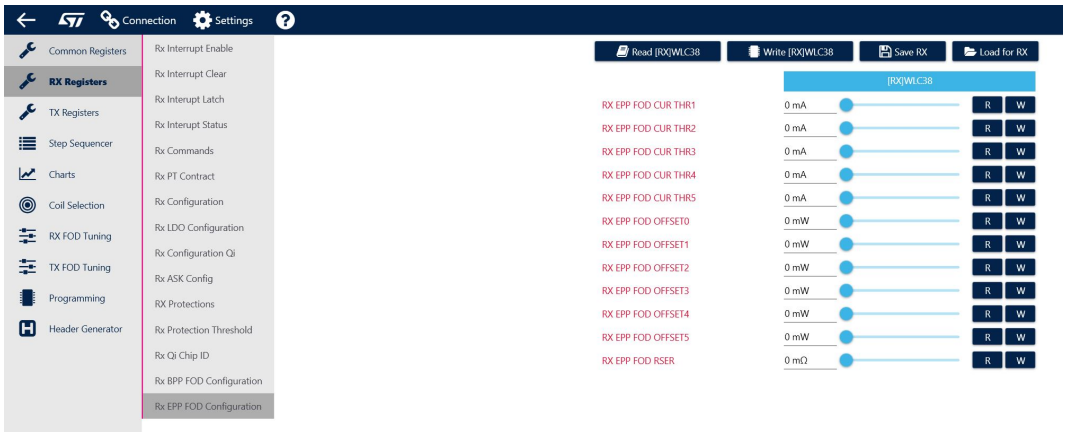
FOD should not be triggered by the misalignment.

9.4.2 EPP

Step 1. Begin by navigating to the Rx EPP FOD Configuration section.

Step 2. Set all the registers to 0 and generate a new Configuration file.

Figure 73. Cleared EPP FOD settings



Step 3. Load the file into the device.

Step 4. Navigate to the FOD tuning tab, switch to EPP mode and capture the data.

The tool allows the user to set a different target ΔP value for low (0-5 W), medium (5-10 W) and high (10-15) power operation.

Step 5. Using reference Qi compliant transmitter the ADC Isns, Prx and Ptx values shall be manually written in the table.

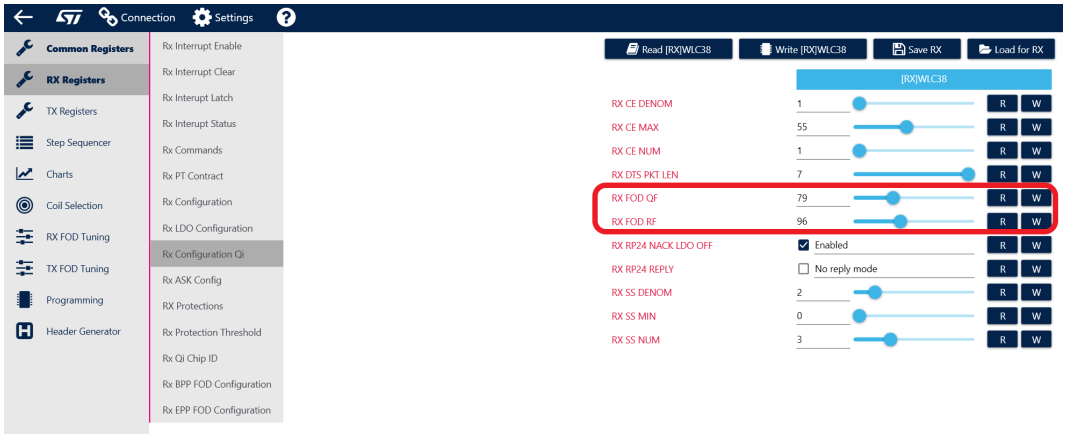
Figure 74. Tuning to obtain a flat EPP FOD curve



Step 6. Follow the same procedure as BPP mode to complete the tuning.

Step 7. The EPP FOD tuning is necessary to adjust the Rf (frequency) and Qf (quality factor). The value used is from a Qi certified EPP transmitter and should be set to the register shown below:

Figure 75. Rf - Qf



9.5 Reference code with STM32 development boards

9.5.1 Hardware requirements

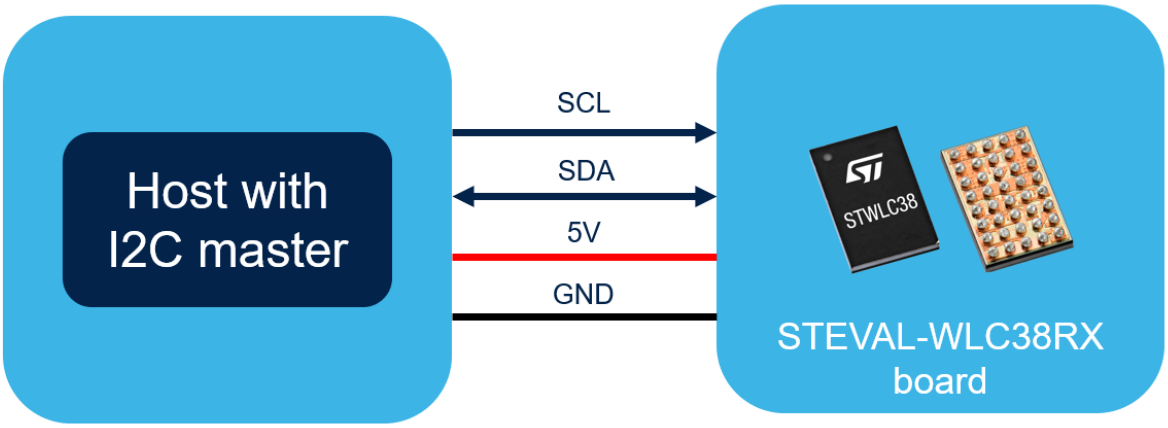
1. STM32 development board (e.g., STM32 Nucleo-144)
2. STEVAL-WLC38RX

9.5.2 Hardware connections

Step 1. Connect the STWLC38 I^2C pins to the master I^2C bus.
These signals require pull-up resistors to work properly.

Step 2. Connect a 5 V power supply to STWLC38 VRECT/VOUT.

Figure 76. Pin connection between host and STWLC38 chip/evaluation board



9.5.3 Software requirements

1. Patch/Configuration data in header (.h) file format
2. STWLC NVM programming reference code
3. STM32CubeIDE

9.5.4 Source files

The source files included in the reference code are listed below.

1. STWLC38.h
   – Provides NVM programming API and holds the structure and register definitions of STWLC38.
2. STWLC38.c
   – Main file which describes the NVM related programming sequences and I²C Write/WriteRead operations.
3. nvm_data.h
   – Patch and Configuration data in header file format to be programmed into the chip.

9.5.5 Reference code porting procedure

Step 1. Create a new project in the STM32CubeIDE with a STM32 development board. Visit the ST website for STM32CubeIDE documentation.

Step 2. Copy *STWLC38.h, STWLC38.c*, and *nvm_data.h* into the main directory (*Core/Src*).

Step 3. Add the following code into *Core/Src/main.c*.

Step 4. Include the Header file

```
/* USER CODE BEGIN Includes */
#include "STWLC38.h"
/* USER CODE END Includes */
```

Step 5. Define platform functions and variables to be used by driver API. See below for the function definitions.

```
    int32_t platform_write(void *phandle, uint8_t *wbuf, int32_t wlen);
    int32_t platform_write_read(void *phandle, uint8_t *wbuf, int32_t wlen, uint8_t
*rbuf, int32_t rlen);
    void platform_delay(uint32_t millisec);
    void *platform_alloc_mem(size_t size);
    void platform_free_mem(void *ptr);
```

Step 6. Declare and assign the platform functions to mandatory interfaces.

```
    /** stwlcxx is the used part number **/
    struct stwlcxx_dev stwlc38 = { 0 };

    stwlc38.bus_write = platform_write;
    stwlc38.bus_write_read = platform_write_read;
    stwlc38.mdelay = HAL_Delay;
    stwlc38.alloc_mem = malloc;
    stwlc38.free_mem = free;
```

Step 7. If user data is needed by the platform functions, initialize the phandle parameter.

```
    struct stmdev_platform platform = {
            .hi2c = &hi2c1,
            .huart = &huart3,
    };
    stwlc38.phandle = &platform;
```

Step 8. Optionally, initialize log parameters and set log_info to 1 to print more information logs.

```
    stwlc38.log = platform_log;
    stwlc38.log_info = 1;
```

Step 9. Execute NVM programming

```
/* Start FW update */
stwlc38_fw_update(&stwlc38, STWLC38_FW_PATCH_CFG, 0);
```

9.5.6 API

9.5.6.1 *stwlc38_write_fwreg*

Table 13. stwlc38_write_fwreg

Name	int32_t stwlc38_write_fwreg(struct stwlc38_dev *dev, uint16_t reg, const uint8_t *data, int32_t len)		
Description	Write device firmware register.		
Parameters	struct stwlc38_dev *	dev	Pointer to device interface.
	uint16_t	reg	Register to write
	uint8_t *	data	Pointer to data to write into register
	int32_t	len	Number of consecutive bytes to write.
Return Value	int32_t		Error status, return 0 if no error.

9.5.6.2 *stwlc38_read_fwreg*

Table 14. stwlc38_read_fwreg

Name	int32_t stwlc38_read_fwreg(struct stwlc38_dev *dev, uint16_t reg, uint8_t *data, int32_t len)		
Description	Read device firmware register		
Parameters	struct stwlc38_dev *	dev	Pointer to device interface.
	uint16_t	reg	Register to read
	uint8_t *	data	Pointer to store data read from register
	int32_t	len	Number of consecutive bytes to read.
Return Value	int32_t		Error status, return 0 if no error.

9.5.6.3 *stwlc38_get_chip_info*

Table 15. stwlc38_get_chip_info

Name	int32_t stwlc38_get_chip_info(struct stwlc38_dev *dev, struct stwlc38_chip_info *info);		
Description	Read device information.		
Parameters	struct stwlc38_dev *	dev	Pointer to device interface.
	struct stwlc38_chip_info *	info	Pointer to device information structure
Return	int32_t		Error status, return 0 if no error.

9.5.6.4 *stwlc38_fw_update*

Table 16. stwlc38_fw_update

Name	int32_t stwlc38_fw_update(struct stwlc38_dev *dev, enum stwlc38_fw_type fw_type, int32_t force_update);		
Description	Update STWLC38 patch and config.		
Parameters	struct stwlc38_dev *	dev	Pointer to device interface.
	enum stwlc38_fw_type	fw_type	Type of firmware data to write
	int32_t	force_update	Set 0 if no force update, otherwise yes

9.5.7 Error codes

Table 17. Error codes and descriptions

Error code	Value	Description
STWLC38_OK	0x00000000U	No error
STWLC38_ERR_BUS_W	0x80000000U	Bus write error
STWLC38_ERR_BUS_WR	0x80000001U	Bus write-read error
STWLC38_ERR_ALLOC_MEM	0x80000002U	Failed to allocate memory
STWLC38_ERR_INVALID_PARAM	0x80000003U	Invalid parameter
STWLC38_ERR_TIMEOUT	0x80000004U	Timeout error
STWLC38_ERR_INVALID_OP_MODE	0x80000005U	Invalid operation mode
STWLC38_ERR_INVALID_CHIP_ID	0x80000006U	Invalid target chip ID
STWLC38_ERR_NVM_ID_MISMATCH	0x80000007U	Compare failed patch or cfg ID
STWLC38_ERR_NVM_DATA_CORRUPTED	0x80000008U	Compare failed patch or cfg data

9.6 Common testing and debugging issues

Table 18. Common issues and causes

Indication of the issue	Possible causes
No VOUT voltage	• VRECT voltage too low – misalignment of the coils too big – insufficient transmitter power • RX protection triggered
LED D7 blinking	• Communication cannot be established – misalignment – other communication problems (e. g. noise)
LED D7 off	• Receiver is not powered – misalignment – transmitter not powered – transmitter protection triggered
Power transfer terminated	• Charging complete • Foreign object detected • Communication error
Excessive IC temperature	• High LDO power loss – LDO drop set too high – Modulation depth too high

10 Board versions

Table 19. **STEVAL-WLC38RX versions**

Finished good	Schematic diagrams	Bill of materials
STEVAL$WLC38RXA (1)	STEVAL$WLC38RXA schematic diagrams	STEVAL$WLC38RXA bill of materials

1. *This code identifies the STEVAL-WLC38RX evaluation board first version.*

11 Qi Certification

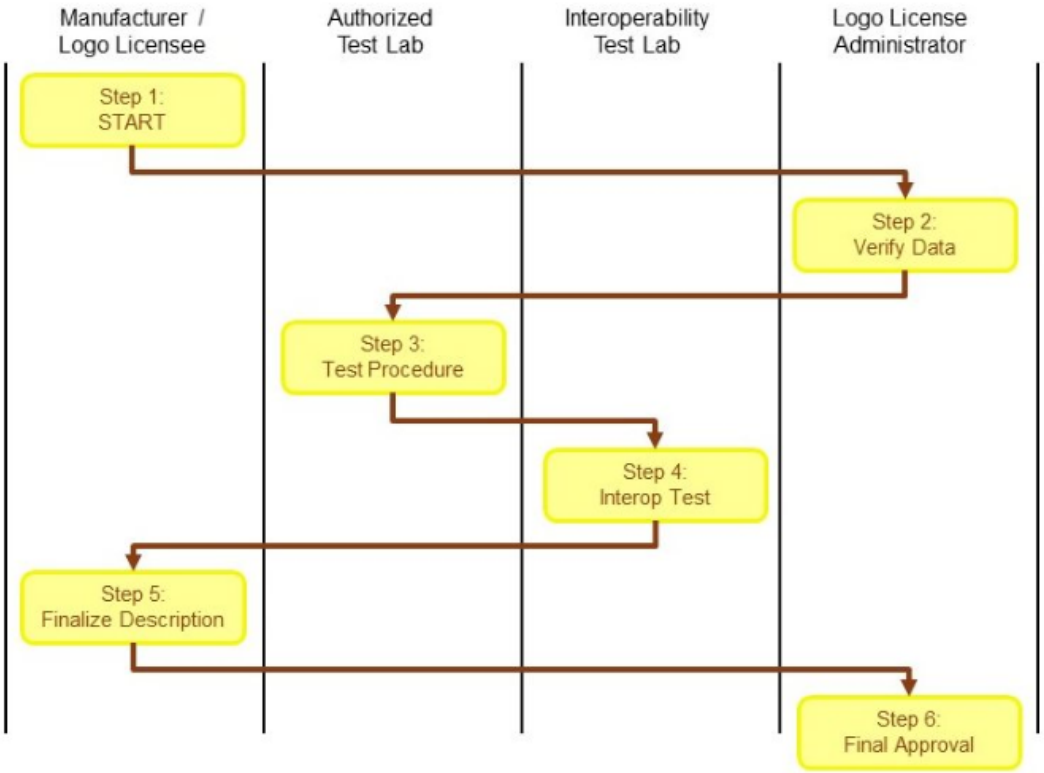
STWLC38 is Qi 1.3 EPP compatible and can therefore be Qi certified. The following section describes the basic requirements necessary for Qi EPP certification.

It should be noted that STWLC38 Extended Power Profile (EPP) Qi compatibility guarantees Baseline Power Profile (BPP) compatibility as EPP is a BPP extension.

The 6 basic steps of the Qi certification process:

1. The registration procedure starts with the product's manufacturer filling in an on-line form, providing basic information about the product, uploading its picture and submitting the necessary self-declaration forms. The manufacturer then selects an Authorized Test Lab (ATL) which will perform the compliance test, as well as selecting an Interoperability Testing Center (IOC) which will perform the interoperability test.
2. The Logo License Administrator (LLA) verifies whether the form is filled in correctly.
3. The Authorized Test Lab (ATL) performs the mandatory compliance tests described in the test specification and releases a test report summary.
4. The Interoperability testing Center (IOC) performs the interoperability test.
5. The product manufacturer finalizes description of the product. Uploads a picture, showing the product as it will be shipped to customers, and provides a type number identifying the product.
6. The Logo License Administrator (LLA) verifies the information is complete and consistent, after which the product is made visible in the product registration database.

Figure 77. Qi certification flowchart



For more information regarding the certification procedure, please visit the Wireless Power Consortium website.

12 Regulatory compliance information

Formal Product Notice Required by FCC:

For evaluation only; not FCC approved for resale

FCC NOTICE - This kit is designed to allow:

(1) Product developers to evaluate electronic components, circuitry, or software associated with the kit to determine

whether to incorporate such items in a finished product and

(2) Software developers to write software applications for use with the end product.

This kit is not a finished product and when assembled may not be resold or otherwise marketed unless all required FCC equipment authorizations are first obtained. Operation is subject to the condition that this product not cause harmful interference to licensed radio stations and that this product accept harmful interference. Unless the assembled kit is designed to operate under part 15, part 18 or part 95 of this chapter, the operator of the kit must operate under the authority of an FCC license holder or must secure an experimental authorization under part 5 of this chapter 3.1.2.

Formal Product Notice Required by Industry Canada

For evaluation purposes only. This kit generates, uses, and can radiate radio frequency energy and has not been tested for compliance with the limits of computing devices pursuant to Industry Canada (IC) rules.

À des fins d'évaluation uniquement. Ce kit génère, utilise et peut émettre de l'énergie radiofréquence et n'a pas été testé pour sa conformité aux limites des appareils informatiques conformément aux règles d'Industrie Canada (IC).

Notice for the European Union

The kit STEVAL-WLC38RX is in conformity with the essential requirements of the Directive 2014/53/EU (RED) and of the Directive 2015/863/EU (RoHS). Applied harmonized standards are listed in the EU Declaration of Conformity.

Notice for the United Kingdom

The kit STEVAL-WLC38RX is in compliance with the UK Radio Equipment Regulations 2017 (UK SI 2017 No. 1206 and amendments) and with the Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 (UK SI 2012 No. 3032 and amendments). Applied standards are listed in the UK Declaration of Conformity.

13 Appendix

Thermal measurements on the Board tested at 25ºC ambient temperature.

Note: The ambient temperature should not exceed 65ºC.

Figure 78. Temperature Before Power Transfer Maximum 25ºC on STEVAL-WBC86TX

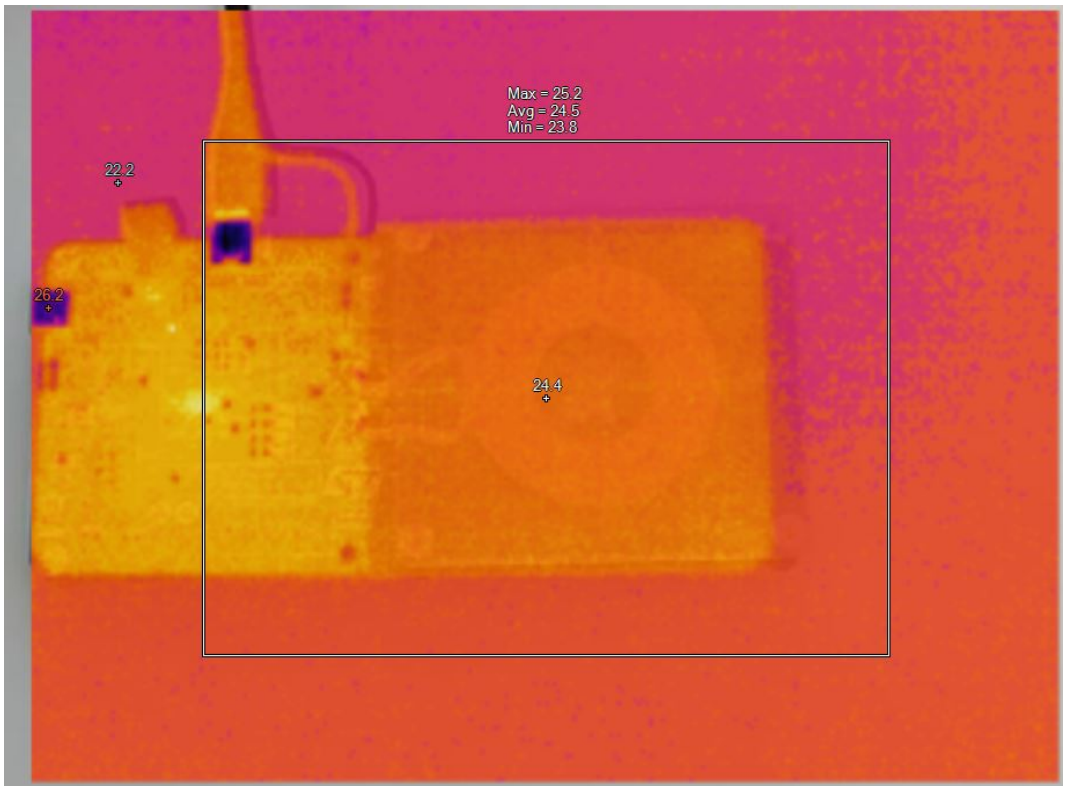


Figure 79. **Maximum Temperature of the Board after 2hours (40.8ºC) – 5W power transfer STEVAL-WLC38RX**

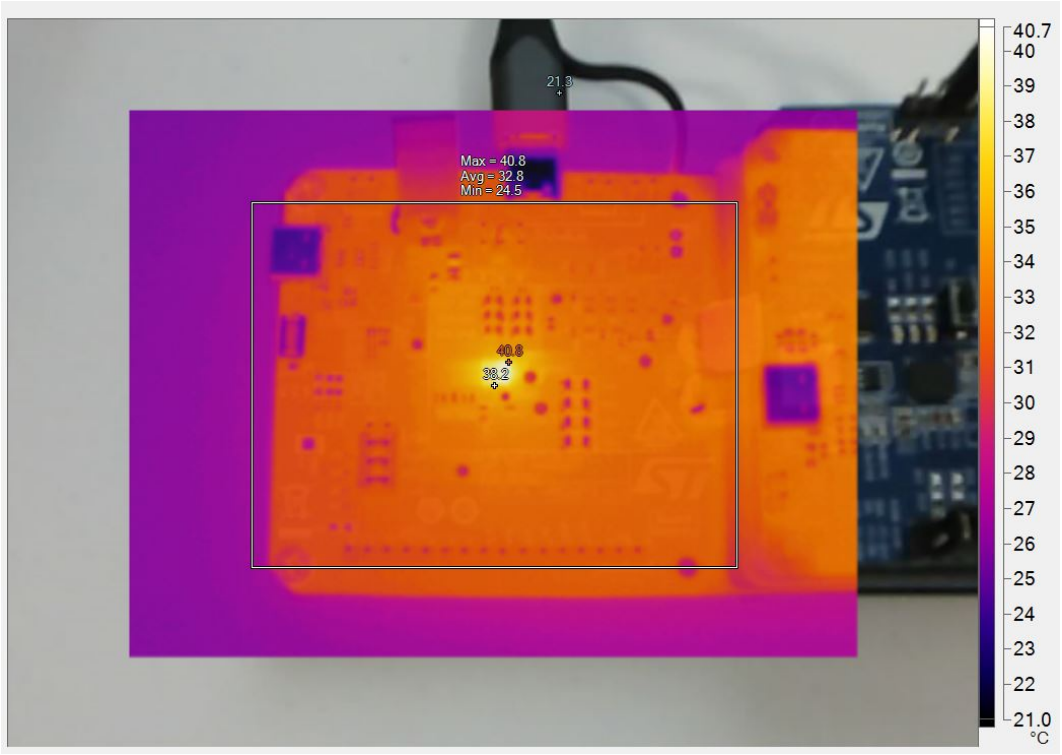


Figure 80. **Maximum Temperature of the Board after 2hours (42.5ºC) – 5W power transfer**

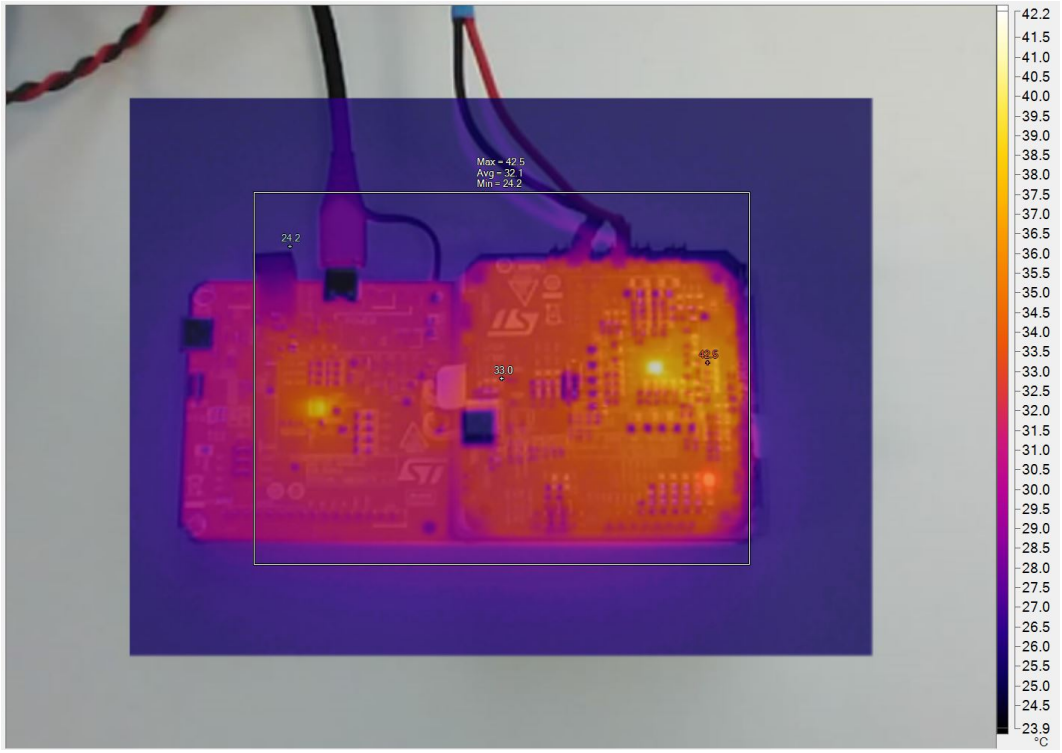
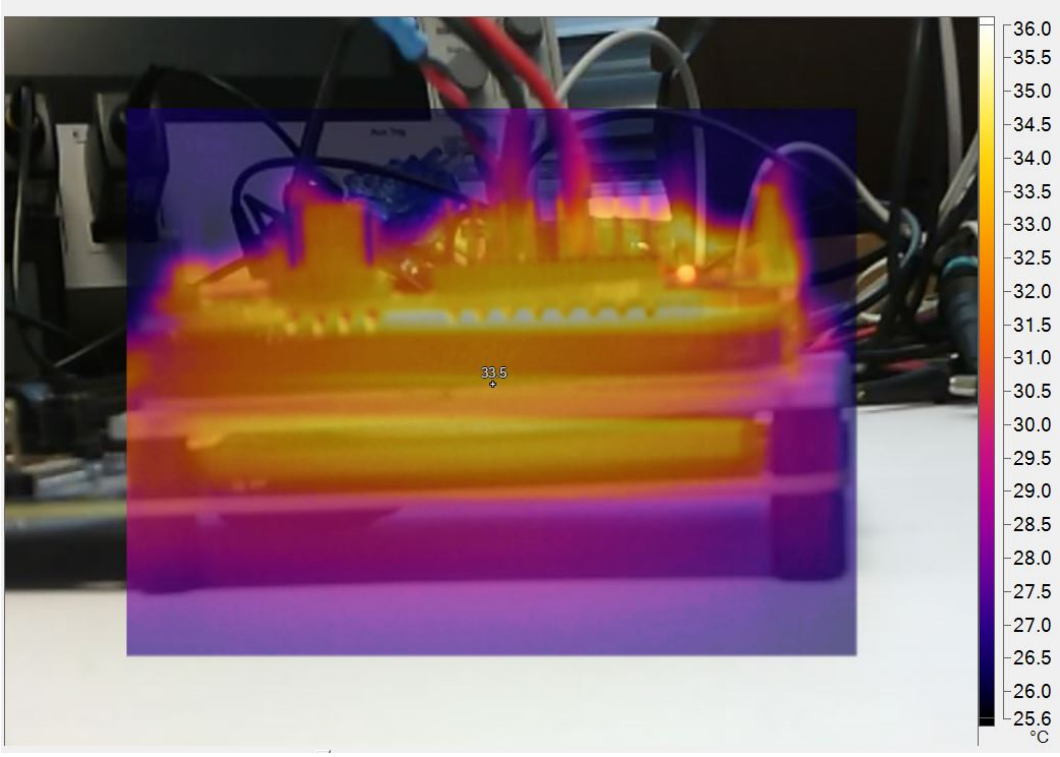


Figure 81. **Maximum Temperature of the Board and Plastic case after 2hours (33.5ºC) – 5W power transfer**



Revision history

Table 20. **Document revision history**

Date	Version	Changes
17-Jul-2023	1	Initial release.
27-Sep-2023	2	Updated Section 6 Schematic diagrams, Section 7 Bill of material (BOM). Minor text changes.

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